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DELAWARE AVENUE SEPARATED BIKEWAY Technical Report



Delaware Avenue Separated Bikeway Technical Report

Newark, Delaware

Recommended by:

Mir A. Wahed,

Mir Wahed, P.E., PTOE, PTP

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Johnson, Mirmiran & Thompson, Inc.
Vice President
Traffic Planning & Engineering





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EXECUTIVE SUMMARY

Delaware Avenue in Newark, Delaware, serves as a major thoroughfare for University of Delaware students traveling through campus. Although the bicycling volume is significant, Delaware Avenue lacks a dedicated bicycle facility for westbound cyclists. Cyclists are often seen riding on sidewalks or against the one-way traffic (contraflow) in the existing bike lane along Delaware Avenue. This cycling behavior is risky and creates an uncomfortable environment for not only the cyclists but also for pedestrians and motorists as well. Thus, a study was conducted in November 2015 that analyzed the feasibility of installing a two-way separated bicycle facility along Delaware Avenue that will accommodate both eastbound and westbound bicyclists. The two-way separated bicycle facility will extend from Orchard Road to the Pomeroy Trail. From the Pomeroy Trail to the east, the eastbound bike lane is along the south side and the westbound bike lane is along the north side of Delaware Avenue. The protected bikeway terminates at Library Avenue.

To determine best practices for signal operation of two-way separated bicycle lanes on urban roadways, JMT conducted research on bicycle facilities and bicycle signals being used in various regions of the country. The research evaluated eight case studies with various intersection design and signal operations. Common approaches, what has worked, and the challenges experienced by agencies are documented in a research summary prepared by JMT. This research summary is included in **Appendix E**.

JMT conducted this traffic analysis as a continuation of the feasibility study to review the implementation of the separated bicycle facility. As part of the analysis, peak hour turning movement counts were collected within the study area, which includes six signalized intersections as well as the major midblock pedestrian crossing, known as The Green. For this analysis, the PM peak hour traffic volumes were used as it is the time of day when the vehicular and pedestrian volumes are highest and would offer a worst-case scenario. VISSIM, a microsimulation traffic software, was used to model the existing PM peak hour traffic conditions for the study area to reflect current traffic operations. This model was then modified to incorporate a two-way separated bicycle facility along Delaware Avenue under three operational alternatives. These alternatives are:

Alternative 1- Permissive with Turn Queue Box: Bicyclists would travel along the corridor during the same signal phases as the vehicles. Any cyclist wishing to cross Delaware Avenue to the south will first pull into the two-stage turn queue box located on the north side of Delaware Avenue, at each signalized intersection, adjacent to the two-way separated bicycle facility where the cyclist will wait for a green signal to proceed south. These cyclists would travel concurrently with the side street traffic. At the Tyre Avenue intersection, two-stage turn queue boxes would be located on

the north and south side of Delaware Avenue to accommodate the eastbound bike lane that is on the south side and the westbound bike lane that is on the north side.

Alternative 2 - Protected Only: Alternative 2 creates a condition where bicyclists are only able to cross at signalized intersections during their own signal phase.

Alternative 3 – Protected and Permissive: Like Alternative 2, Alternative 3 provides an exclusive bicycle phase. However, Alternative 3 also allows cyclists to navigate intersections during vehicular signal phases.

Each of the alternatives were modeled and compared to each other and to the existing model. The results of the evaluation and comparisons are summarized within the report.

Based on the vehicular and bicycle levels of service and delay results as well as travel times and queues, and follow-up discussions with DelDOT, and other participating stakeholders, Alternative 3 is recommended. The protected and permissive characteristics of Alternative 3 provides the most options for bicyclists while reducing delays for the bicyclists and providing a shorter vehicular travel time (from Orchard Road to Tyre Avenue) when compared to Alternative 2. Allowing bicyclists more opportunities to navigate through the intersections increase signal compliance and decrease sporadic occurrences when bicyclists execute movements during their red signal phase. This will create more predictable bicycle-vehicle interactions at intersections, leading to safer conditions for all users.

Furthermore, an evaluation of the signal operations for Alternative 3 was also conducted. Under Alternative 3, bicyclists will be given an advanced protected bicycle signal phase followed by a permissive signal phase that will be shared with through motor vehicles with a green ball. Two options for signal operations during the shared signal phase was reviewed for bicyclists and left turning vehicles. The first option would have bicyclists see a green bicycle signal face while motor vehicles in the left lane see a flashing yellow left turn arrow signal face. The second option would also have motor vehicles in the left lane see a flashing yellow left turn arrow signal face but bicyclists see a flashing yellow bicycle signal face.

Both options would have flashing yellow left turn arrow signal faces indicating to vehicles that they should yield to bicyclists. However, the second option is recommended as the flashing yellow bicycle signal face communicates to bicyclists that they are not operating during a protected phase, leading to more cautious movements at the signalized intersections.

INTRODUCTION

The State of Delaware has been a national leader in advancing safe, accessible and enjoyable bicycling. In 2015, Delaware was named the 3rd most Bike Friendly State in the U.S., the highest ranked state on the east coast. This ranking is based on successes in a variety of categories including bicycle legislation, enforcement, policies, programs, infrastructure, and education. Delaware Avenue, located in the City of Newark, has been prioritized by the Newark Bike Plan as a desirable location for a two-way bicycle facility. As part of the state's ongoing plan to improve bicycle infrastructure, the Delaware Department of Transportation (DelDOT) agreed that this location should be investigated as a candidate for the installation of a two-way separated bicycle facility. This technical report analyzes the implementation of a two-way separated bicycle facility along Delaware Avenue between Orchard Road and Library Avenue from a traffic operational perspective.

Background

Delaware Avenue is located in the City of Newark, along a 15-mile regional bicycle route (Route NC-3) where it provides access to and from two major urban centers in New Castle County and connects to White Clay Creek State Park. The route also provides access to major highways, statewide bicycle routes, regional bicycle routes, and recreational bicycle connectors.

Study Area

The study area, which is shown in **Figure 1** below, is the portion of Delaware Avenue between Orchard Road and Library Avenue, located in the heart of the University of Delaware campus. The study area includes six signalized intersections, numerous commercial entrances and a major midblock pedestrian crossing, known as The Green.

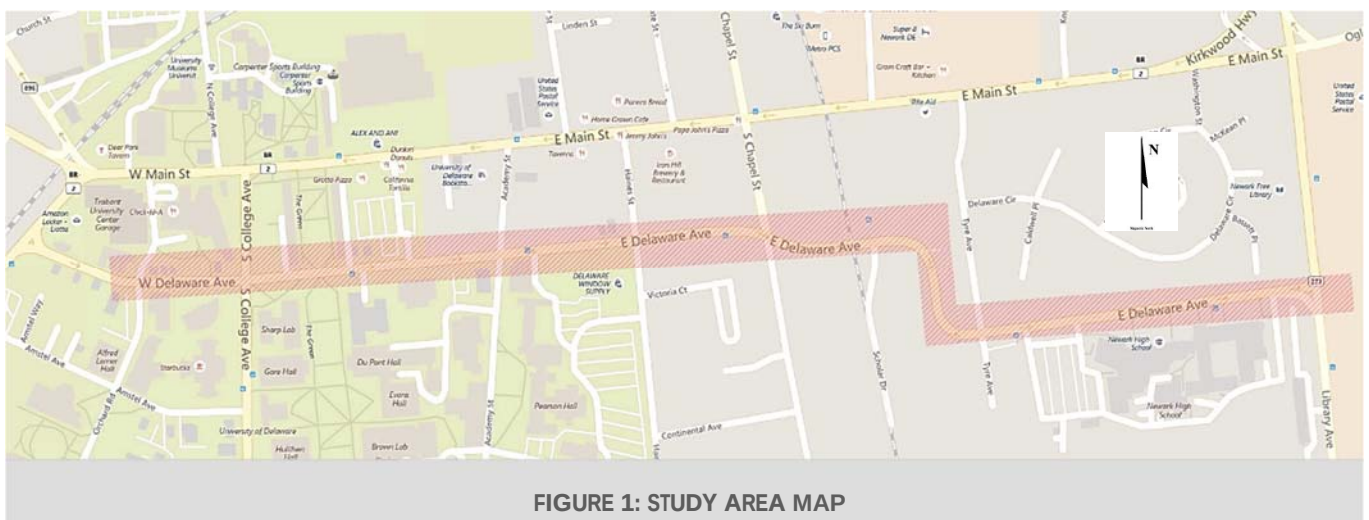


FIGURE 1: STUDY AREA MAP

Purpose

Delaware Avenue serves as a major thoroughfare for University of Delaware students traveling through campus. Although the bicycling volume is significant, Delaware Avenue lacks a dedicated bicycle facility for westbound cyclists. Cyclists are often seen riding on sidewalks or against the one-way traffic (contraflow) in the existing bike lane along Delaware Avenue. This cycling behavior is risky and creates an uncomfortable environment for not only the cyclists but pedestrians and motorists as well.



Photo 1: WB Cyclist in Sidewalk

A feasibility study was conducted in November 2015 that analyzed the feasibility of installing a two-way separated bicycle facility along Delaware Avenue that will accommodate both eastbound and westbound bicycle travel. The study identified potential solutions to conflict areas within the corridor, many of the recommendations were based on the guidelines provided by National Association of City Transportation Officials (NACTO), FHWA's Separated Bike Lane Planning and Design Guide, and other federal design guidelines. This feasibility study also included a summary of the extensive community involvement effort and public opinion documented during the project.

This technical report takes the process one step further by analyzing the existing conditions and traffic volumes associated with implementing a two-way separated bicycle facility. VISSIM microsimulation software was used to analyze how effective the two-way separated bicycle facility would be within the project area.

EXISTING ROADWAY CONDITIONS

Delaware Avenue is a 1.2-mile arterial roadway with two eastbound lanes that connect South Main Street to Library Avenue. It is part of a paired one-way road network through downtown Newark with East Main Street serving westbound traffic. A bike lane, varying from 3' to 6', is striped along the right side of the road to accommodate eastbound bicycle traffic. The speed limit is posted at 25 mph throughout the corridor. Traffic volumes along the roadway vary from 7,500 to 10,000 ADT.

Existing conditions at each of the project intersections as well as The Green are as follows:

- **Orchard Road** - At the intersection of Delaware Avenue and Orchard Road, Delaware Avenue is a one-way street with two eastbound vehicular travel lanes and an eastbound bike lane located on the south side of the street. To the south, Orchard Road is a two-way street with one northbound and one southbound lane. To the north, Orchard Road is a one-way southbound street providing access from a parking garage and Trabant University Center to Delaware Avenue. The intersection is signalized and operates under a three-phase operation with each of the eastbound, northbound, and southbound movements having a protected phase. Additionally, there are pedestrian phase movements on the north, south, and western legs of the intersection.
- **South College Avenue** - At the intersection of Delaware Avenue and South College Avenue, Delaware Avenue is one-way street with two eastbound vehicular travel lanes and an eastbound bike lane located on the south side of the street. At the intersection, the two eastbound lanes are designated as one through-left lane and one shared through-right lane. To the south, South College Avenue is a two-way street with one northbound and one southbound lane. At the intersection, the northbound lane widens to two lanes with one through lane and one right-turn only lane. To the north, South College Avenue is two-way street with one northbound and one southbound lane. At the intersection, the southbound lane widens to two lanes with one through lane and one left-turn lane. The intersection is signalized and operates under a four-phase operation: eastbound; southbound with protected left; northbound and southbound with permissive southbound left; and a pedestrian only phase. The pedestrian phase is an exclusive pedestrian phase that allows pedestrians on all movements to cross the four crosswalks that cover each leg of the intersection. During this exclusive pedestrian phase, pedestrians can also execute diagonal crossing maneuvers.
- **The Green** - At the University of Delaware, The Green is a large landscaped pedestrian friendly area that spans the length of the campus and is surrounded by University buildings. At The Green's current intersection with Delaware Avenue, an unsignalized midblock pedestrian crosswalk exists with associated mid-block crossing traffic signing. Delaware Avenue is a one-way street with two eastbound vehicular travel lanes and an eastbound bike lane located on the south side. This intersection is proposed to be signalized to facilitate controlled pedestrian movements.
- **Galleria Entrance** - The Galleria is a commercial development with retail and restaurant uses. At the intersection of Delaware Avenue and the entrance to the Galleria, Delaware Avenue is one-way street with two eastbound vehicular travel lanes and an eastbound bike lane located on the south side of the street. The entrance creates a T-intersection to

the north with one lane in each direction. The entrance is atypical in that the entering lane is to the west, and the exiting lane is to the east. The existing intersection is unsignalized.

- **Academy Street** - At the intersection of Delaware Avenue and Academy Street, Delaware Avenue is a one-way street with two eastbound vehicular travel lanes and an eastbound bike lane located on the south side of the street. At the intersection, the two eastbound lanes are designated as one through-left lane and one shared through-right lane. To the south, Academy Street is a two-way street with one northbound and one southbound lane. At the intersection, the northbound lane widens to two lanes with one through lane and one right-turn lane. To the north, Academy Street is a two-way street with one northbound and one southbound lane. At the intersection, the southbound movement widens to two lanes with one through lane and one left-turn lane. The intersection is signalized and operates under a three-phase operation: southbound with protected left; north and southbound with permissive southbound left; and eastbound movement. The pedestrian movements operate with each of the corresponding traffic movements. There are existing crosswalks on all four legs of the intersection.
- **Haines Street** - At the intersection of Delaware Avenue and Haines Street, Delaware Avenue is a one-way street with two eastbound vehicular travel lanes and an eastbound bike lane located on the south side of the street. To the north and to the south, Haines Street is a two-way street with one northbound and one southbound lane. There are no turn lanes at the intersection. The intersection is currently unsignalized with the northbound and southbound Haines Street approaches as stop controlled. There are crosswalks on all four legs of the intersection; however, there are no pedestrian signal accommodations.
- **South Chapel Street** - At the intersection of Delaware Avenue and South Chapel Street, Delaware Avenue is a one-way street with two eastbound vehicular travel lanes and an eastbound bike lane located on the south side of the street. At the intersection, the two through eastbound lanes operate as one through lane and one shared through/right-turn lane. An eastbound left-turn lane is also provided at the intersection. To the south, South Chapel Street is a two-way street with one northbound and one southbound lane. At the intersection, the northbound lane widens to two lanes with one through lane and one channelized right-turn lane. To the north, South Chapel Street is a two-way street with one northbound and one southbound lane. At the intersection, the southbound movement widens to two lanes with one through lane and one left-turn lane. The intersection is signalized and operates under a three-phase operation: southbound with protected left; northbound and southbound with permissive southbound left; and eastbound movement. The pedestrian movements operate with each of the corresponding traffic movements. There are existing crosswalks on all four legs of the intersection.

- **Pomeroy Trail** – Just east of the intersection with South Chapel Street, the Newark and Pomeroy Trail crosses Delaware Avenue. The Pomeroy Trail offers urban connections to the University of Delaware campus, Newark Shopping Center, DART Transit Hub and the downtown area, as well as the recreational amenities of White Clay Creek State Park. The current intersection with Delaware Avenue is signalized with pedestrian push button signal activation. At the intersection with the Pomeroy Trail, Delaware Avenue is a one-way street with two eastbound vehicular travel lanes as well as bike lanes located on both sides of Delaware Avenue. The south side bike lane is a continuation of the one present along the project corridor while the bike lane on the north side began at the intersection of Delaware Avenue and South Chapel Street. This signal operates under a two-phase operation with one phase for the eastbound vehicular traffic and the second phase as an exclusive phase for the trail's pedestrian traffic.
- **Tyre Avenue** – At the intersection of Delaware Avenue and Tyre Avenue, Delaware Avenue is a one-way street with two eastbound vehicular travel lanes and eastbound bike lane located on the south side of the road. At the intersection, the two through eastbound lanes operate as one shared through/left-turn lane and one shared through/right-turn lane. To the north and to the south, Tyre Avenue is a two-way street with one northbound and one southbound lane. There are no northbound or southbound exclusive turn lanes. The intersection is signalized with advanced signalization located just west of the intersection for the eastbound movement to provide advance warning of the signal given the limited sight distance due to an existing curve approaching the Tyre Avenue intersection.

DATA COLLECTION

Existing Volumes

Peak hour turning movement counts were conducted along the project corridor on March 8, 2016 during the hours of 7:00-9:00 AM and 4:00-6:00 PM at the following project intersections along Delaware Avenue: Orchard Road, South College Avenue, The Green, Galleria Entrance, Academy Street, Haines Street, South Chapel Street, Pomeroy Trail, and Tyre Avenue. These peak hour turning movement counts included all vehicular movements as well as pedestrians and bicycles. The vehicular existing 2016 peak hour turning movement counts are shown in **Figure 2** on Page 12. Detailed turning movement counts for each intersection are included in **Appendix A**. For this analysis, PM peak hour traffic volumes were used as it is the time of day when the vehicular and pedestrian volumes are highest and would offer a worst-case scenario.

Projected Volumes

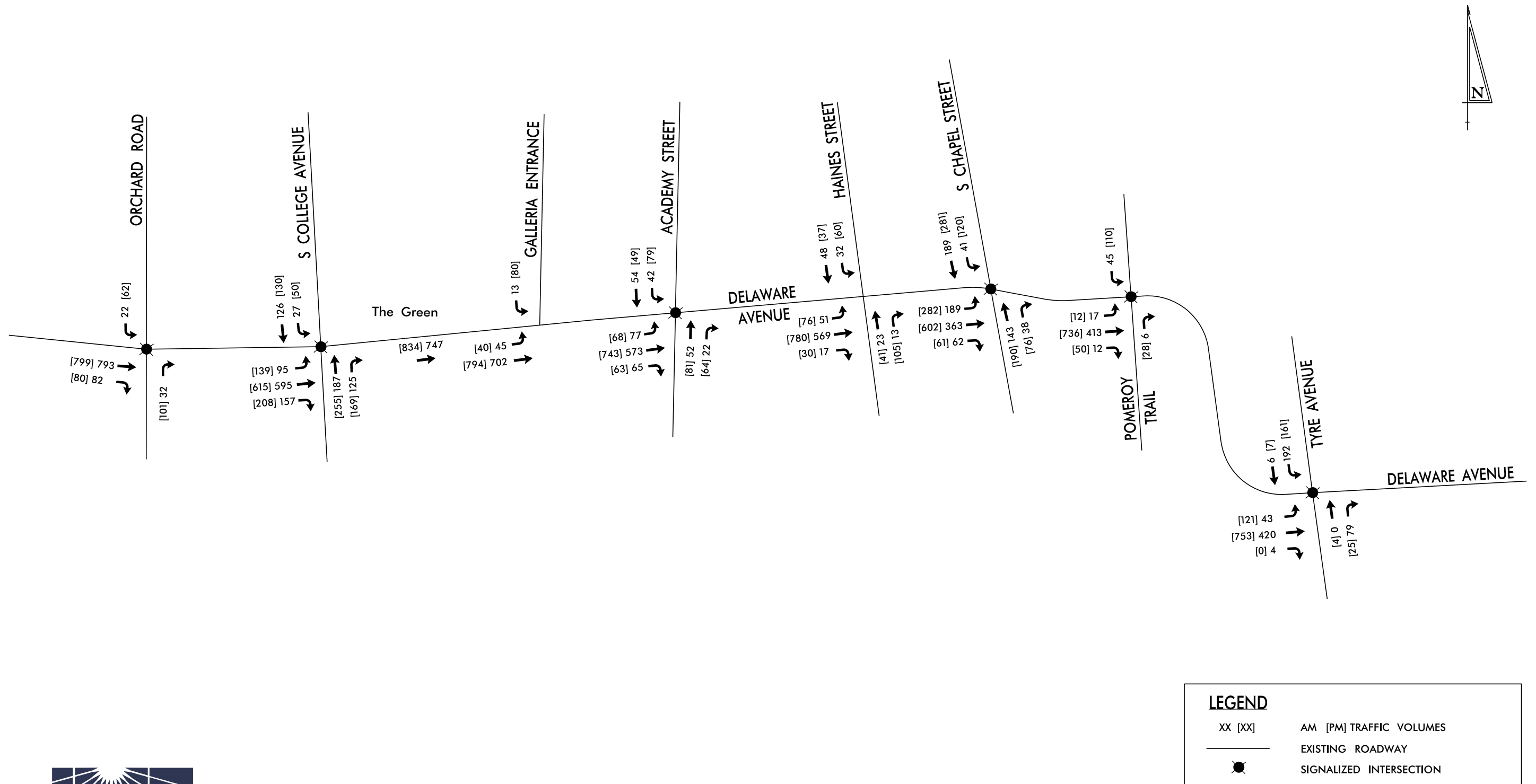
For the analysis of the project corridor, projected 2016 traffic volumes are based off the 2016 peak hour turning movement counts observed in the field on March 8, 2016, only modified slightly to balance traffic volumes between intersections. Due to the proximity of the University of Delaware campus and the potential growth of bicycle users due to the construction of the two-way separated bicycle facility, bicycle volumes were doubled and balanced for the purpose of the traffic analysis. The 2016 projected volumes used for this analysis are shown in **Figures 3, 4, and 5** on pages 13,14 and 15.



2016 WEEKDAY AM AND PM TRAFFIC VOLUMES

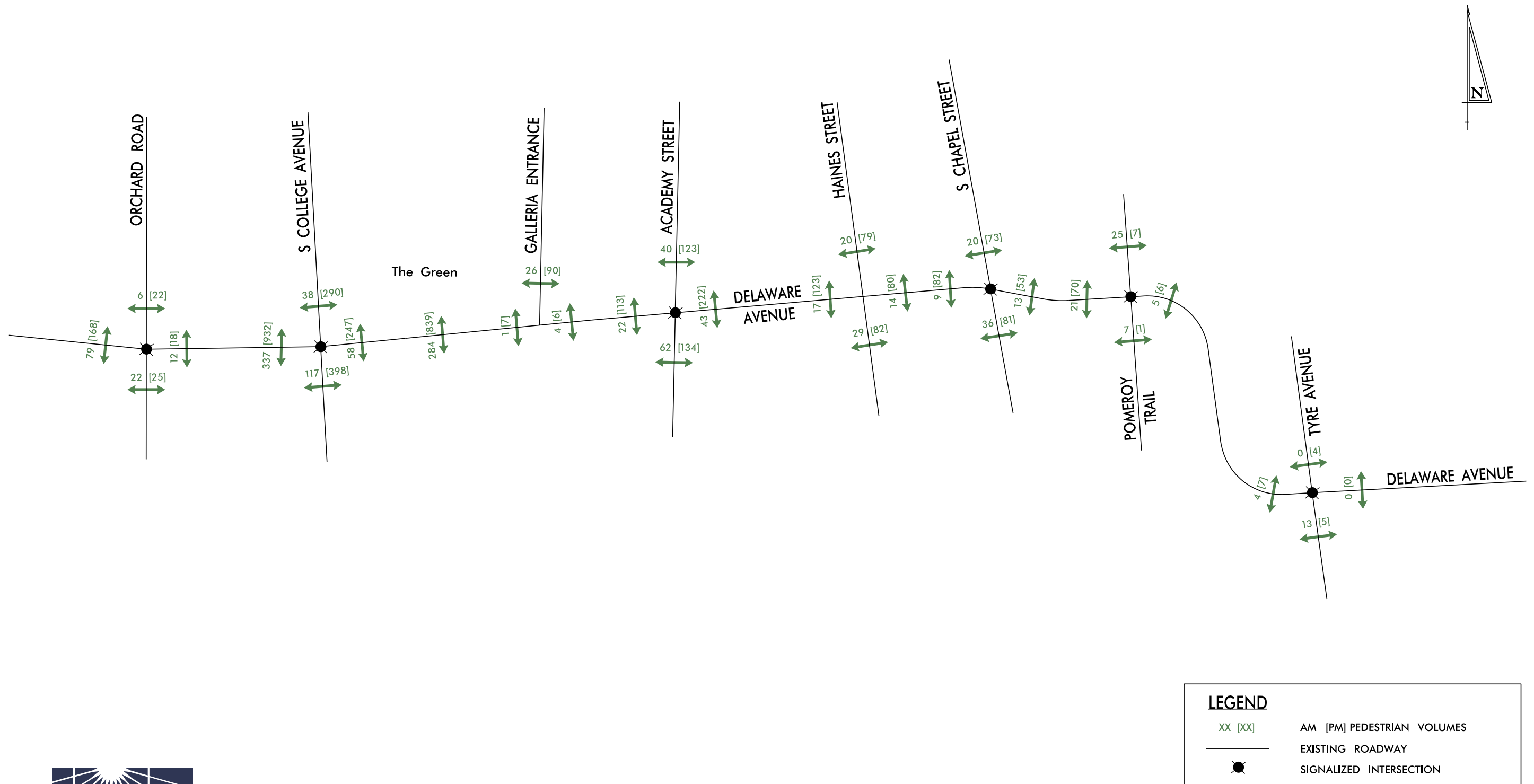
DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

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2016 WEEKDAY AM AND PM PEDESTRIAN VOLUMES

DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT



PROJECTED BIKE VOLUMES

DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

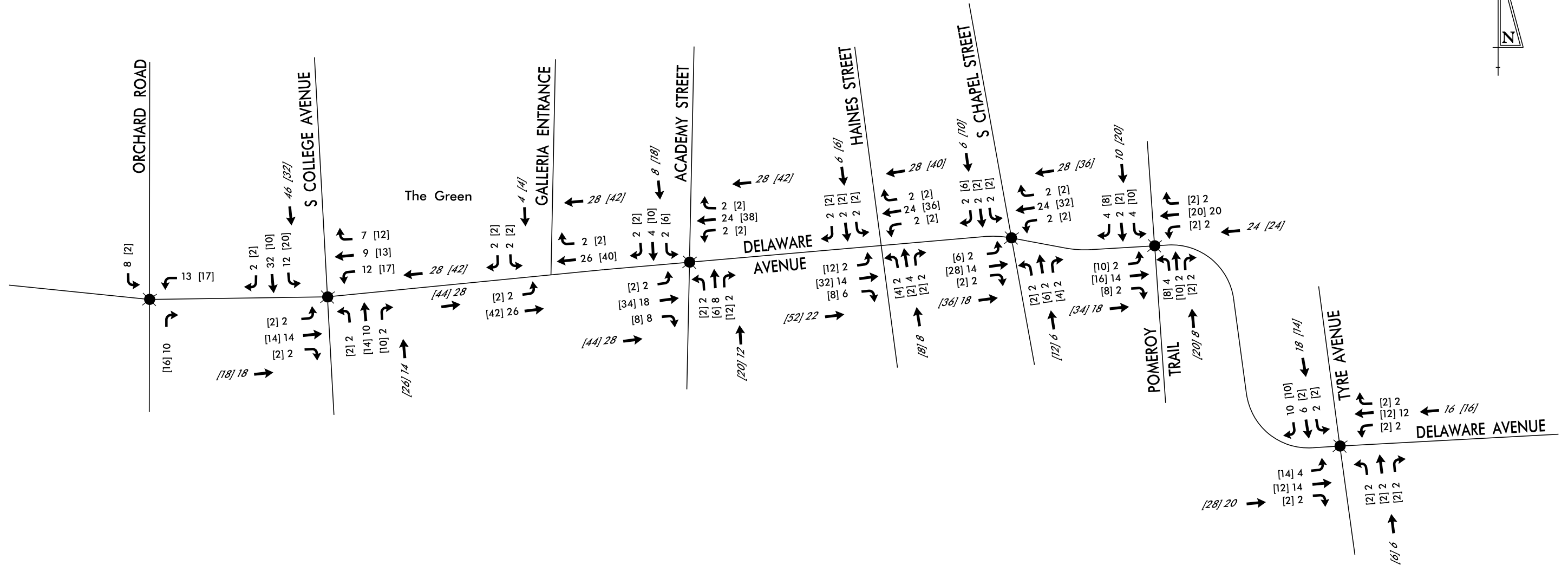


FIGURE 5 FEBRUARY 2018

ALTERNATIVES

As part of the Feasibility Study (November 2015), various options were evaluated with regards to the location of the two-way separated bicycle facility. The Feasibility Study determined that locating the two-way separated bicycle facility along the north side of Delaware Avenue from Orchard Road to the Pomeroy Trail Crossing would be the ideal solution.

The facility would terminate west of the Orchard Road intersection. This termination point was established due to the complexity of the South Main Street intersection with Delaware Avenue. At the termination point, cyclists traveling west along Delaware Avenue would be encouraged to turn left towards Amstel Avenue to access South Main Street. Conversely, cyclists traveling north on South Main Street would be encouraged to turn right on Amstel Avenue to access the separated bicycle facility at Orchard Road.

The Feasibility Study also identified splitting the separated bicycle facility into two one-way separated bike lanes from the Pomeroy Trail Crossing to Library Avenue. From the Pomeroy Trail Crossing to Library Avenue the eastbound separated bike lane is striped along the south side of Delaware Avenue and the westbound separated bike lane is striped along the north side of Delaware Avenue.

Once the alignment of the two-way separated bicycle facility was established, JMT evaluated various options for the operation of the bicycle facility, particularly at intersections along the project corridor. JMT conducted research on bicycle facilities and bicycle signals being used throughout the country to determine best practices for signal operations of two-way separated bike lanes on urban roadways. Out of this research, the team decided to analyze the operation of a two-way separated bicycle facility along Delaware Avenue within the project limits under three proposed alternatives. Each of the three alternative designs and operations are described in detail below. Graphical figures depicting the operation of each Alternative can be found in **Appendix B**.

Alternative 1: Permissive with Turn Queue Box

Alternative 1 implements a two-stage turn queue box throughout the project corridor. With this alternative, cyclists can travel along Delaware Avenue concurrently with vehicles and a two-stage turn queue box would be located in the roadway on the northwest corner of each signalized intersection to assist cyclists conducting turning movements that conflict with vehicles. Specifically, any cyclist wishing to cross Delaware Avenue to the south will utilize the turn queue box located on the north side of Delaware Avenue adjacent to the two-way separated bicycle facility where the cyclist will wait for a green signal allowing the south movement. These cyclists will run concurrently with the side street traffic. Note, at the Tyre Avenue intersection, two-stage

turn queue boxes would be located on the north and south side of Delaware Avenue as the eastbound bike lane would be on the south side of the roadway and the westbound bike lane would be on the north side of the roadway.

Alternative 2: Protected Only

Alternative 2 creates a condition where any cyclist wishing to cross Delaware Avenue can do so only during an exclusive bicycle signal phase at a signalized intersection. This exclusive bicycle phase will be an all way stop to vehicular traffic and allow for bicycle traffic along Delaware Avenue to cross whatever direction they choose within the intersection. During the vehicular signal phases, cyclists traveling along Delaware Avenue will receive the red signal.

Alternative 3: Protected and Permissive

Like Alternative 2, Alternative 3 provides an exclusive bicycle phase. However, Alternative 3 also allows for permissive movements by the cyclists. That is, cyclists will be allowed to cross Delaware Avenue while vehicular traffic is in motion. This allows cyclists more opportunity to cross Delaware Avenue in lower traffic and non-peak hour conditions. However, during the permissive phase, cyclists conducting eastbound right turn and westbound left turn movements would be required to yield to vehicles.

ANALYSIS METHODOLOGY

VISSIM

VISSIM, a microsimulation traffic software, was used to model the study area to reflect current traffic operations. This software allows each mode of transportation (car, bicycle, pedestrian, etc.) to be simulated individually, including a signal controlled bike lane. This provides for greater control when it comes to more accurately simulating the actions of all roadway users. JMT reviewed both the AM and PM peak hour vehicular, pedestrian, and bicycle traffic and determined to use the PM peak hour volumes for the *VISSIM* model to represent the worst case scenario.

Calibration Techniques

It is common practice to calibrate any microsimulation model after one is initially created to generate the most reliable output. Calibration in this case means to:

1. Create a model for existing 2016 PM peak hour conditions;
2. Measure or obtain real time key factors in the field; and
3. Compare the output of the model with those variables measured in the field.

There are several factors that can be modified within the model to best simulate existing conditions. These include:

Signal Timings

The model used existing Delaware Department of Transportation (DelDOT) provided signal cycle lengths and offsets. Signal split times were based on field observations. Signal operation and phasing was also field verified.

Travel Times

Observed field travel times for vehicular traffic were conducted for the project corridor. This consisted of a team member driving the project corridor during the PM peak hours while video recording the travel and documenting the travel time along the project corridor as a whole. These travel times were used as a baseline of comparison for the travel times produced by the VISSIM model. If there was a notable degree of discrepancy between the model output and the conditions observed/measured in the field, then the VISSIM model was modified/updated. The models were calibrated to be within 10% of the field collected travel times. **Appendix C** contains the calibration tables.

Volumes

All the models (existing and the alternatives) used the proposed 2016 traffic volumes. The models were calibrated to be within 10% of the proposed volumes. **Appendix C** contains the calibration tables.

Queues

The queue lengths produced by the Existing 2016 model were field verified and were found to be similar to the queues observed.

For the bicycle traffic, the default parameters provided within VISSIM were used to model the driver behaviors. Additionally, typical bicyclist speeds at intersections as published in the Design Guidance for Bicycle Signal Heads, prepared by the National Association of City Transportation Officials (NACTO) were incorporated into the model.

ANALYSIS RESULTS

Comparison of Alternatives

Once the existing model for the 2016 PM peak hour was generated and calibrated, three different PM peak hour models were created based on the alternatives discussed before. These models reflect the operational condition defined by each alternative. All three alternative traffic models were compared with the existing condition traffic model based on various measurable outputs. Signal cycle lengths of 120 seconds were used for the existing and alternative models. The analysis results of each comparison are shown in the following tables.

Vehicular LOS and Delay Results

The *Highway Capacity Manual* (HCM) 2010 defines the Level of Service (LOS) for signalized and unsignalized intersections as a function of the average vehicle control delay. LOS may be calculated per movement or per approach for any intersection configuration and is defined based on delay in seconds and assigned a corresponding letter "A" through "F". The delay output from VISSIM and the corresponding LOS letter from HCM 2010 for each intersection along the project corridor for all four traffic models are summarized in **Table 1** below.

**Table 1: Vehicular Intersection Delay and LOS
PM Peak Hour**

Intersection	Existing		Alternative 1 (Permissive with Turn Queue Box)		Alternative 2 (Protected Only)		Alternative 3 (Protected and Permissive)	
	Delay (s)	LOS	Delay (s)	LOS	Delay (s)	LOS	Delay (s)	LOS
Orchard St	22.0	C	17.6	B	18.3	B	18.5	B
S College Ave	49.9	D	47.3	D	62.1	E	65.4	E
Academy St	25.8	C	26.1	C	24.3	C	24.5	C
S Chapel St	29.4	C	30.1	C	36.1	D	28.1	C
Tyre Ave	18.5	B	19.1	B	17.0	B	18.0	B

Overall, the difference between the levels of service for each alternative at each intersection is relatively minor. At the intersection with South College Avenue, Alternative 1 would experience improvement in delay while Alternatives 2 and 3 would experience a 12 second and a 15 second increase in delay changing the overall intersection LOS. The intersection with South Chapel Street is the only other intersection where the alternatives show notable change from existing conditions with Alternative 2 increasing the anticipated delay by nearly 7 seconds. At Orchard Street, the anticipated delay for all the alternatives would remain within 2 seconds of existing conditions. At the intersections with Academy Street and Tyre Avenue, the delay for all the alternatives would be within 1.6 seconds when compared to existing conditions.

Bicycle Delay Results

As part of the VISSIM model reporting, the model will calculate the expected delay, similar to the vehicular delay presented in Table 1. These results are presented for each alternative in **Table 2** on page 21.

It is important to note that under existing conditions, there is only a defined eastbound movement for cyclists. As a result, eastbound right turns were modeled as free flow movements since the bike lane is located on the south side of Delaware Avenue and therefore, right turns could be negotiated without vehicle conflict. As shown in Table 2, Alternative 2 poses the most delay to cyclists as they are forced to wait for a protected signalized phase at all intersections. Alternatives 1 and 3 allow the cyclists more flexibility, with Alternative 3 providing the most opportunity for cyclists to complete their desired maneuver. Thus, for the most part, Alternative 3 has the lowest delay per intersection and per movement.

**Table 2: Bicycle Delay
PM Peak Hour**

Intersection	Movement	Existing	Alternative 1 (Permissive with Turn Queue Box)	Alternative 2 (Protected Only)	Alternative 3 (Protected and Permissive)
		Delay (s)	Delay (s)	Delay (s)	Delay (s)
Orchard St	WBL	-	6.4	26.6	8.2
	EBT	5.8	-	-	-
S College Ave	WBL	-	60.1	82.7	29.2
	WBT	-	28.4	68.1	29.8
	WBR	-	38.3	35.7	11.1
	EBL	78.9	39.1	44.8	26.2
	EBT	32.6	26.4	62.2	19.6
	EBR	-	66.4	12.2	8.2
Academy St	WBL	-	66.5	50.9	5.7
	WBT	-	28.5	55.8	16.7
	WBR	-	30.0	20.7	20.9
	EBL	-	32.1	23.9	29.8
	EBT	22.9	18.9	59.0	13.5
	EBR	16.2	58.2	83.9	8.6
S Chapel St	WBL	-	2.7	14.2	22.5
	WBT	-	14.7	53.9	22.2
	WBR	-	64.0	58.4	15.6
	EBL	20.3	17.0	48.4	28.0
	EBT	22.3	18.7	42.9	28.5
	EBR	2.5	15.9	10.4	12.6
Tyre Ave	WBL	-	65.7	66.1	9.0
	WBT	-	4.0	63.9	6.4
	WBR	-	0.3	5.2	0.6
	EBL	5.1	65.7	61.9	9.5
	EBT	0.5	5.5	54.7	11.2
	EBR	-	0.7	16.3	1.1

Queue Results

As part of the VISSIM reporting, the model provides 95th percentile queue lengths for each approach at an intersection. These results are presented for each alternative in **Table 3** on below.

**Table 3: Vehicular 95th Percentile Queue
PM Peak Hour**

Intersection	Approach	PM 95th Percentile Queues (ft)			
		Existing	Alternative 1 (Permissive with Turn Queue Box)	Alternative 2 (Protected Only)	Alternative 3 (Protected and Permissive)
Orchard Rd	NB	202	154	127	131
	SB	97	80	80	77
	EB	246	205	216	205
S College Ave	NB	333	357	358	363
	SB	141	145	156	168
	EB	491	473	457	500
Academy St	NBT	90	91	107	105
	NBR	90	91	107	105
	SBL	81	83	93	93
	SBT	81	83	94	94
	EB	383	389	311	354
Chapel St	NB	254	266	260	298
	SBL	237	204	207	137
	SBT	236	204	207	137
	EB	324	292	363	293
Tyre Ave	NB	23	51	35	38
	SB	220	234	195	197
	EB	0	0	0	0

Given the different alternatives, the anticipated queue lengths for the various intersections and approaches either improve (get shorter) or increase by approximately 50 feet or less. At the intersection with Chapel Street, the queue for the eastbound movement increases by 39 feet with Alternative 2. This is likely related to the limited vehicle storage along the eastbound left-turn lane. Also, at the intersection with Chapel Street, the queue for the northbound movement increases by 44 feet with Alternative 3. This is likely related to the permissive condition allowing for more opportunity for bikes and pedestrians to utilize the island for refuge and then to cross the channelized right-turn lane.

Travel Time Results

Similar to the queueing report, VISSIM also calculates travel times. These travel time results are presented for each alternative in **Table 4** below and are from Orchard Road to Tyre Avenue.

**Table 4: Travel Times for the Complete Project Corridor
PM Peak Hour**

From Orchard Road to Tyre Avenue	Alternatives			
	Existing	Alternative 1 (Permissive with Turn Queue Box)	Alternative 2 (Protected Only)	Alternative 3 (Protected and Permissive)
Vehicular Travel Time (min)	4.3	3.7	3.9	3.7
Cyclists EB Travel Time (min)	5.2	5.4	7.0	5.9
Cyclists WB Travel Time (min)	N/A	5.7	7.5	5.6

As indicated in Table 4, Alternatives 1, 2, and 3 would reduce the vehicular travel times along the corridor when compared to existing conditions. This is due to the presence of an existing unsignalized midblock pedestrian crossing at The Green. However, in all the alternatives, The Green was signalized which reduces the overall travel times as vehicles only must stop for a red signal rather than yield for every pedestrian. The model incorporates this improvement as The Green is expected to be signalized prior to the installation of the separated bike lane.

For the eastbound cyclists, travel times for Alternatives 1 and 3 are comparable to existing conditions since cyclists can continue to travel along Delaware Avenue concurrently with vehicles during the vehicular green signal phase. Alternative 2 increases the eastbound travel time for cyclists as they receive the red signal phase during the eastbound and westbound vehicular green signal phases. With regards to westbound cyclists, their travel times are consistent with those of the eastbound cyclists all while being provided a safer alternative to what they have currently available under existing conditions.

Pros and Cons of Each Alternative

Each alternative was analyzed and a list of pros and cons for each is shown below:

Alternative 1 - Permissive with two-stage turn queue box

Pros:

- Safer option for bicyclists who are executing turning movements at the signalized intersections.
- Shorter vehicular travel time from Orchard Road to Tyre Avenue when compared to Alternative 2.
- Shorter bicyclists travel time when compared to Alternative 2.
- Signalized intersections would operate at the same LOS as existing conditions.
- Minimum impact to vehicular operation.

Cons:

- Bicyclists may not utilize the turn queue boxes and utilize a gap in traffic instead.
- Size of turn queue box may not be sufficient if high cyclist volumes utilizing box are present.
- Northbound vehicles conflict with northbound left turning bicyclists.
- Stop bars will have to be located further from the intersection to allow placement of turn queue box.

Alternative 2 – Protected only

Pros:

- Beneficial when significant left-turn volumes are present and/or high opposing traffic volumes.
- Safer option in cases when it is difficult for bicyclists to make judgement of acceptable gaps.
- Some bicyclists feel safer executing left turn movements with a protected arrow.

Cons:

- Bicyclists may not obey the signal and execute the turn once they find a suitable gap, creating unpredictability.
- Bicyclists delay is the longest.
- Bicycle-protected phase takes away from the green time of the vehicular phases.
- Reduces intersection operational efficiency.
- Level of service at S. Chapel Street intersection degrades from LOS C to LOS D.
- Longer vehicular and bicyclists travel time when compared to the other alternatives.

Alternative 3 – Protected and Permissive

Pros:

- Reduces delays for the bicyclists when compared to Alternative 2 since left turning bicyclists may have the opportunity to execute the turns once an acceptable gap is found.
- Shorter vehicular travel time from Orchard Road to Tyre Avenue when compared to Alternative 2.
- Shorter travel time for bicyclists when compared to Alternative 2.
- Should enhance bicycle signal compliance.

Cons:

- May not be beneficial for westbound bicyclists in case of high opposing traffic volume.
- Level of service at S. College Avenue intersection degrades from LOS D to LOS E.
- A potential for increased conflict between vehicles and bicyclists.

RECOMMENDATIONS

The purpose of this technical report was to determine the impact of adding a two-way separated bicycle facility to Delaware Avenue between Orchard Road and Library Avenue as well as to determine which of the three operational Alternatives would work the best for this project corridor. In order to determine which alternative is best suited for the project corridor, JMT evaluated the study area based on several factors, some of which are summarized below.

Intersection Delay

Table 1 indicates that the anticipated vehicular delay and levels of service between the three alternatives and existing conditions was minimal. The largest change in delay was at South College Avenue under Alternative 3. The overall intersection LOS did change from a LOS D to a LOS E with an increase of delay of approximately 16 seconds. Additionally, Alternative 2 had approximately 12 more seconds of delay at the South College Avenue intersection. However, all other changes at each intersection for each alternative were less than 7 seconds. Therefore, with regards to delay, the two-way separated bicycle facility has minimal impact on the project corridor, and Alternatives 1, 2 and 3 produce similar vehicular delay results.

Bicycle Delay

With regards to cyclist delay, as shown in Table 2, Alternative 2 poses the most delay to cyclists as they are forced to wait for a protected signalized phase at all intersections. Alternatives 1 and 3 allow the cyclists more flexibility with Alternative 3 providing the most opportunity for cyclists to complete their desired maneuver and thus resulting in the lowest delay per intersection and per movement.

Queue

The queuing results shown in Table 3 indicate that there is minimal difference between the queueing anticipated from the alternatives when compared to the existing conditions. Only a few instances were greater than one car length. Therefore, with regards to vehicular queueing, the two-way separated bicycle facility has minimal impact on the project corridor.

Travel Time

The travel time results shown in Table 4 indicates that all alternatives will improve the vehicular travel time along the project corridor especially due to the installation of a new signal at The Green. With regards to cyclists, Alternative 2 has the highest travel times, due to the limited signal phases under which they are allowed to maneuver along Delaware Avenue. The travel times for Alternatives 1 and 3 are comparable to the existing conditions for both eastbound and westbound cyclists since cyclists can continue to travel along Delaware Avenue concurrently with vehicles during the vehicular green signal phase.

Based on the vehicular and bicycle levels of service and delay results as well as travel times and queues, and follow-up discussions with DelDOT, and other participating stakeholders, Alternative 3 is recommended. The protected and permissive characteristics of Alternative 3 provides the most options for bicyclists while reducing delays for the bicyclists and providing a shorter vehicular travel time (from Orchard Road to Tyre Avenue) when compared to Alternative 2.

Furthermore, an evaluation of the signal operations for Alternative 3 was also conducted. Under Alternative 3, bicyclists will be given an advanced protected bicycle signal phase followed by a permissive signal phase that will be shared with motor vehicles. Two options for signal operations during the shared signal phase was reviewed. **Appendix D** contains graphical depictions of the signal operation for each of these two options. As part of this evaluation, the location of the exclusive pedestrian phase within the phasing sequence was also examined. It was determined that having the exclusive pedestrian phase prior to the protected bicycle phase generated comparable results to having the exclusive pedestrian phase after the protected bicycle phase. Therefore, it is recommended the exclusive pedestrian phase take place prior to the protected bicycle phase.

The first option for the signal operations during the shared signal phase, which received FHWA approval in the City of Spartanburg in South Carolina, would have bicyclists see a green bicycle signal face while motor vehicles in the left lane see a flashing yellow left turn arrow signal face. The second option, which received FHWA approval in the City of Saint Paul in Minnesota, would also have motor vehicles in the left lane see a flashing yellow left turn arrow signal face but bicyclists see a flashing yellow bicycle signal face.

Both options would have flashing yellow left turn arrow signal faces indicating to vehicles that they should yield to bicyclists. However, the second option is recommended as the flashing yellow bicycle signal face communicates to bicyclists that they are not operating during a protected phase, leading to more cautious movements at the signalized intersections.

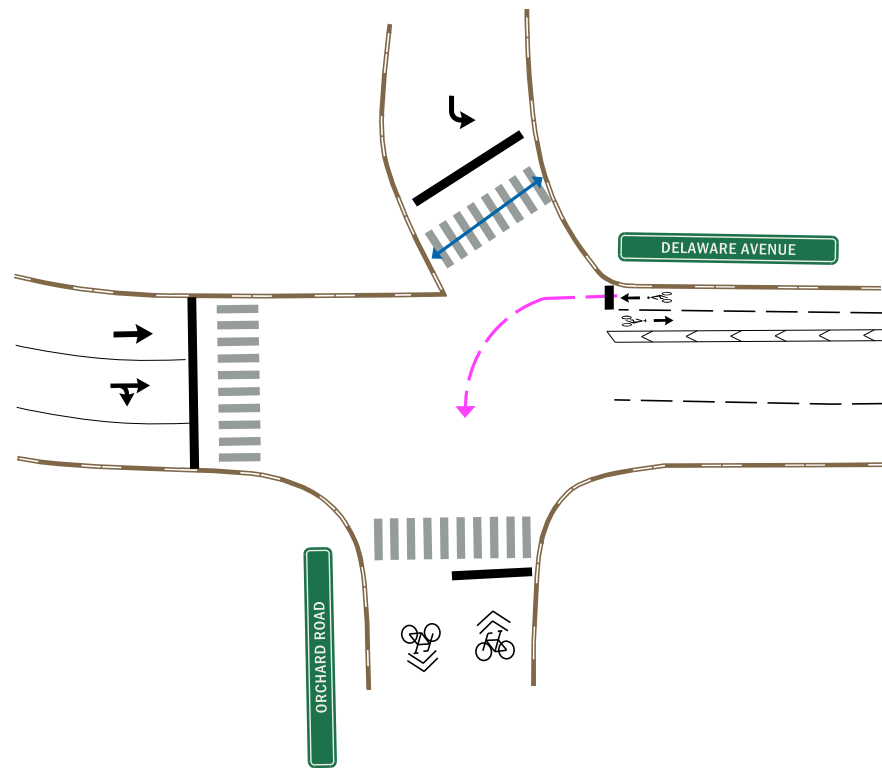
The recommended signal phasing at the Orchard Road, S. College Avenue, Academy Street, S. Chapel Street, and Tyre Avenue intersections are depicted on Figures 6, 7, 8, 9, and 10, respectively.

On November 29, 2017, DelDOT received FHWA approval on the recommended signal phasing (second option). As part of the FHWA approval, a before and after evaluation will be required of the bicycle signal operations. Therefore, it is recommended that DelDOT and/or the City of Newark monitor this corridor after construction and perform an assessment to evaluate the operation and performance of the bicycle signals to ensure safety and efficiency as well as evaluate the motorist/bicyclist interaction and behavior.

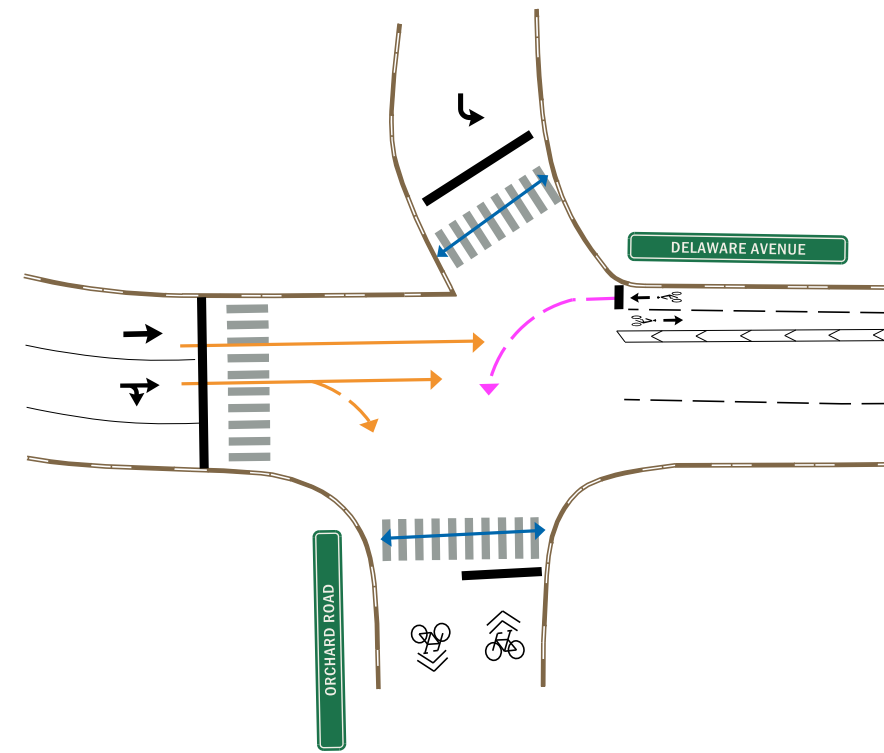
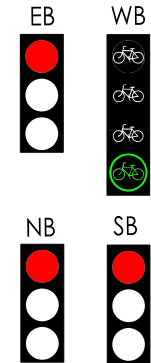
DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

DELAWARE AVENUE/ORCHARD ROAD - RECOMMENDED SIGNAL PHASING

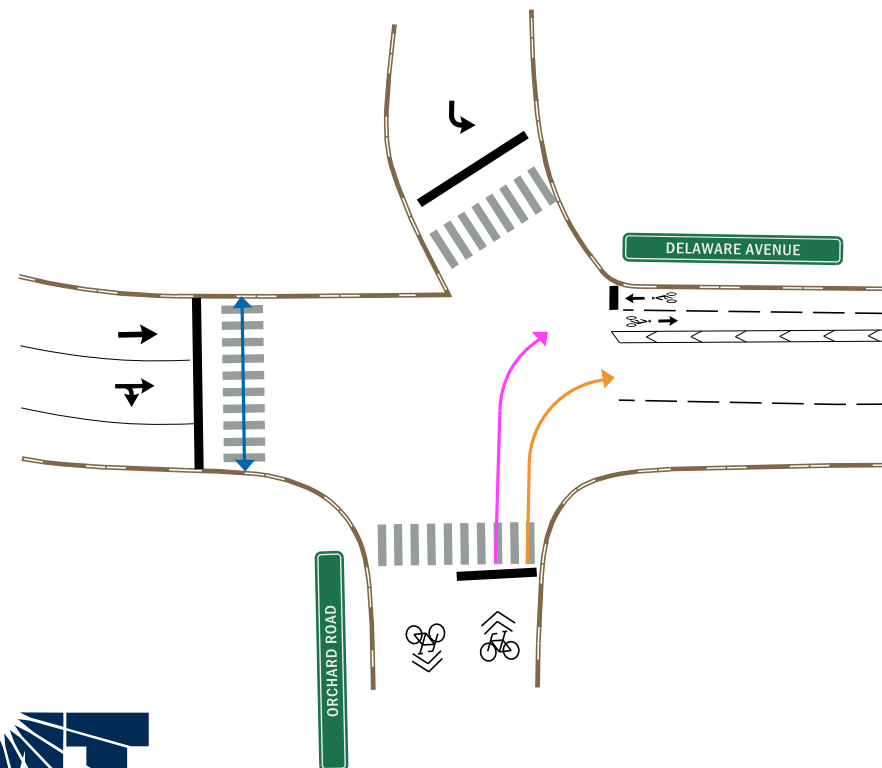
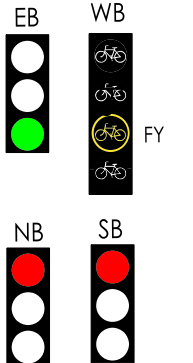
28



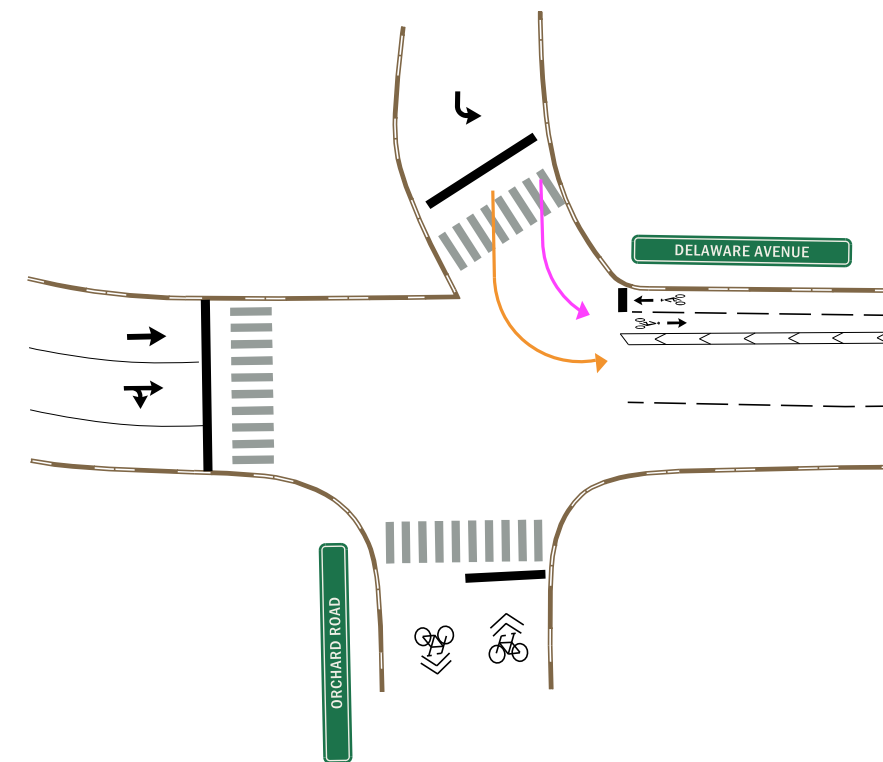
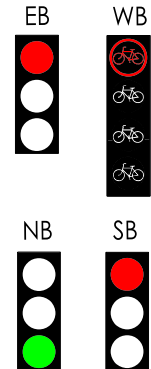
PHASE 1



PHASE 2



PHASE 3



PHASE 4

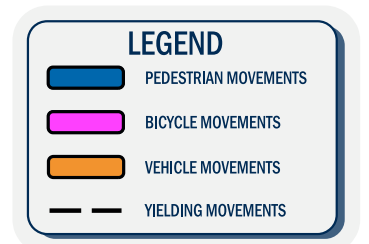
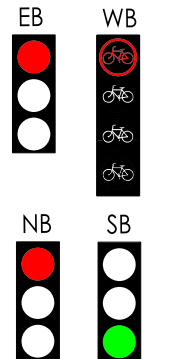


FIGURE 6

FEBRUARY 2018

DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

DELAWARE AVENUE/S. COLLEGE AVENUE - RECOMMENDED SIGNAL PHASING

29

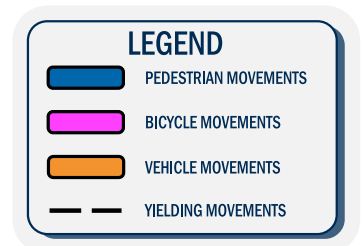
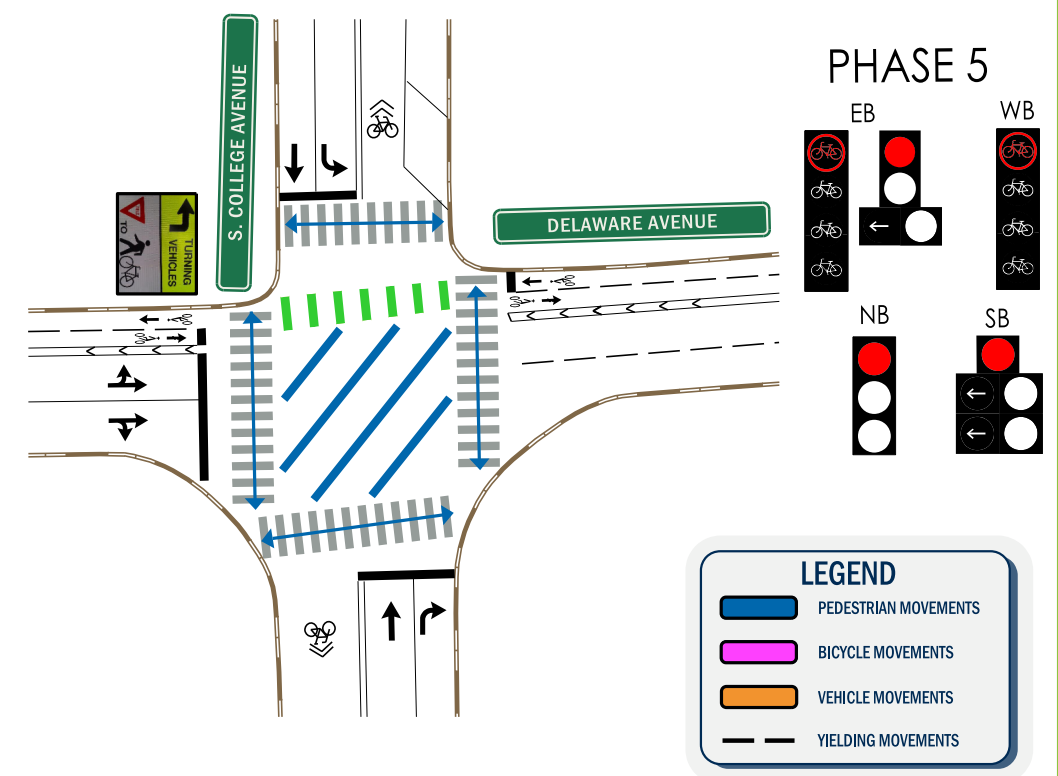
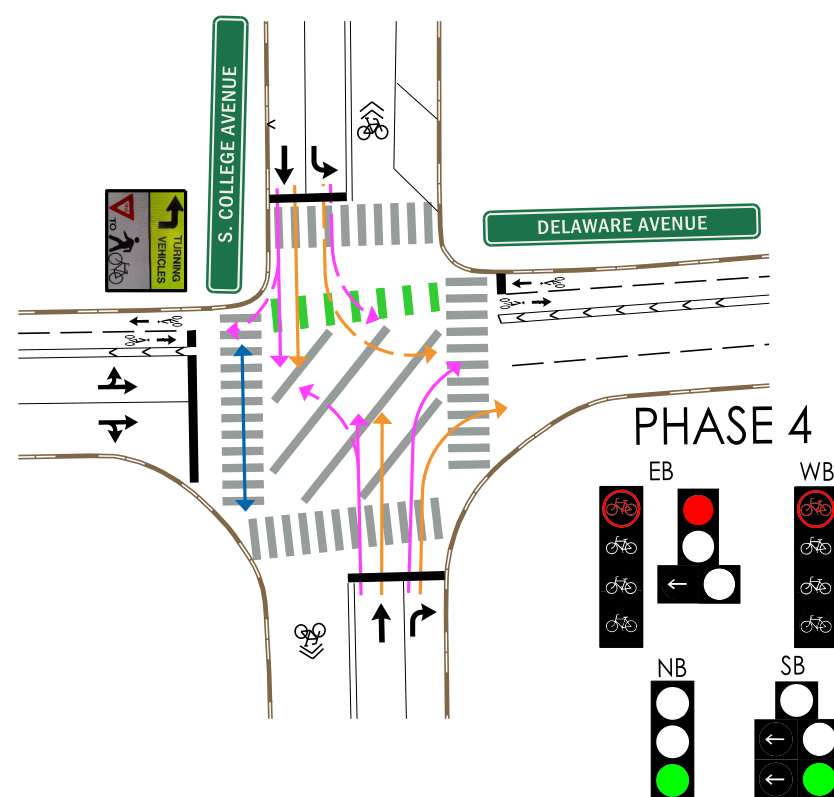
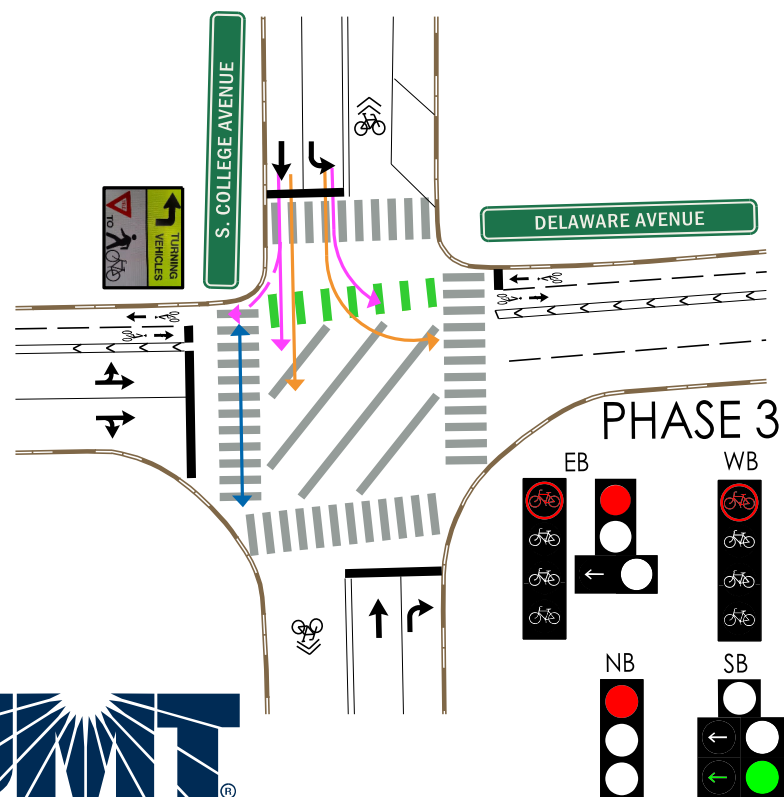
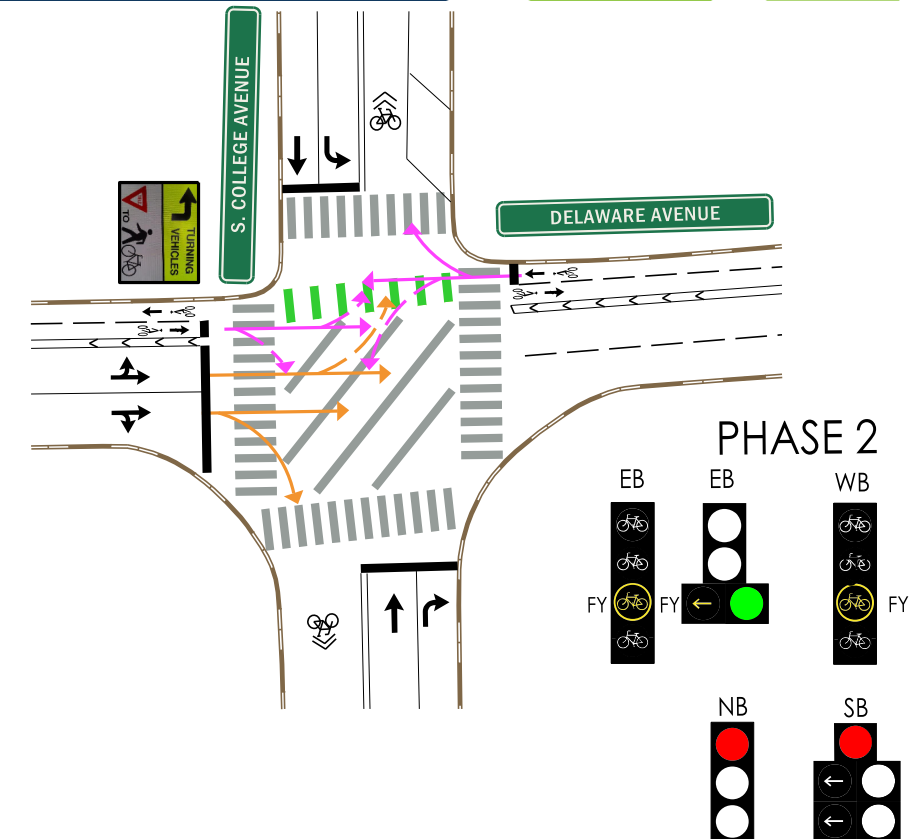
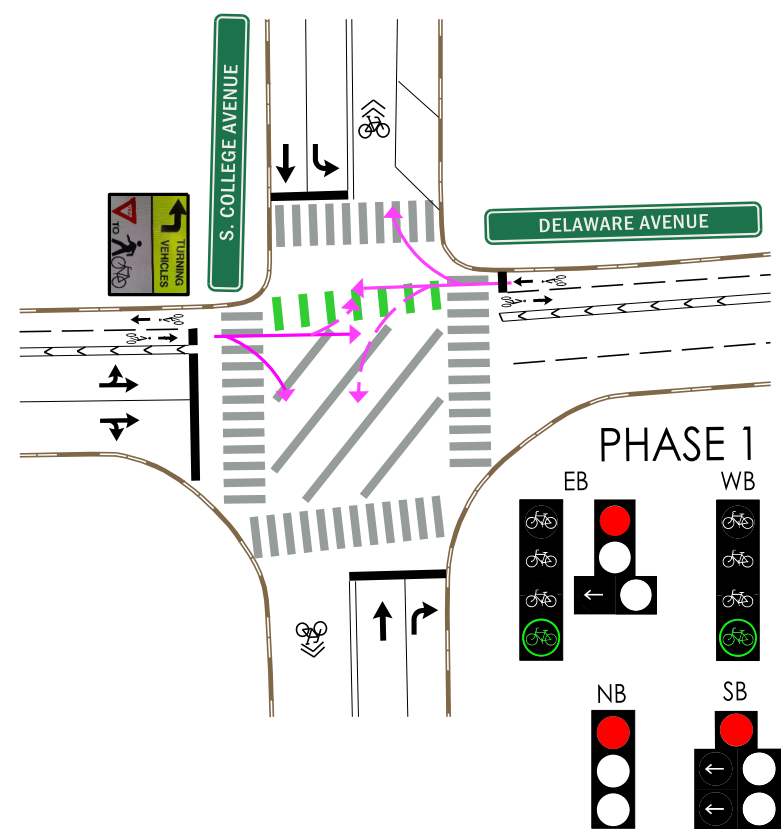


FIGURE 7

FEBRUARY 2018

DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

DELAWARE AVENUE/ACADEMY STREET - RECOMMENDED SIGNAL PHASING

30

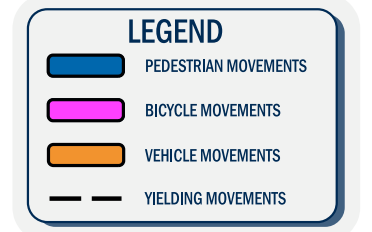
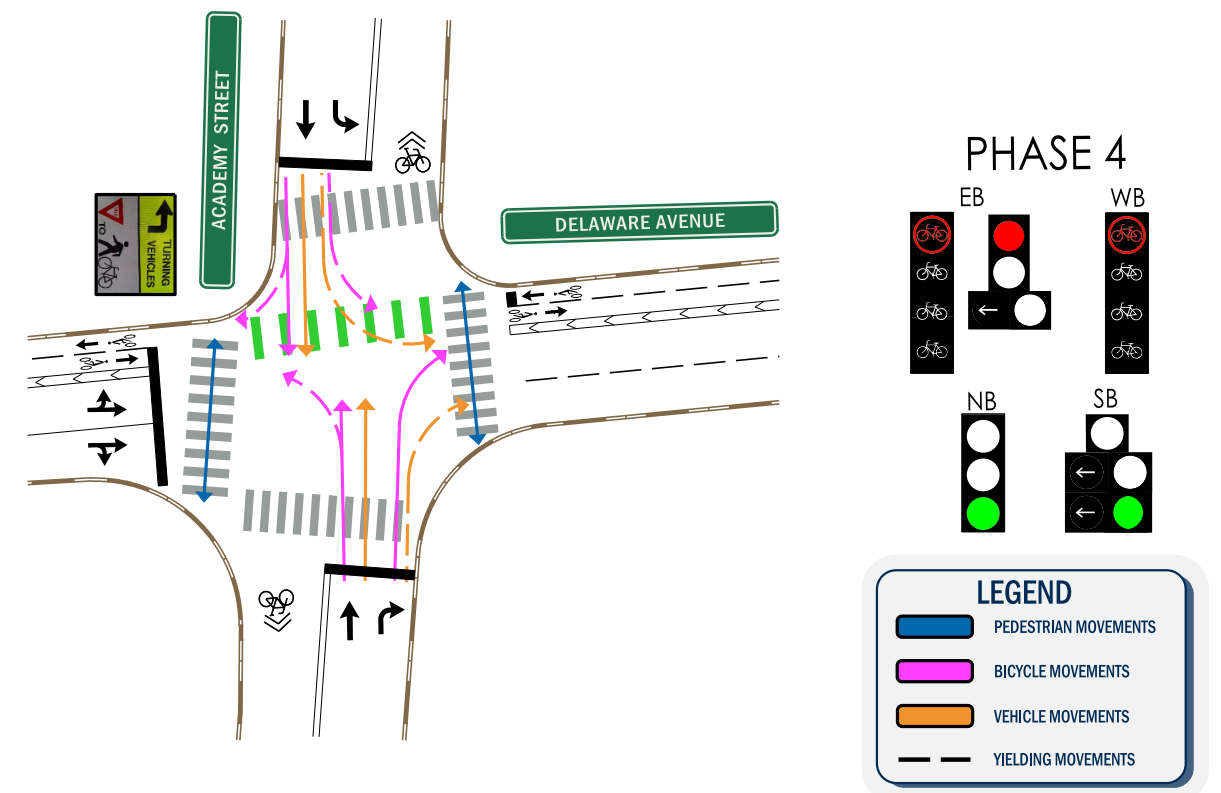
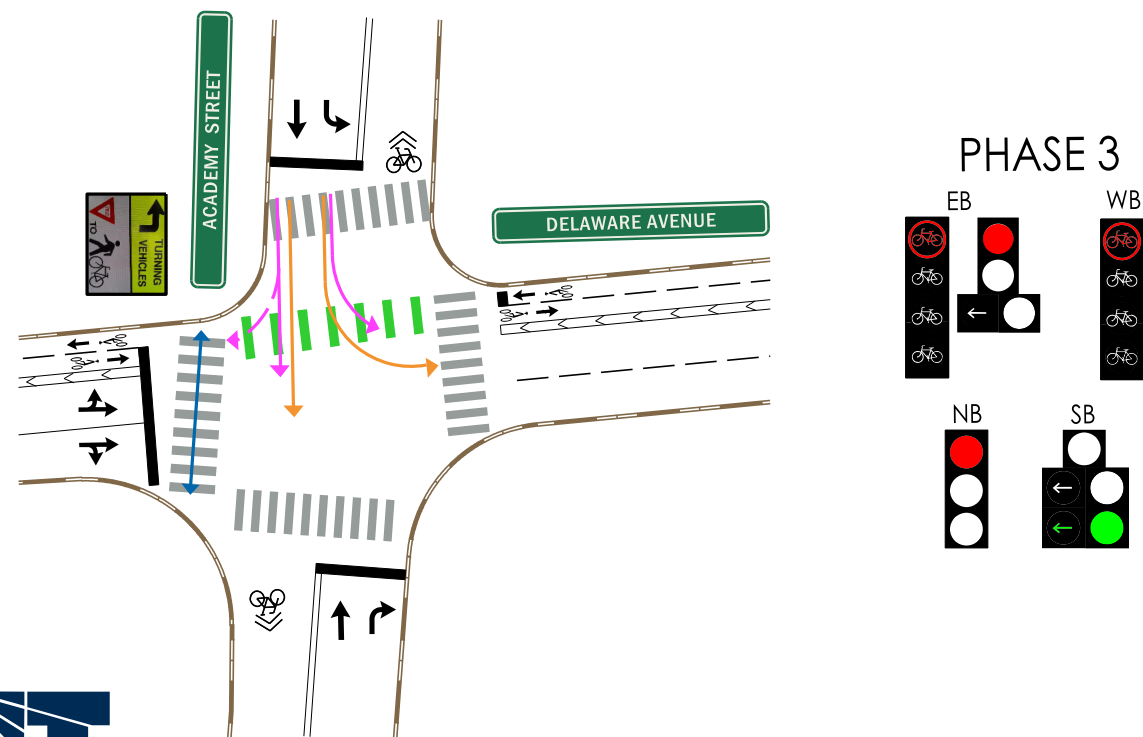
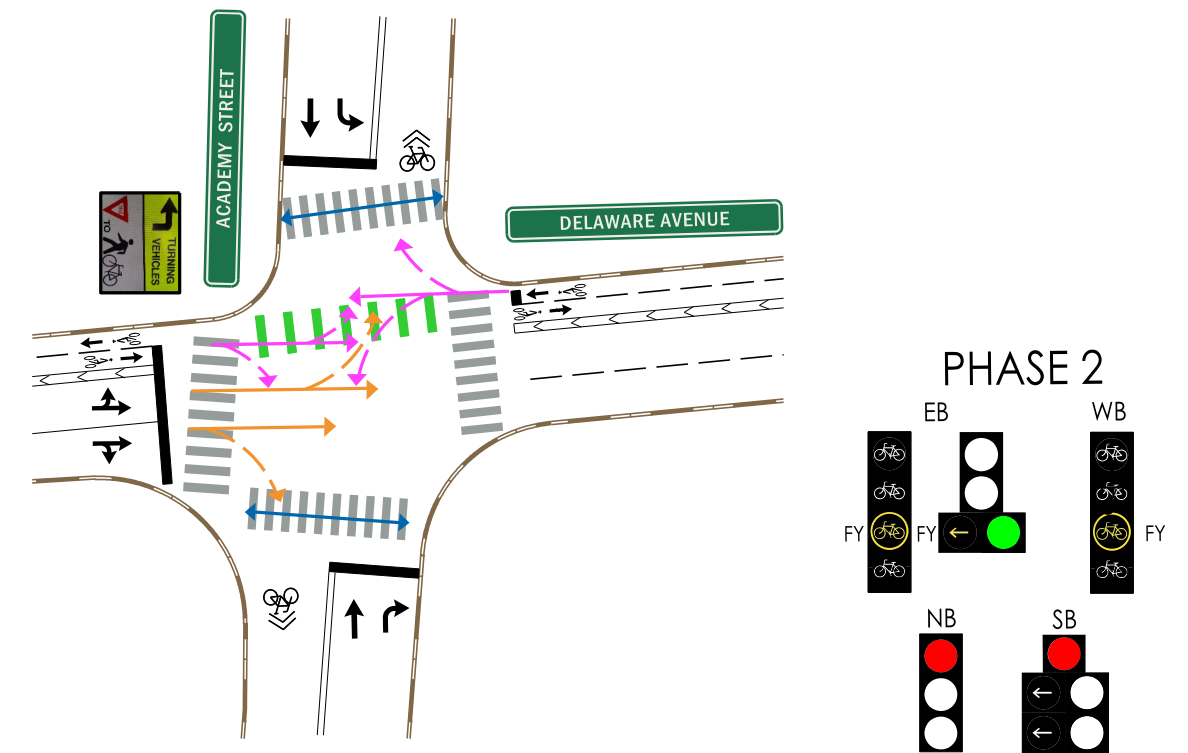
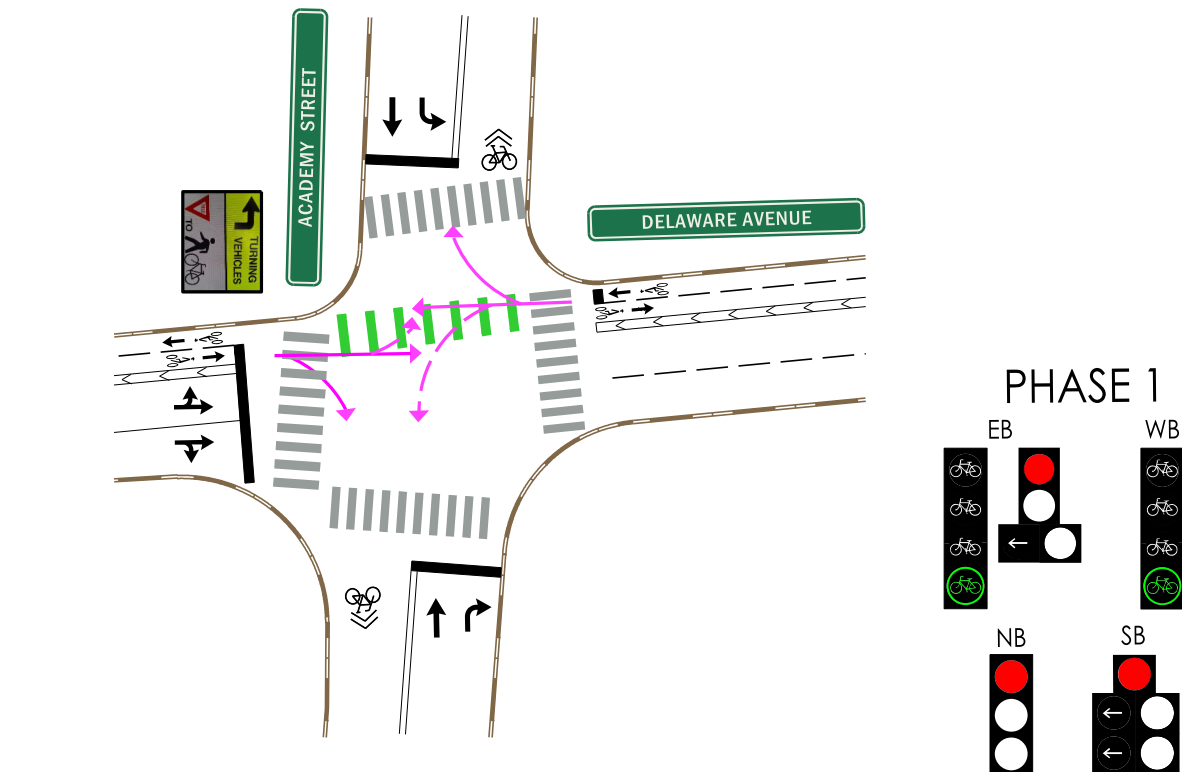


FIGURE 8

FEBRUARY 2018

DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

DELAWARE AVENUE/S. CHAPEL STREET - RECOMMENDED SIGNAL PHASING

31

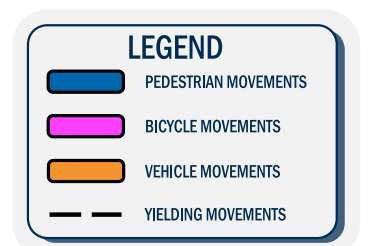
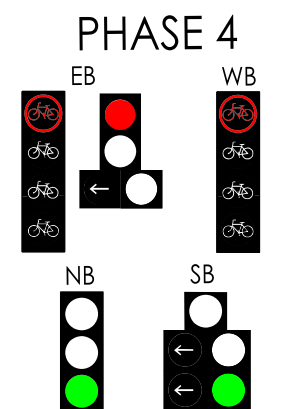
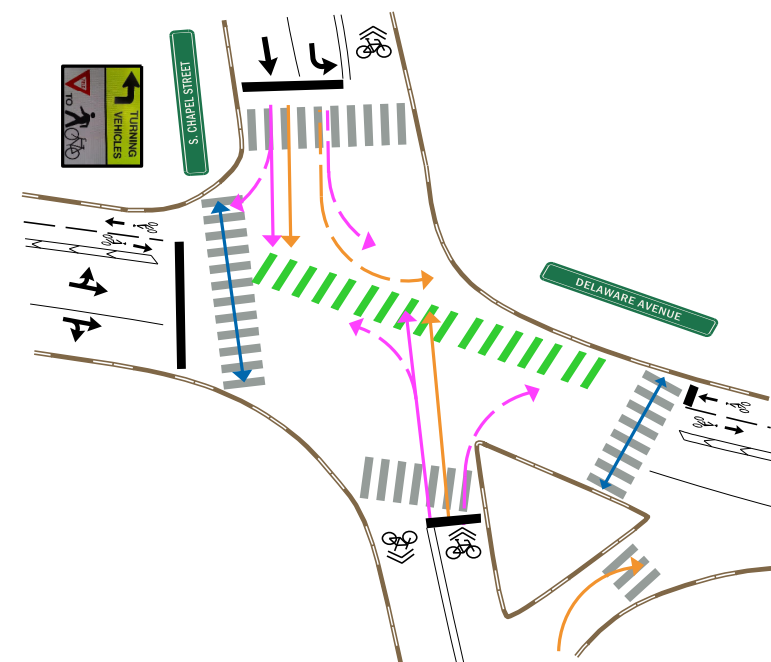
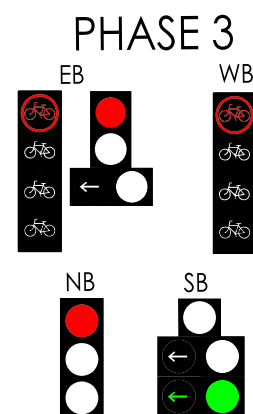
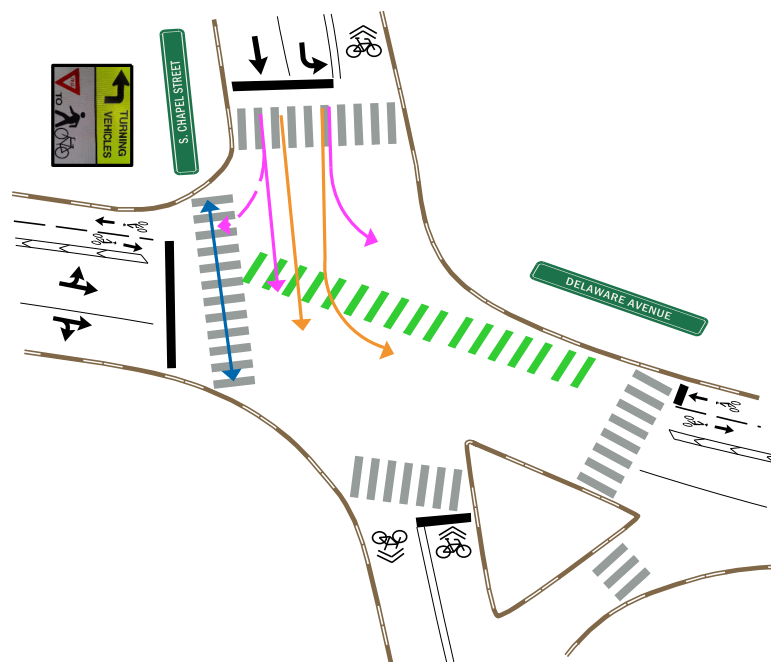
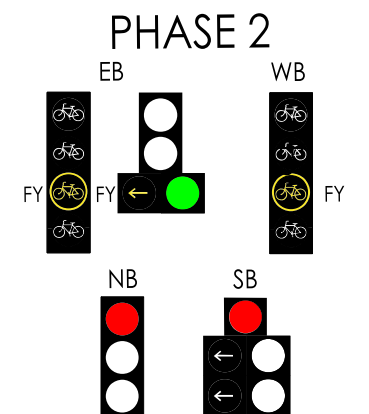
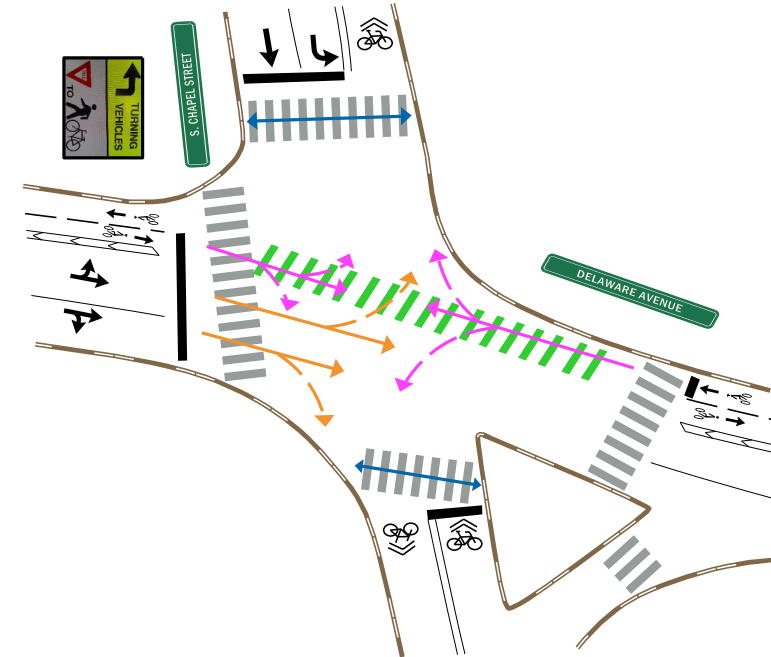
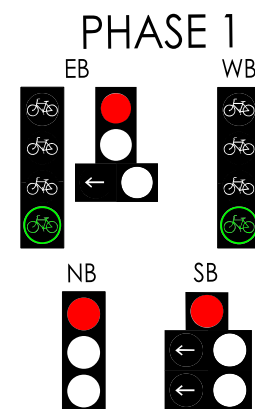
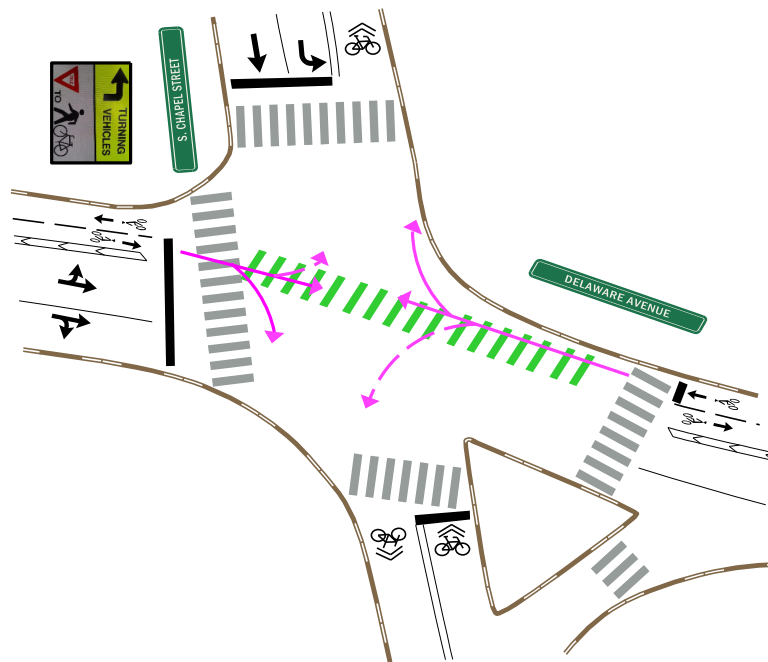


FIGURE 9

FEBRUARY 2018

DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

DELAWARE AVENUE/TYRE AVENUE - RECOMMENDED SIGNAL PHASING

32

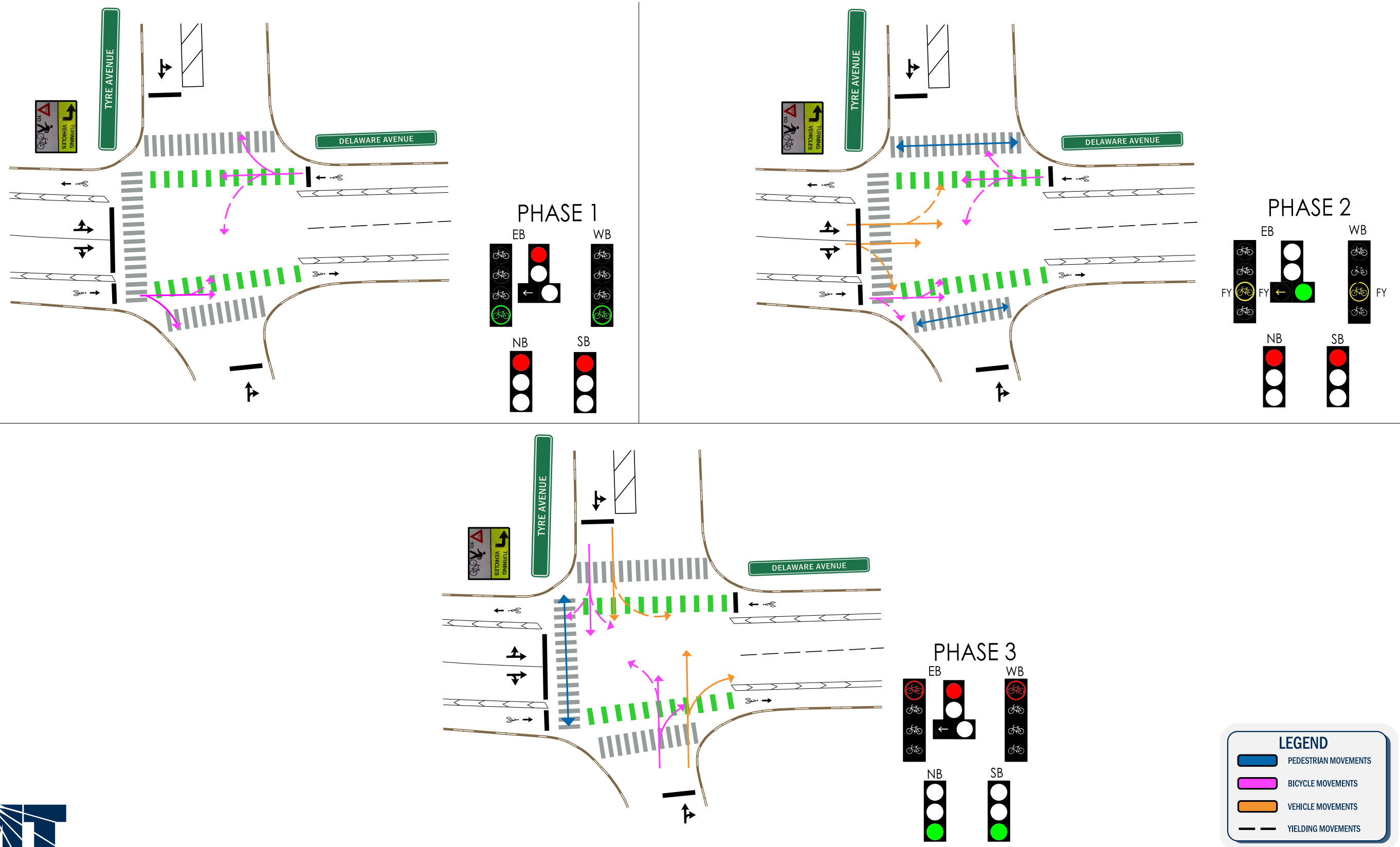


FIGURE 10

FEBRUARY 2018

DELAWARE AVENUE/TYRE AVENUE - FEBRUARY 2018





APPENDIX



APPENDIX A

DETAILED TURNING MOVEMENT COUNTS

File Name : Delaware_Ave_at_Orchard_Rd_AM_297711_03-08-2016

Site Code : Location 1 AM

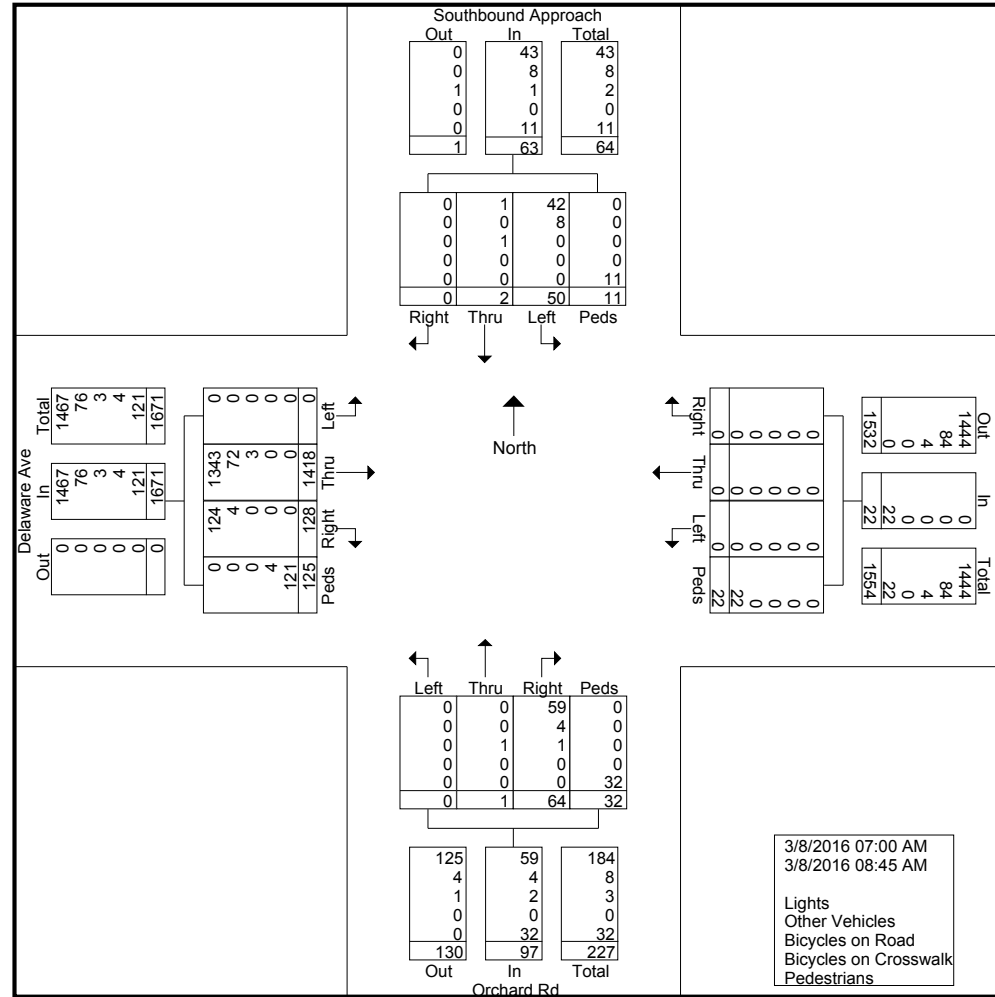
Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

	Southbound Approach From North					Delaware Ave From East					Orchard Rd From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
07:00 AM	0	1	3	1	5	0	0	0	0	0	7	0	0	0	7	1	156	0	2	159	171
07:15 AM	0	0	6	2	8	0	0	0	1	1	2	0	0	1	3	13	140	0	3	156	168
07:30 AM	0	0	2	1	3	0	0	0	1	1	9	0	0	1	10	19	200	0	9	228	242
07:45 AM	0	0	8	3	11	0	0	0	5	5	6	0	0	15	21	25	217	0	40	282	319
Total	0	1	19	7	27	0	0	0	7	7	24	0	0	17	41	58	713	0	54	825	900
08:00 AM	0	0	5	0	5	0	0	0	4	4	9	0	0	4	13	24	194	0	10	228	250
08:15 AM	0	0	7	2	9	0	0	0	2	2	8	1	0	2	11	14	171	0	20	205	227
08:30 AM	0	1	4	2	7	0	0	0	2	2	10	0	0	7	17	20	156	0	19	195	221
08:45 AM	0	0	15	0	15	0	0	0	7	7	13	0	0	2	15	12	184	0	22	218	255
Total	0	1	31	4	36	0	0	0	15	15	40	1	0	15	56	70	705	0	71	846	953
Grand Total	0	2	50	11	63	0	0	0	22	22	64	1	0	32	97	128	1418	0	125	1671	1853
Apprch %	0	3.2	79.4	17.5		0	0	0	100		66	1	0	33		7.7	84.9	0	7.5		
Total %	0	0.1	2.7	0.6	3.4	0	0	0	1.2	1.2	3.5	0.1	0	1.7	5.2	6.9	76.5	0	6.7	90.2	
Lights	0	1	42	0	43	0	0	0	0	0	59	0	0	0	59	124	1343	0	0	1467	1569
% Lights	0	50	84	0	68.3	0	0	0	0	0	92.2	0	0	0	60.8	96.9	94.7	0	0	87.8	84.7
Other Vehicles	0	0	8	0	8	0	0	0	0	0	4	0	0	0	4	4	72	0	0	76	88
% Other Vehicles	0	0	16	0	12.7	0	0	0	0	0	6.2	0	0	0	4.1	3.1	5.1	0	0	4.5	4.7
Bicycles on Road	0	1	0	0	1	0	0	0	0	0	1	1	0	0	2	0	3	0	0	3	6
% Bicycles on Road	0	50	0	0	1.6	0	0	0	0	0	1.6	100	0	0	2.1	0	0.2	0	0	0.2	0.3
Bicycles on Crosswalk	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	4
% Bicycles on Crosswalk	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3.2	0.2	0.2
Pedestrians	0	0	0	11	11	0	0	0	22	22	0	0	0	32	32	0	0	0	121	121	186
% Pedestrians	0	0	0	100	17.5	0	0	0	100	100	0	0	0	100	33	0	0	0	96.8	7.2	10

Page No : 2



File Name : Delaware_Ave_at_Orchard_Rd_AM_297711_03-08-2016

Site Code : Location 1 AM

Start Date : 3/8/2016

Page No : 3

	Southbound Approach From North					Delaware Ave From East					Orchard Rd From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 07:30 AM																					
07:30 AM	0	0	2	1	3	0	0	0	1	1	9	0	0	1	10	19	200	0	9	228	242
07:45 AM	0	0	8	3	11	0	0	0	5	5	6	0	0	15	21	25	217	0	40	282	319
08:00 AM	0	0	5	0	5	0	0	0	4	4	9	0	0	4	13	24	194	0	10	228	250
08:15 AM	0	0	7	2	9	0	0	0	2	2	8	1	0	2	11	14	171	0	20	205	227
Total Volume	0	0	22	6	28	0	0	0	12	12	32	1	0	22	55	82	782	0	79	943	1038
% App. Total	0	0	78.6	21.4		0	0	0	100		58.2	1.8	0	40		8.7	82.9	0	8.4		
PHF	.000	.000	.688	.500	.636	.000	.000	.000	.600	.600	.889	.250	.000	.367	.655	.820	.901	.000	.494	.836	.813

File Name : Delaware_Ave_at_Orchard_Rd_PM_297711_03-08-2016

Site Code : Location 1 PM

Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

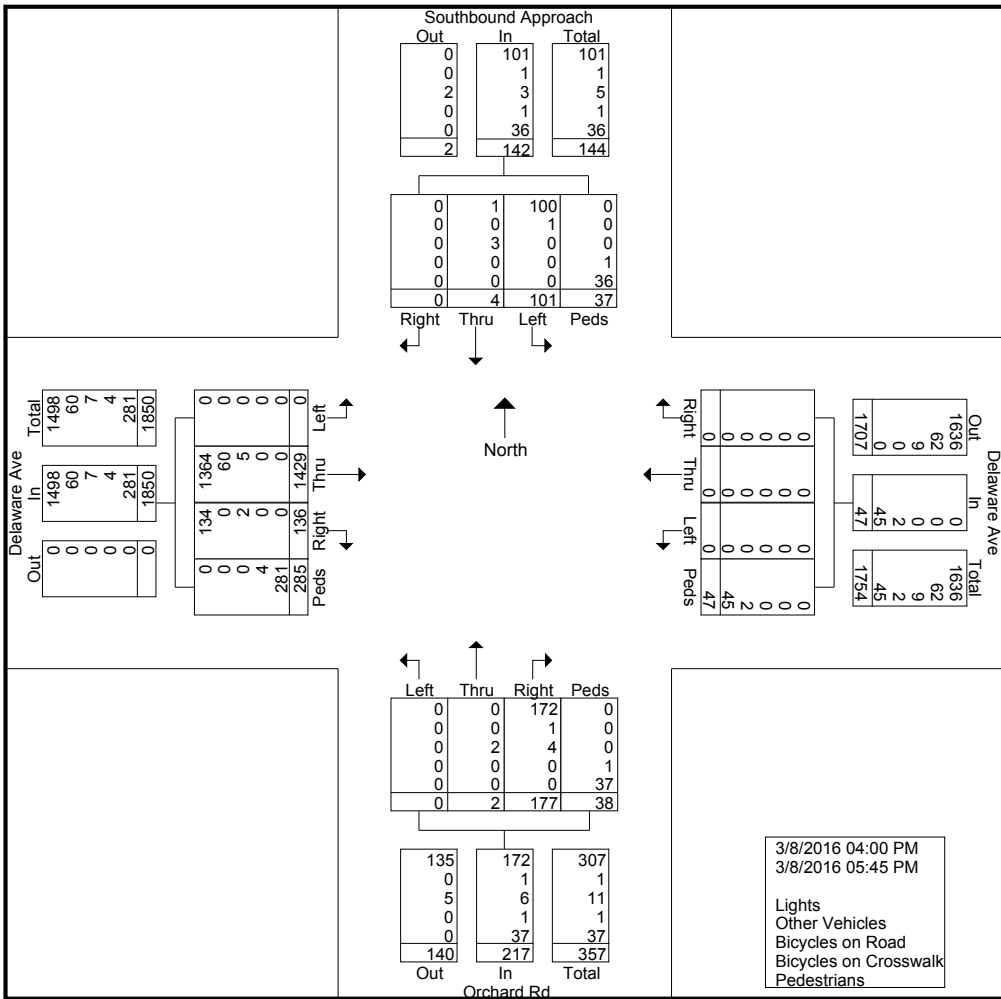
	Southbound Approach From North					Delaware Ave From East					Orchard Rd From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
04:00 PM	0	0	10	3	13	0	0	0	3	3	11	0	0	3	14	11	151	0	18	180	210
04:15 PM	0	0	3	2	5	0	0	0	1	1	20	1	0	2	23	5	166	0	22	193	222
04:30 PM	0	2	16	4	22	0	0	0	7	7	23	0	0	4	27	16	164	0	43	223	279
04:45 PM	0	1	10	6	17	0	0	0	8	8	19	0	0	2	21	24	177	0	63	264	310
Total	0	3	39	15	57	0	0	0	19	19	73	1	0	11	85	56	658	0	146	860	1021
05:00 PM	0	0	20	5	25	0	0	0	2	2	30	0	0	11	41	18	194	0	48	260	328
05:15 PM	0	0	18	7	25	0	0	0	5	5	29	0	0	5	34	21	205	0	22	248	312
05:30 PM	0	0	17	4	21	0	0	0	3	3	25	1	0	7	33	26	181	0	35	242	299
05:45 PM	0	1	7	6	14	0	0	0	18	18	20	0	0	4	24	15	191	0	34	240	296
Total	0	1	62	22	85	0	0	0	28	28	104	1	0	27	132	80	771	0	139	990	1235
Grand Total	0	4	101	37	142	0	0	0	47	47	177	2	0	38	217	136	1429	0	285	1850	2256
Apprch %	0	2.8	71.1	26.1		0	0	0	100		81.6	0.9	0	17.5		7.4	77.2	0	15.4		
Total %	0	0.2	4.5	1.6	6.3	0	0	0	2.1	2.1	7.8	0.1	0	1.7	9.6	6	63.3	0	12.6	82	
Lights	0	1	100	0	101	0	0	0	0	0	172	0	0	0	172	134	1364	0	0	1498	1771
% Lights	0	25	99	0	71.1	0	0	0	0	0	97.2	0	0	0	79.3	98.5	95.5	0	0	81	78.5
Other Vehicles	0	0	1	0	1	0	0	0	0	0	1	0	0	0	1	0	60	0	0	60	62
% Other Vehicles	0	0	1	0	0.7	0	0	0	0	0	0.6	0	0	0	0.5	0	4.2	0	0	3.2	2.7
Bicycles on Road	0	3	0	0	3	0	0	0	0	0	4	2	0	0	6	2	5	0	0	7	16
% Bicycles on Road	0	75	0	0	2.1	0	0	0	0	0	2.3	100	0	0	2.8	1.5	0.3	0	0	0.4	0.7
Bicycles on Crosswalk	0	0	0	1	1	0	0	0	2	2	0	0	0	1	1	0	0	0	4	4	8
% Bicycles on Crosswalk	0	0	0	2.7	0.7	0	0	0	4.3	4.3	0	0	0	2.6	0.5	0	0	0	1.4	0.2	0.4
Pedestrians	0	0	0	36	36	0	0	0	45	45	0	0	0	37	37	0	0	0	281	281	399
% Pedestrians	0	0	0	97.3	25.4	0	0	0	95.7	95.7	0	0	0	97.4	17.1	0	0	0	98.6	15.2	17.7

File Name : Delaware_Ave_at_Orchard_Rd_PM_297711_03-08-2016

Site Code : Location 1 PM

Start Date : 3/8/2016

Page No : 2



File Name : Delaware_Ave_at_Orchard_Rd_PM_297711_03-08-2016

Site Code : Location 1 PM

Start Date : 3/8/2016

Page No : 3

	Southbound Approach From North					Delaware Ave From East					Orchard Rd From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 04:45 PM																					
04:45 PM	0	1	10	6	17	0	0	0	8	8	19	0	0	2	21	24	177	0	63	264	310
05:00 PM	0	0	20	5	25	0	0	0	2	2	30	0	0	11	41	18	194	0	48	260	328
05:15 PM	0	0	18	7	25	0	0	0	5	5	29	0	0	5	34	21	205	0	22	248	312
05:30 PM	0	0	17	4	21	0	0	0	3	3	25	1	0	7	33	26	181	0	35	242	299
Total Volume	0	1	65	22	88	0	0	0	18	18	103	1	0	25	129	89	757	0	168	1014	1249
% App. Total	0	1.1	73.9	25		0	0	0	100		79.8	0.8	0	19.4		8.8	74.7	0	16.6		
PHF	.000	.250	.813	.786	.880	.000	.000	.000	.563	.563	.858	.250	.000	.568	.787	.856	.923	.000	.667	.960	.952

File Name : Delaware_Ave_at_S_College_Ave_AM_297604_03-08-2016

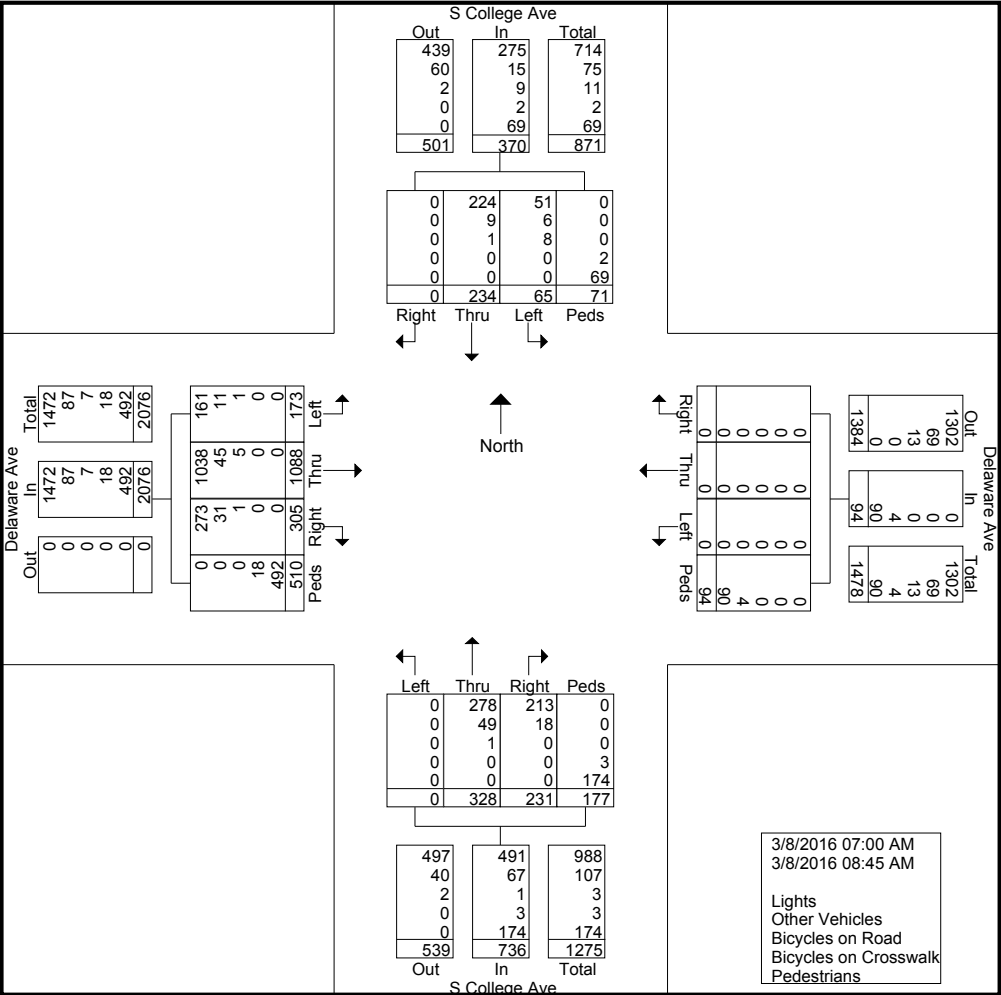
Site Code : Location 2 AM

Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

	S College Ave From North					Delaware Ave From East					S College Ave From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
07:00 AM	0	15	5	1	21	0	0	0	1	1	30	18	0	5	53	40	122	12	5	179	254
07:15 AM	0	34	12	3	49	0	0	0	7	7	22	32	0	7	61	33	92	11	12	148	265
07:30 AM	0	20	6	7	33	0	0	0	5	5	30	39	0	21	90	44	165	25	52	286	414
07:45 AM	0	40	11	17	68	0	0	0	38	38	36	46	0	65	147	45	153	23	208	429	682
Total	0	109	34	28	171	0	0	0	51	51	118	135	0	98	351	162	532	71	277	1042	1615
08:00 AM	0	30	8	6	44	0	0	0	9	9	32	53	0	15	100	37	159	30	34	260	413
08:15 AM	0	36	8	8	52	0	0	0	6	6	27	50	0	16	93	32	123	17	43	215	366
08:30 AM	0	32	6	12	50	0	0	0	10	10	23	42	0	22	87	38	122	29	43	232	379
08:45 AM	0	27	9	17	53	0	0	0	18	18	31	48	0	26	105	36	152	26	113	327	503
Total	0	125	31	43	199	0	0	0	43	43	113	193	0	79	385	143	556	102	233	1034	1661
Grand Total	0	234	65	71	370	0	0	0	94	94	231	328	0	177	736	305	1088	173	510	2076	3276
Apprch %	0	63.2	17.6	19.2		0	0	0	100		31.4	44.6	0	24		14.7	52.4	8.3	24.6		
Total %	0	7.1	2	2.2	11.3	0	0	0	2.9	2.9	7.1	10	0	5.4	22.5	9.3	33.2	5.3	15.6	63.4	
Lights	0	224	51	0	275	0	0	0	0	0	213	278	0	0	491	273	1038	161	0	1472	2238
% Lights	0	95.7	78.5	0	74.3	0	0	0	0	0	92.2	84.8	0	0	66.7	89.5	95.4	93.1	0	70.9	68.3
Other Vehicles	0	9	6	0	15	0	0	0	0	0	18	49	0	0	67	31	45	11	0	87	169
% Other Vehicles	0	3.8	9.2	0	4.1	0	0	0	0	0	7.8	14.9	0	0	9.1	10.2	4.1	6.4	0	4.2	5.2
Bicycles on Road	0	1	8	0	9	0	0	0	0	0	0	1	0	0	1	1	5	1	0	7	17
% Bicycles on Road	0	0.4	12.3	0	2.4	0	0	0	0	0	0	0.3	0	0	0.1	0.3	0.5	0.6	0	0.3	0.5
Bicycles on Crosswalk	0	0	0	2	2	0	0	0	4	4	0	0	0	3	3	0	0	0	18	18	27
% Bicycles on Crosswalk	0	0	0	2.8	0.5	0	0	0	4.3	4.3	0	0	0	1.7	0.4	0	0	0	3.5	0.9	0.8
Pedestrians	0	0	0	69	69	0	0	0	90	90	0	0	0	174	174	0	0	0	492	492	825
% Pedestrians	0	0	0	97.2	18.6	0	0	0	95.7	95.7	0	0	0	98.3	23.6	0	0	0	96.5	23.7	25.2



File Name : Delaware_Ave_at_S_College_Ave_AM_297604_03-08-2016

Site Code : Location 2 AM

Start Date : 3/8/2016

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	S College Ave From North					Delaware Ave From East					S College Ave From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 07:30 AM																					
07:30 AM	0	20	6	7	33	0	0	0	5	5	30	39	0	21	90	44	165	25	52	286	414
07:45 AM	0	40	11	17	68	0	0	0	38	38	36	46	0	65	147	45	153	23	208	429	682
08:00 AM	0	30	8	6	44	0	0	0	9	9	32	53	0	15	100	37	159	30	34	260	413
08:15 AM	0	36	8	8	52	0	0	0	6	6	27	50	0	16	93	32	123	17	43	215	366
Total Volume	0	126	33	38	197	0	0	0	58	58	125	188	0	117	430	158	600	95	337	1190	1875
% App. Total	0	64	16.8	19.3		0	0	0	100		29.1	43.7	0	27.2		13.3	50.4	8	28.3		
PHF	.000	.788	.750	.559	.724	.000	.000	.000	.382	.382	.868	.887	.000	.450	.731	.878	.909	.792	.405	.693	.687

File Name : Delaware_Ave_at_S_College_Ave_PM_297605_03-08-2016

Site Code : Location 2 PM

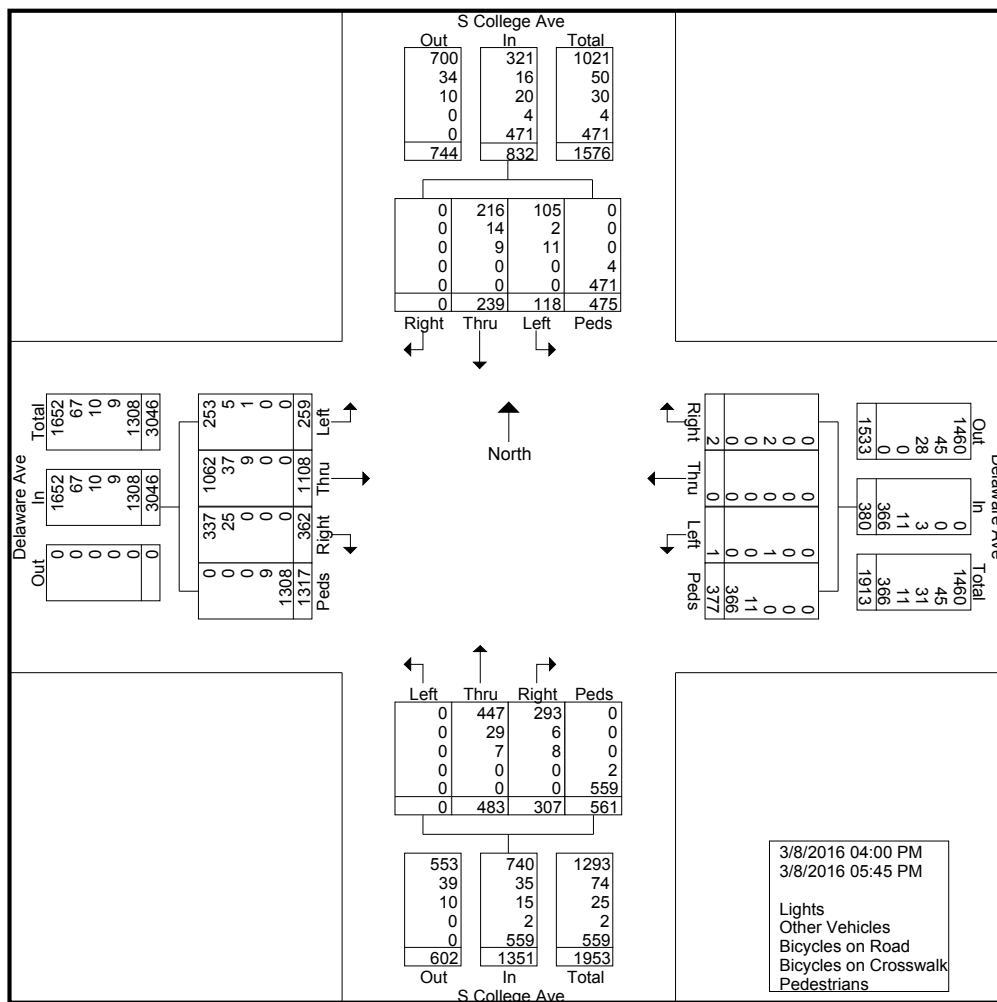
Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

	S College Ave From North					Delaware Ave From East					S College Ave From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
04:00 PM	0	26	10	42	78	0	0	1	41	42	24	50	0	34	108	38	131	16	80	265	493
04:15 PM	0	29	15	54	98	0	0	0	31	31	45	53	0	70	168	28	124	25	162	339	636
04:30 PM	0	28	18	85	131	0	0	0	110	110	37	53	0	120	210	52	130	33	226	441	892
04:45 PM	0	22	17	107	146	0	0	0	80	80	36	68	0	171	275	36	105	45	430	616	1117
Total	0	105	60	288	453	0	0	1	262	263	142	224	0	395	761	154	490	119	898	1661	3138
05:00 PM	0	25	12	44	81	1	0	0	26	27	39	64	0	37	140	57	188	34	114	393	641
05:15 PM	0	27	16	43	86	0	0	0	25	25	48	65	0	24	137	59	132	44	80	315	563
05:30 PM	0	38	15	43	96	1	0	0	34	35	37	64	0	26	127	47	167	31	72	317	575
05:45 PM	0	44	15	57	116	0	0	0	30	30	41	66	0	79	186	45	131	31	153	360	692
Total	0	134	58	187	379	2	0	0	115	117	165	259	0	166	590	208	618	140	419	1385	2471
Grand Total	0	239	118	475	832	2	0	1	377	380	307	483	0	561	1351	362	1108	259	1317	3046	5609
Apprch %	0	28.7	14.2	57.1		0.5	0	0.3	99.2		22.7	35.8	0	41.5		11.9	36.4	8.5	43.2		
Total %	0	4.3	2.1	8.5	14.8	0	0	0	6.7	6.8	5.5	8.6	0	10	24.1	6.5	19.8	4.6	23.5	54.3	
Lights	0	216	105	0	321	0	0	0	0	0	293	447	0	0	740	337	1062	253	0	1652	2713
% Lights	0	90.4	89	0	38.6	0	0	0	0	0	95.4	92.5	0	0	54.8	93.1	95.8	97.7	0	54.2	48.4
Other Vehicles	0	14	2	0	16	0	0	0	0	0	6	29	0	0	35	25	37	5	0	67	118
% Other Vehicles	0	5.9	1.7	0	1.9	0	0	0	0	0	2	6	0	0	2.6	6.9	3.3	1.9	0	2.2	2.1
Bicycles on Road	0	9	11	0	20	2	0	1	0	3	8	7	0	0	15	0	9	1	0	10	48
% Bicycles on Road	0	3.8	9.3	0	2.4	100	0	100	0	0.8	2.6	1.4	0	0	1.1	0	0.8	0.4	0	0.3	0.9
Bicycles on Crosswalk	0	0	0	4	4	0	0	0	11	11	0	0	0	2	2	0	0	0	9	9	26
% Bicycles on Crosswalk	0	0	0	0.8	0.5	0	0	0	2.9	2.9	0	0	0	0.4	0.1	0	0	0	0.7	0.3	0.5
Pedestrians	0	0	0	471	471	0	0	0	366	366	0	0	0	559	559	0	0	0	1308	1308	2704
% Pedestrians	0	0	0	99.2	56.6	0	0	0	97.1	96.3	0	0	0	99.6	41.4	0	0	0	99.3	42.9	48.2

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File Name : Delaware_Ave_at_S_College_Ave_PM_297605_03-08-2016

Site Code : Location 2 PM

Start Date : 3/8/2016

Page No : 3

	S College Ave From North					Delaware Ave From East					S College Ave From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 04:15 PM																					
04:15 PM	0	29	15	54	98	0	0	0	31	31	45	53	0	70	168	28	124	25	162	339	636
04:30 PM	0	28	18	85	131	0	0	0	110	110	37	53	0	120	210	52	130	33	226	441	892
04:45 PM	0	22	17	107	146	0	0	0	80	80	36	68	0	171	275	36	105	45	430	616	1117
05:00 PM	0	25	12	44	81	1	0	0	26	27	39	64	0	37	140	57	188	34	114	393	641
Total Volume	0	104	62	290	456	1	0	0	247	248	157	238	0	398	793	173	547	137	932	1789	3286
% App. Total	0	22.8	13.6	63.6		0.4	0	0	99.6		19.8	30	0	50.2		9.7	30.6	7.7	52.1		
PHF	.000	.897	.861	.678	.781	.250	.000	.000	.561	.564	.872	.875	.000	.582	.721	.759	.727	.761	.542	.726	.735

File Name : Delaware_Ave_at_Galleria_Ent_AM_297606_03-08-2016

Site Code : Location 4 AM

Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

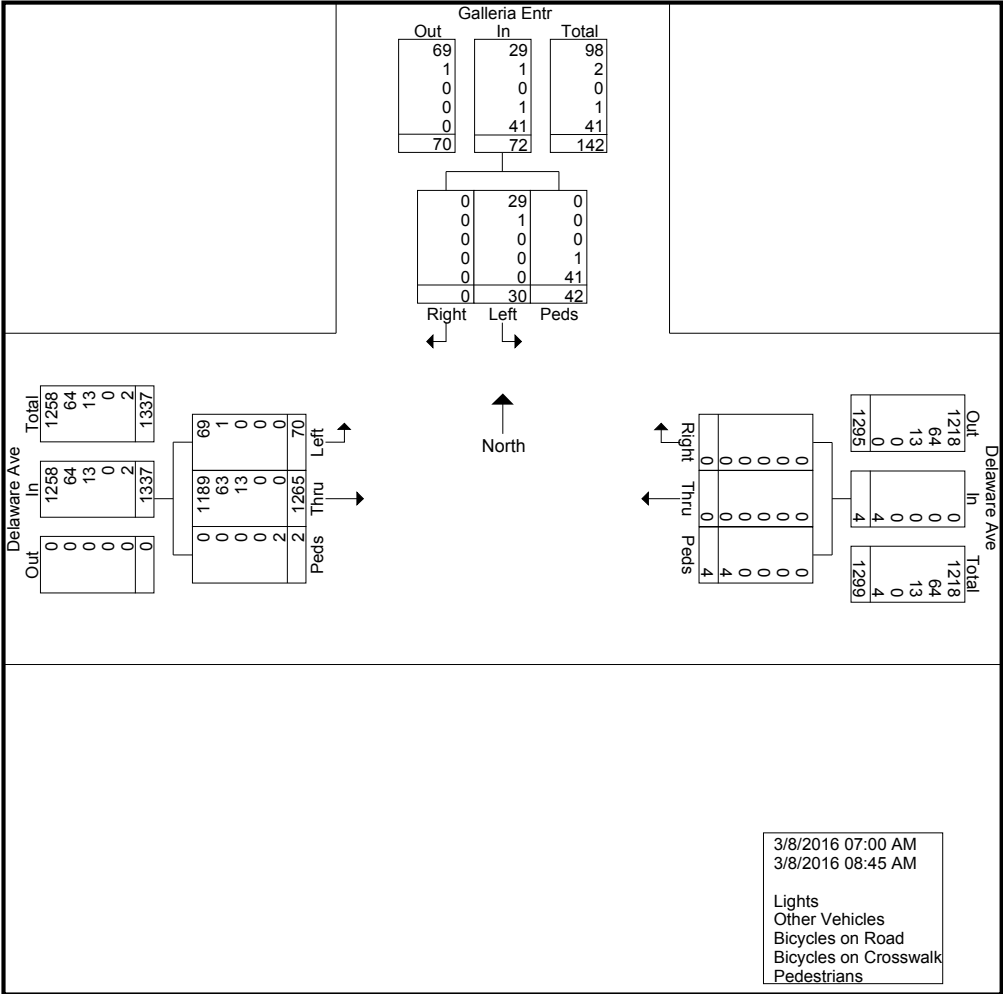
	Galleria Entr Southbound				Delaware Ave Westbound				Delaware Ave Eastbound				
Start Time	Left	Right	Peds	App. Total	Thru	Right	Peds	App. Total	Left	Thru	Peds	App. Total	Int. Total
07:00 AM	4	0	1	5	0	0	0	0	4	149	0	153	158
07:15 AM	5	0	3	8	0	0	0	0	4	120	0	124	132
07:30 AM	2	0	2	4	0	0	2	2	6	186	0	192	198
07:45 AM	4	0	11	15	0	0	1	1	17	187	0	204	220
Total	15	0	17	32	0	0	3	3	31	642	0	673	708
08:00 AM	4	0	5	9	0	0	0	0	14	178	1	193	202
08:15 AM	3	0	8	11	0	0	1	1	8	151	0	159	171
08:30 AM	3	0	7	10	0	0	0	0	4	140	0	144	154
08:45 AM	5	0	5	10	0	0	0	0	13	154	1	168	178
Total	15	0	25	40	0	0	1	1	39	623	2	664	705
Grand Total	30	0	42	72	0	0	4	4	70	1265	2	1337	1413
Apprch %	41.7	0	58.3		0	0	100		5.2	94.6	0.1		
Total %	2.1	0	3	5.1	0	0	0.3	0.3	5	89.5	0.1	94.6	
Lights	29	0	0	29	0	0	0	0	69	1189	0	1258	1287
% Lights	96.7	0	0	40.3	0	0	0	0	98.6	94	0	94.1	91.1
Other Vehicles	1	0	0	1	0	0	0	0	1	63	0	64	65
% Other Vehicles	3.3	0	0	1.4	0	0	0	0	1.4	5	0	4.8	4.6
Bicycles on Road	0	0	0	0	0	0	0	0	0	13	0	13	13
% Bicycles on Road	0	0	0	0	0	0	0	0	0	1	0	1	0.9
Bicycles on Crosswalk	0	0	1	1	0	0	0	0	0	0	0	0	1
% Bicycles on Crosswalk	0	0	2.4	1.4	0	0	0	0	0	0	0	0	0.1
Pedestrians	0	0	41	41	0	0	4	4	0	0	2	2	47
% Pedestrians	0	0	97.6	56.9	0	0	100	100	0	0	100	0.1	3.3

File Name : Delaware_Ave_at_Galleria_Ent_AM_297606_03-08-2016

Site Code : Location 4 AM

Start Date : 3/8/2016

Page No : 2



File Name : Delaware_Ave_at_Galleria_Ent_AM_297606_03-08-2016

Site Code : Location 4 AM

Start Date : 3/8/2016

Page No : 3

	Galleria Entr Southbound				Delaware Ave Westbound				Delaware Ave Eastbound				
Start Time	Left	Right	Peds	App. Total	Thru	Right	Peds	App. Total	Left	Thru	Peds	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1													
Peak Hour for Entire Intersection Begins at 07:30 AM													
07:30 AM	2	0	2	4	0	0	2	2	6	186	0	192	198
07:45 AM	4	0	11	15	0	0	1	1	17	187	0	204	220
08:00 AM	4	0	5	9	0	0	0	0	14	178	1	193	202
08:15 AM	3	0	8	11	0	0	1	1	8	151	0	159	171
Total Volume	13	0	26	39	0	0	4	4	45	702	1	748	791
% App. Total	33.3	0	66.7		0	0	100		6	93.9	0.1		
PHF	.813	.000	.591	.650	.000	.000	.500	.500	.662	.939	.250	.917	.899
Lights	12	0	0	12	0	0	0	0	44	664	0	708	720
% Lights	92.3	0	0	30.8	0	0	0	0	97.8	94.6	0	94.7	91.0
Other Vehicles	1	0	0	1	0	0	0	0	1	31	0	32	33
% Other Vehicles	7.7	0	0	2.6	0	0	0	0	2.2	4.4	0	4.3	4.2
Bicycles on Road	0	0	0	0	0	0	0	0	0	7	0	7	7
% Bicycles on Road	0	0	0	0	0	0	0	0	0	1.0	0	0.9	0.9
Bicycles on Crosswalk	0	0	0	0	0	0	0	0	0	0	0	0	0
% Bicycles on Crosswalk	0	0	0	0	0	0	0	0	0	0	0	0	0
Pedestrians	0	0	26	26	0	0	4	4	0	0	1	1	31
% Pedestrians	0	0	100	66.7	0	0	100	100	0	0	100	0.1	3.9

File Name : Delaware_Ave_at_Galleria_Entr_PM_297607_03-08-2016

Site Code : Location 4 PM

Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

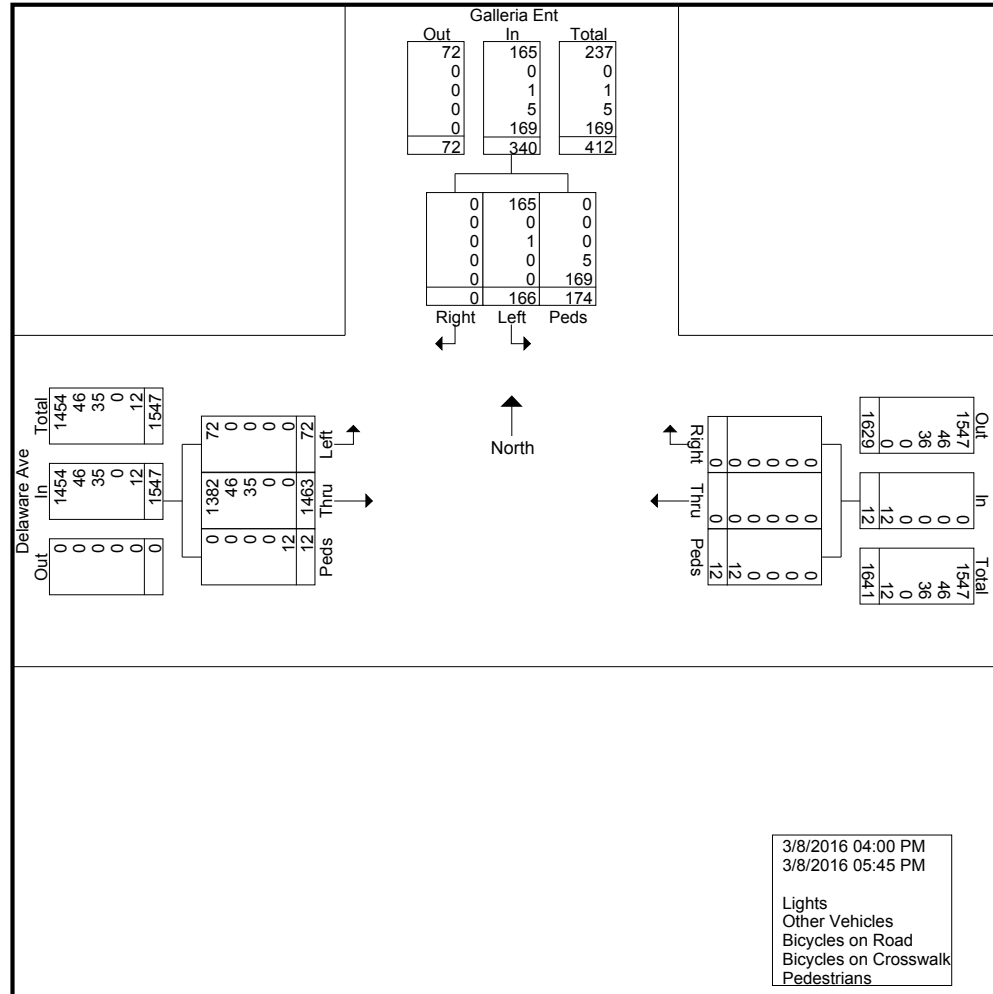
	Galleria Ent From North				Delaware Ave From East				Delaware Ave From West				
Start Time	Right	Left	Peds	App. Total	Right	Thru	Peds	App. Total	Thru	Left	Peds	App. Total	Int. Total
04:00 PM	0	15	8	23	0	0	2	2	153	4	0	157	182
04:15 PM	0	17	27	44	0	0	3	3	171	9	0	180	227
04:30 PM	0	21	24	45	0	0	1	1	165	10	1	176	222
04:45 PM	0	37	49	86	0	0	3	3	165	9	2	176	265
Total	0	90	108	198	0	0	9	9	654	32	3	689	896
05:00 PM	0	24	14	38	0	0	2	2	237	5	2	244	284
05:15 PM	0	15	17	32	0	0	1	1	189	9	3	201	234
05:30 PM	0	15	10	25	0	0	0	0	207	8	0	215	240
05:45 PM	0	22	25	47	0	0	0	0	176	18	4	198	245
Total	0	76	66	142	0	0	3	3	809	40	9	858	1003
Grand Total	0	166	174	340	0	0	12	12	1463	72	12	1547	1899
Apprch %	0	48.8	51.2		0	0	100		94.6	4.7	0.8		
Total %	0	8.7	9.2	17.9	0	0	0.6	0.6	77	3.8	0.6	81.5	
Lights	0	165	0	165	0	0	0	0	1382	72	0	1454	1619
% Lights	0	99.4	0	48.5	0	0	0	0	94.5	100	0	94	85.3
Other Vehicles	0	0	0	0	0	0	0	0	46	0	0	46	46
% Other Vehicles	0	0	0	0	0	0	0	0	3.1	0	0	3	2.4
Bicycles on Road	0	1	0	1	0	0	0	0	35	0	0	35	36
% Bicycles on Road	0	0.6	0	0.3	0	0	0	0	2.4	0	0	2.3	1.9
Bicycles on Crosswalk	0	0	5	5	0	0	0	0	0	0	0	0	5
% Bicycles on Crosswalk	0	0	2.9	1.5	0	0	0	0	0	0	0	0	0.3
Pedestrians	0	0	169	169	0	0	12	12	0	0	12	12	193
% Pedestrians	0	0	97.1	49.7	0	0	100	100	0	0	100	0.8	10.2

File Name : Delaware_Ave_at_Galleria_Entr_PM_297607_03-08-2016

Site Code : Location 4 PM

Start Date : 3/8/2016

Page No : 2



File Name : Delaware_Ave_at_Galleria_Entr_PM_297607_03-08-2016

Site Code : Location 4 PM

Start Date : 3/8/2016

Page No : 3

	Galleria Ent From North				Delaware Ave From East				Delaware Ave From West				
Start Time	Right	Left	Peds	App. Total	Right	Thru	Peds	App. Total	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1													
Peak Hour for Entire Intersection Begins at 04:45 PM													
04:45 PM	0	37	49	86	0	0	3	3	165	9	2	176	265
05:00 PM	0	24	14	38	0	0	2	2	237	5	2	244	284
05:15 PM	0	15	17	32	0	0	1	1	189	9	3	201	234
05:30 PM	0	15	10	25	0	0	0	0	207	8	0	215	240
Total Volume	0	91	90	181	0	0	6	6	798	31	7	836	1023
% App. Total	0	50.3	49.7		0	0	100		95.5	3.7	0.8		
PHF	.000	.615	.459	.526	.000	.000	.500	.500	.842	.861	.583	.857	.901

File Name : Delaware_Ave_at_Academy_St_AM_297608_03-08-2016

Site Code : Location 5 AM

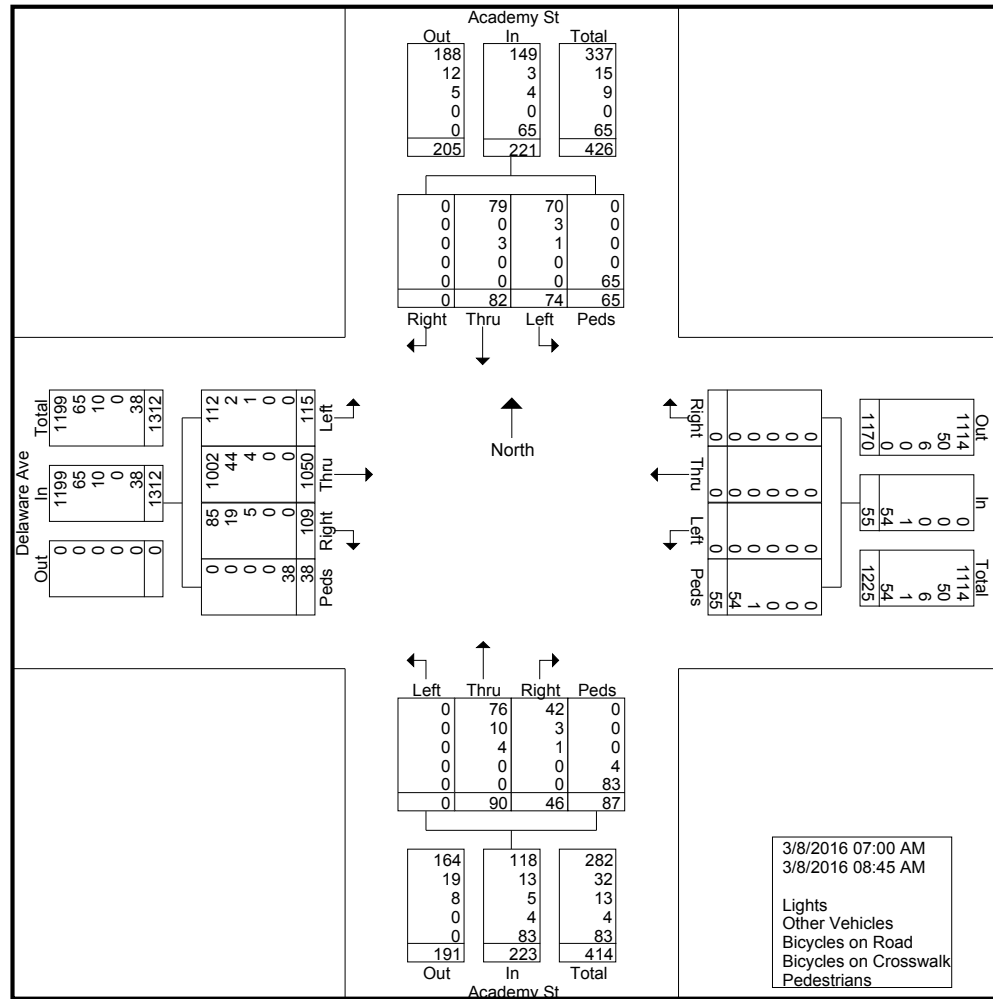
Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

	Academy St From North					Delaware Ave From East					Academy St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
07:00 AM	0	2	4	2	8	0	0	0	0	0	5	3	0	2	10	8	127	7	1	143	161
07:15 AM	0	6	6	5	17	0	0	0	0	0	1	8	0	1	10	7	101	12	1	121	148
07:30 AM	0	10	6	5	21	0	0	0	5	5	3	12	0	12	27	14	165	9	3	191	244
07:45 AM	0	15	15	17	47	0	0	0	14	14	9	11	0	34	54	15	147	16	7	185	300
Total	0	33	31	29	93	0	0	0	19	19	18	34	0	49	101	44	540	44	12	640	853
08:00 AM	0	17	11	5	33	0	0	0	14	14	8	21	0	7	36	17	147	35	2	201	284
08:15 AM	0	14	11	13	38	0	0	0	10	10	2	11	0	9	22	20	114	14	10	158	228
08:30 AM	0	7	12	8	27	0	0	0	3	3	10	13	0	7	30	13	129	9	3	154	214
08:45 AM	0	11	9	10	30	0	0	0	9	9	8	11	0	15	34	15	120	13	11	159	232
Total	0	49	43	36	128	0	0	0	36	36	28	56	0	38	122	65	510	71	26	672	958
Grand Total	0	82	74	65	221	0	0	0	55	55	46	90	0	87	223	109	1050	115	38	1312	1811
Apprch %	0	37.1	33.5	29.4		0	0	0	100		20.6	40.4	0	39		8.3	80	8.8	2.9		
Total %	0	4.5	4.1	3.6	12.2	0	0	0	3	3	2.5	5	0	4.8	12.3	6	58	6.4	2.1	72.4	
Lights	0	79	70	0	149	0	0	0	0	0	42	76	0	0	118	85	1002	112	0	1199	1466
% Lights	0	96.3	94.6	0	67.4	0	0	0	0	0	91.3	84.4	0	0	52.9	78	95.4	97.4	0	91.4	80.9
Other Vehicles	0	0	3	0	3	0	0	0	0	0	3	10	0	0	13	19	44	2	0	65	81
% Other Vehicles	0	0	4.1	0	1.4	0	0	0	0	0	6.5	11.1	0	0	5.8	17.4	4.2	1.7	0	5	4.5
Bicycles on Road	0	3	1	0	4	0	0	0	0	0	1	4	0	0	5	5	4	1	0	10	19
% Bicycles on Road	0	3.7	1.4	0	1.8	0	0	0	0	0	2.2	4.4	0	0	2.2	4.6	0.4	0.9	0	0.8	1
Bicycles on Crosswalk	0	0	0	0	0	0	0	0	1	1	0	0	0	4	4	0	0	0	0	0	5
% Bicycles on Crosswalk	0	0	0	0	0	0	0	0	1.8	1.8	0	0	0	4.6	1.8	0	0	0	0	0	0.3
Pedestrians	0	0	0	65	65	0	0	0	54	54	0	0	0	83	83	0	0	0	38	38	240
% Pedestrians	0	0	0	100	29.4	0	0	0	98.2	98.2	0	0	0	95.4	37.2	0	0	0	100	2.9	13.3

Page No : 2



File Name : Delaware_Ave_at_Academy_St_AM_297608_03-08-2016

Site Code : Location 5 AM

Start Date : 3/8/2016

Page No : 3

	Academy St From North					Delaware Ave From East					Academy St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 07:30 AM																					
07:30 AM	0	10	6	5	21	0	0	0	5	5	3	12	0	12	27	14	165	9	3	191	244
07:45 AM	0	15	15	17	47	0	0	0	14	14	9	11	0	34	54	15	147	16	7	185	300
08:00 AM	0	17	11	5	33	0	0	0	14	14	8	21	0	7	36	17	147	35	2	201	284
08:15 AM	0	14	11	13	38	0	0	0	10	10	2	11	0	9	22	20	114	14	10	158	228
Total Volume	0	56	43	40	139	0	0	0	43	43	22	55	0	62	139	66	573	74	22	735	1056
% App. Total	0	40.3	30.9	28.8		0	0	0	100		15.8	39.6	0	44.6		9	78	10.1	3		
PHF	.000	.824	.717	.588	.739	.000	.000	.000	.768	.768	.611	.655	.000	.456	.644	.825	.868	.529	.550	.914	.880

File Name : delaware_ave_at_academy_st_pm_297609_03-08-2016

Site Code : Location 5 PM

Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

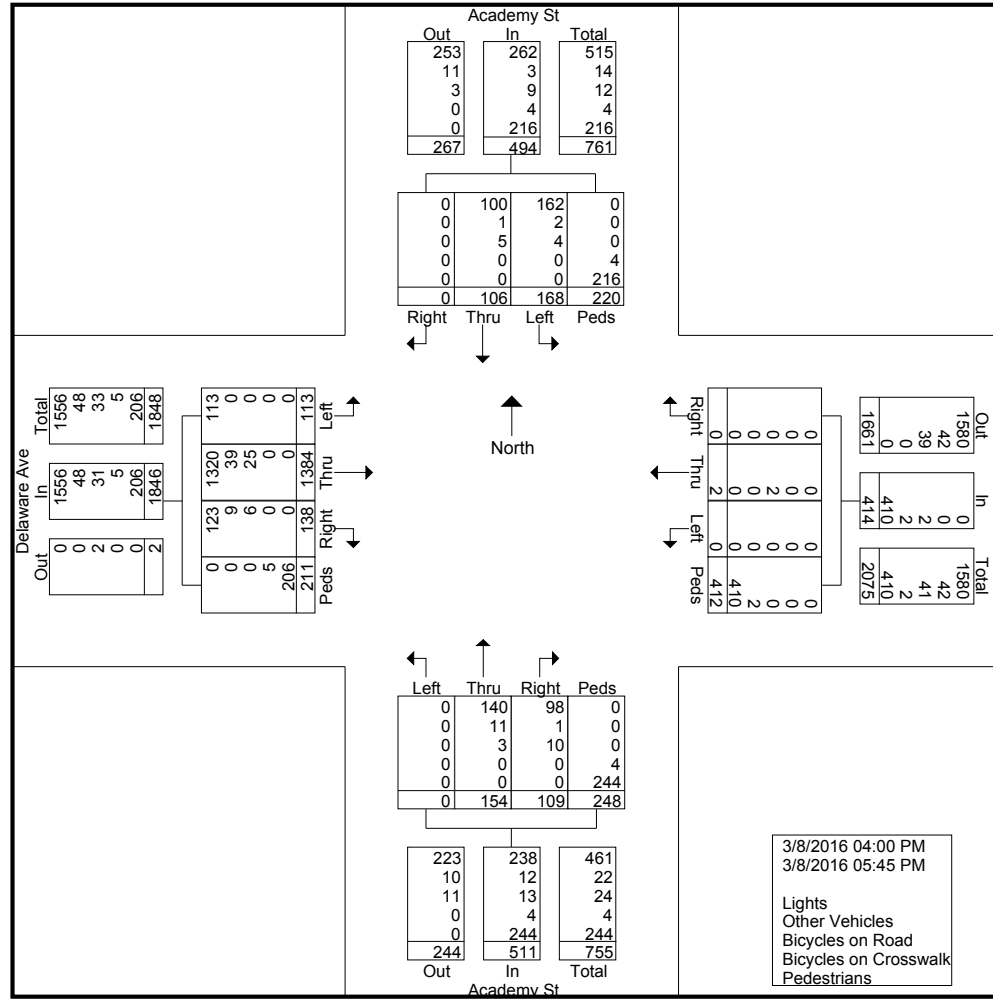
	Academy St From North					Delaware Ave From East					Academy St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
04:00 PM	0	6	25	29	60	0	0	0	34	34	5	13	0	23	41	15	156	12	18	201	336
04:15 PM	0	14	23	22	59	0	2	0	45	47	7	16	0	24	47	17	148	10	16	191	344
04:30 PM	0	17	20	28	65	0	0	0	48	48	10	23	0	34	67	18	175	16	29	238	418
04:45 PM	0	19	21	52	92	0	0	0	72	72	19	20	0	69	108	23	152	7	43	225	497
Total	0	56	89	131	276	0	2	0	199	201	41	72	0	150	263	73	631	45	106	855	1595
05:00 PM	0	12	18	23	53	0	0	0	49	49	19	17	0	17	53	24	229	20	31	304	459
05:15 PM	0	9	22	20	51	0	0	0	53	53	18	22	0	14	54	15	155	14	10	194	352
05:30 PM	0	13	17	17	47	0	0	0	54	54	14	17	0	20	51	11	208	17	27	263	415
05:45 PM	0	16	22	29	67	0	0	0	57	57	17	26	0	47	90	15	161	17	37	230	444
Total	0	50	79	89	218	0	0	0	213	213	68	82	0	98	248	65	753	68	105	991	1670
Grand Total	0	106	168	220	494	0	2	0	412	414	109	154	0	248	511	138	1384	113	211	1846	3265
Apprch %	0	21.5	34	44.5		0	0.5	0	99.5		21.3	30.1	0	48.5		7.5	75	6.1	11.4		
Total %	0	3.2	5.1	6.7	15.1	0	0.1	0	12.6	12.7	3.3	4.7	0	7.6	15.7	4.2	42.4	3.5	6.5	56.5	
Lights	0	100	162	0	262	0	0	0	0	0	98	140	0	0	238	123	1320	113	0	1556	2056
% Lights	0	94.3	96.4	0	53	0	0	0	0	0	89.9	90.9	0	0	46.6	89.1	95.4	100	0	84.3	63
Other Vehicles	0	1	2	0	3	0	0	0	0	0	1	11	0	0	12	9	39	0	0	48	63
% Other Vehicles	0	0.9	1.2	0	0.6	0	0	0	0	0	0.9	7.1	0	0	2.3	6.5	2.8	0	0	2.6	1.9
Bicycles on Road	0	5	4	0	9	0	2	0	0	2	10	3	0	0	13	6	25	0	0	31	55
% Bicycles on Road	0	4.7	2.4	0	1.8	0	100	0	0	0.5	9.2	1.9	0	0	2.5	4.3	1.8	0	0	1.7	1.7
Bicycles on Crosswalk	0	0	0	4	4	0	0	0	2	2	0	0	0	4	4	0	0	0	5	5	15
% Bicycles on Crosswalk	0	0	0	1.8	0.8	0	0	0	0.5	0.5	0	0	0	1.6	0.8	0	0	0	2.4	0.3	0.5
Pedestrians	0	0	0	216	216	0	0	0	410	410	0	0	0	244	244	0	0	0	206	206	1076
% Pedestrians	0	0	0	98.2	43.7	0	0	0	99.5	99	0	0	0	98.4	47.7	0	0	0	97.6	11.2	33

File Name : delaware_ave_at_academy_st_pm_297609_03-08-2016

Site Code : Location 5 PM

Start Date : 3/8/2016

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File Name : delaware_ave_at_academy_st_pm_297609_03-08-2016

Site Code : Location 5 PM

Start Date : 3/8/2016

Page No : 3

	Academy St From North					Delaware Ave From East					Academy St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 04:30 PM																					
04:30 PM	0	17	20	28	65	0	0	0	48	48	10	23	0	34	67	18	175	16	29	238	418
04:45 PM	0	19	21	52	92	0	0	0	72	72	19	20	0	69	108	23	152	7	43	225	497
05:00 PM	0	12	18	23	53	0	0	0	49	49	19	17	0	17	53	24	229	20	31	304	459
05:15 PM	0	9	22	20	51	0	0	0	53	53	18	22	0	14	54	15	155	14	10	194	352
Total Volume	0	57	81	123	261	0	0	0	222	222	66	82	0	134	282	80	711	57	113	961	1726
% App. Total	0	21.8	31	47.1		0	0	0	100		23.4	29.1	0	47.5		8.3	74	5.9	11.8		
PHF	.000	.750	.920	.591	.709	.000	.000	.000	.771	.771	.868	.891	.000	.486	.653	.833	.776	.713	.657	.790	.868

File Name : Delaware_Ave_at_Haines_St_AM_297610_03-08-2016

Site Code : Location 6 AM

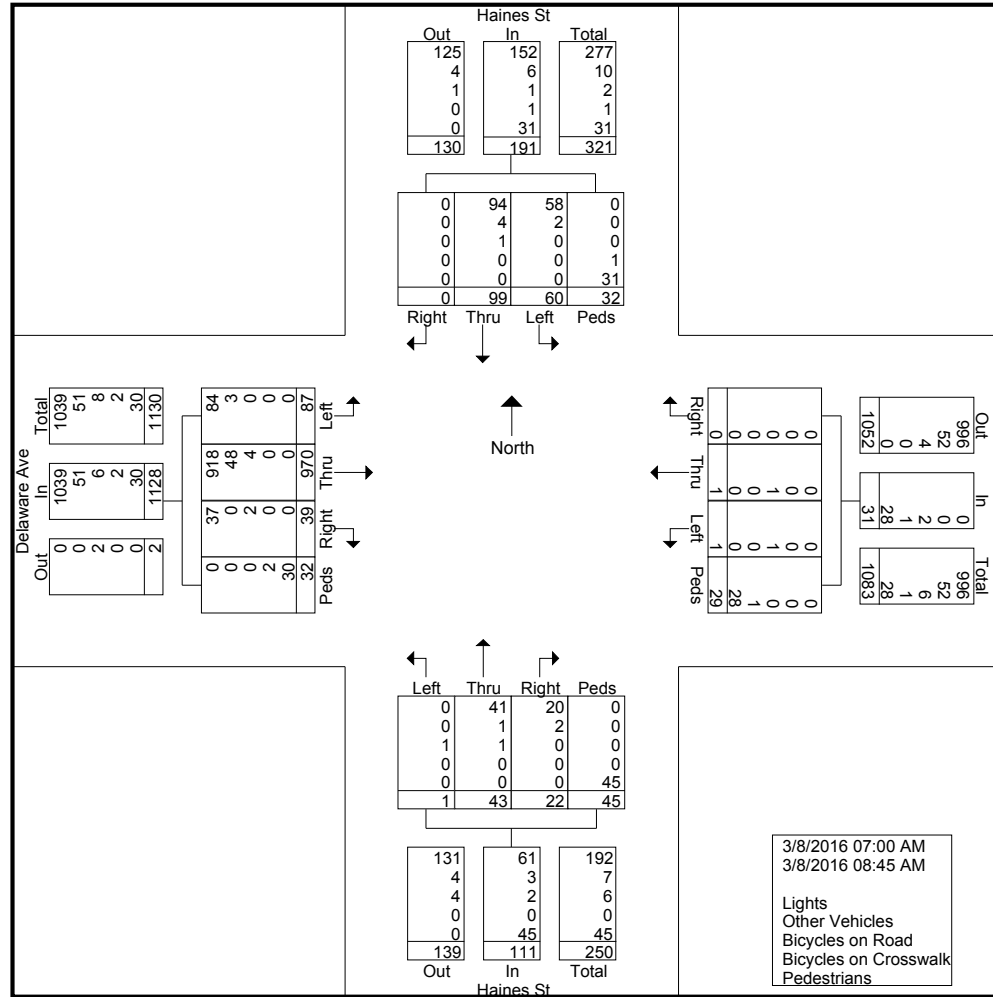
Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

	Haines St From North					Delaware Ave From East					Haines St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
07:00 AM	0	9	8	0	17	0	0	0	1	1	2	2	0	1	5	3	125	4	1	133	156
07:15 AM	0	18	7	3	28	0	0	0	1	1	3	4	0	0	7	8	93	6	4	111	147
07:30 AM	0	4	6	7	17	0	1	0	2	3	4	7	0	8	19	3	145	11	6	165	204
07:45 AM	0	9	7	6	22	0	0	1	3	4	2	7	1	13	23	9	136	13	5	163	212
Total	0	40	28	16	84	0	1	1	7	9	11	20	1	22	54	23	499	34	16	572	719
08:00 AM	0	15	4	3	22	0	0	0	4	4	3	4	0	4	11	4	131	15	4	154	191
08:15 AM	0	20	15	4	39	0	0	0	5	5	4	6	0	4	14	3	108	12	2	125	183
08:30 AM	0	17	4	6	27	0	0	0	7	7	0	2	0	6	8	6	121	13	3	143	185
08:45 AM	0	7	9	3	19	0	0	0	6	6	4	11	0	9	24	3	111	13	7	134	183
Total	0	59	32	16	107	0	0	0	22	22	11	23	0	23	57	16	471	53	16	556	742
Grand Total	0	99	60	32	191	0	1	1	29	31	22	43	1	45	111	39	970	87	32	1128	1461
Apprch %	0	51.8	31.4	16.8		0	3.2	3.2	93.5		19.8	38.7	0.9	40.5		3.5	86	7.7	2.8		
Total %	0	6.8	4.1	2.2	13.1	0	0.1	0.1	2	2.1	1.5	2.9	0.1	3.1	7.6	2.7	66.4	6	2.2	77.2	
Lights	0	94	58	0	152	0	0	0	0	0	20	41	0	0	61	37	918	84	0	1039	1252
% Lights	0	94.9	96.7	0	79.6	0	0	0	0	0	90.9	95.3	0	0	55	94.9	94.6	96.6	0	92.1	85.7
Other Vehicles	0	4	2	0	6	0	0	0	0	0	2	1	0	0	3	0	48	3	0	51	60
% Other Vehicles	0	4	3.3	0	3.1	0	0	0	0	0	9.1	2.3	0	0	2.7	0	4.9	3.4	0	4.5	4.1
Bicycles on Road	0	1	0	0	1	0	1	1	0	2	0	1	1	0	2	2	4	0	0	6	11
% Bicycles on Road	0	1	0	0	0.5	0	100	100	0	6.5	0	2.3	100	0	1.8	5.1	0.4	0	0	0.5	0.8
Bicycles on Crosswalk	0	0	0	1	1	0	0	0	1	1	0	0	0	0	0	0	0	0	2	2	4
% Bicycles on Crosswalk	0	0	0	3.1	0.5	0	0	0	3.4	3.2	0	0	0	0	0	0	0	0	6.2	0.2	0.3
Pedestrians	0	0	0	31	31	0	0	0	28	28	0	0	0	45	45	0	0	0	30	30	134
% Pedestrians	0	0	0	96.9	16.2	0	0	0	96.6	90.3	0	0	0	100	40.5	0	0	0	93.8	2.7	9.2

File Name : Delaware_Ave_at_Haines_St_AM_297610_03-08-2016
 Site Code : Location 6 AM
 Start Date : 3/8/2016
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File Name : Delaware_Ave_at_Haines_St_AM_297610_03-08-2016

Site Code : Location 6 AM

Start Date : 3/8/2016

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	Haines St From North					Delaware Ave From East					Haines St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 07:30 AM																					
07:30 AM	0	4	6	7	17	0	1	0	2	3	4	7	0	8	19	3	145	11	6	165	204
07:45 AM	0	9	7	6	22	0	0	1	3	4	2	7	1	13	23	9	136	13	5	163	212
08:00 AM	0	15	4	3	22	0	0	0	4	4	3	4	0	4	11	4	131	15	4	154	191
08:15 AM	0	20	15	4	39	0	0	0	5	5	4	6	0	4	14	3	108	12	2	125	183
Total Volume	0	48	32	20	100	0	1	1	14	16	13	24	1	29	67	19	520	51	17	607	790
% App. Total	0	48	32	20		0	6.2	6.2	87.5		19.4	35.8	1.5	43.3		3.1	85.7	8.4	2.8		
PHF	.000	.600	.533	.714	.641	.000	.250	.250	.700	.800	.813	.857	.250	.558	.728	.528	.897	.850	.708	.920	.932

File Name : Delaware_Ave_at_Haines_St_PM_297611_03-08-2016

Site Code : Location 6 PM

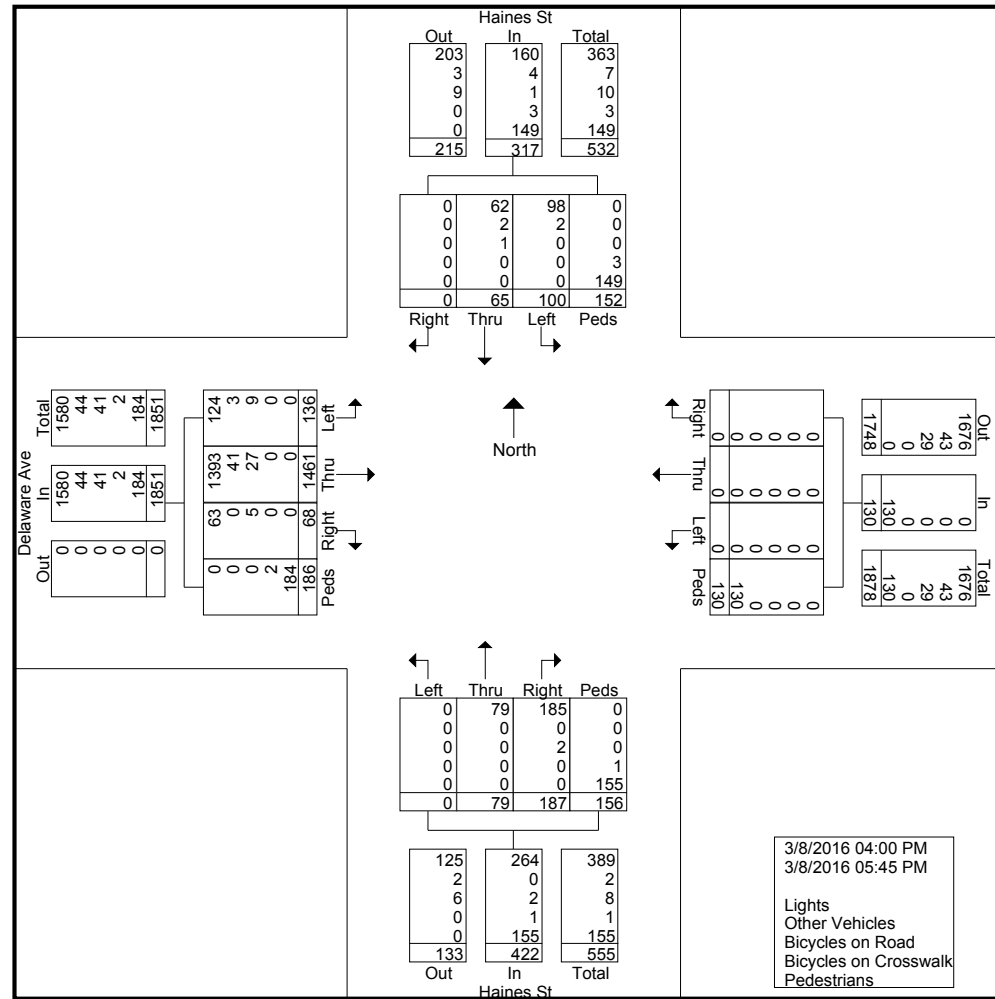
Start Date : 3/8/2016

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Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

	Haines St From North					Delaware Ave From East					Haines St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
04:00 PM	0	10	15	17	42	0	0	0	14	14	14	4	0	8	26	6	168	11	11	196	278
04:15 PM	0	7	6	18	31	0	0	0	17	17	14	15	0	11	40	8	166	12	12	198	286
04:30 PM	0	7	13	17	37	0	0	0	7	7	35	8	0	19	62	11	170	16	17	214	320
04:45 PM	0	4	13	35	52	0	0	0	25	25	29	11	0	35	75	12	163	15	43	233	385
Total	0	28	47	87	162	0	0	0	63	63	92	38	0	73	203	37	667	54	83	841	1269
05:00 PM	0	13	17	13	43	0	0	0	20	20	24	8	0	17	49	7	226	20	40	293	405
05:15 PM	0	9	18	14	41	0	0	0	22	22	22	13	0	12	47	6	184	19	21	230	340
05:30 PM	0	8	12	17	37	0	0	0	13	13	19	9	0	18	46	6	200	24	19	249	345
05:45 PM	0	7	6	21	34	0	0	0	12	12	30	11	0	36	77	12	184	19	23	238	361
Total	0	37	53	65	155	0	0	0	67	67	95	41	0	83	219	31	794	82	103	1010	1451
Grand Total	0	65	100	152	317	0	0	0	130	130	187	79	0	156	422	68	1461	136	186	1851	2720
Apprch %	0	20.5	31.5	47.9		0	0	0	100		44.3	18.7	0	37		3.7	78.9	7.3	10		
Total %	0	2.4	3.7	5.6	11.7	0	0	0	4.8	4.8	6.9	2.9	0	5.7	15.5	2.5	53.7	5	6.8	68.1	
Lights	0	62	98	0	160	0	0	0	0	0	185	79	0	0	264	63	1393	124	0	1580	2004
% Lights	0	95.4	98	0	50.5	0	0	0	0	0	98.9	100	0	0	62.6	92.6	95.3	91.2	0	85.4	73.7
Other Vehicles	0	2	2	0	4	0	0	0	0	0	0	0	0	0	0	0	41	3	0	44	48
% Other Vehicles	0	3.1	2	0	1.3	0	0	0	0	0	0	0	0	0	0	0	2.8	2.2	0	2.4	1.8
Bicycles on Road	0	1	0	0	1	0	0	0	0	0	2	0	0	0	2	5	27	9	0	41	44
% Bicycles on Road	0	1.5	0	0	0.3	0	0	0	0	0	1.1	0	0	0	0.5	7.4	1.8	6.6	0	2.2	1.6
Bicycles on Crosswalk	0	0	0	3	3	0	0	0	0	0	0	0	0	1	1	0	0	0	2	2	6
% Bicycles on Crosswalk	0	0	0	2	0.9	0	0	0	0	0	0	0	0	0.6	0.2	0	0	0	1.1	0.1	0.2
Pedestrians	0	0	0	149	149	0	0	0	130	130	0	0	0	155	155	0	0	0	184	184	618
% Pedestrians	0	0	0	98	47	0	0	0	100	100	0	0	0	99.4	36.7	0	0	0	98.9	9.9	22.7

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File Name : Delaware_Ave_at_Haines_St_PM_297611_03-08-2016

Site Code : Location 6 PM

Start Date : 3/8/2016

Page No : 3

	Haines St From North					Delaware Ave From East					Haines St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 04:45 PM																					
04:45 PM	0	4	13	35	52	0	0	0	25	25	29	11	0	35	75	12	163	15	43	233	385
05:00 PM	0	13	17	13	43	0	0	0	20	20	24	8	0	17	49	7	226	20	40	293	405
05:15 PM	0	9	18	14	41	0	0	0	22	22	22	13	0	12	47	6	184	19	21	230	340
05:30 PM	0	8	12	17	37	0	0	0	13	13	19	9	0	18	46	6	200	24	19	249	345
Total Volume	0	34	60	79	173	0	0	0	80	80	94	41	0	82	217	31	773	78	123	1005	1475
% App. Total	0	19.7	34.7	45.7		0	0	0	100		43.3	18.9	0	37.8		3.1	76.9	7.8	12.2		
PHF	.000	.654	.833	.564	.832	.000	.000	.000	.800	.800	.810	.788	.000	.586	.723	.646	.855	.813	.715	.858	.910

File Name : Delaware_Ave_at_S_Chapel_St_AM_297613_03-08-2016

Site Code : Location 7 AM

Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

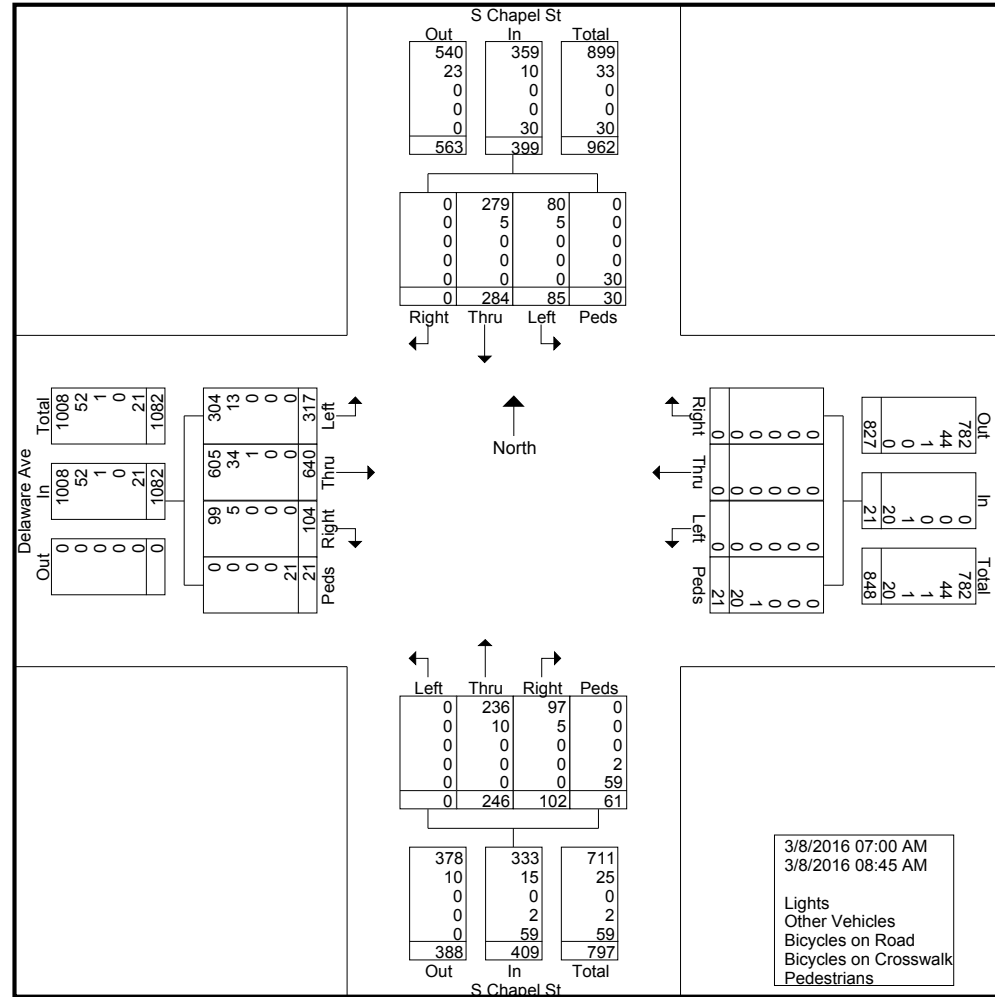
	S Chapel St From North					Delaware Ave From East					S Chapel St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
07:00 AM	0	16	12	1	29	0	0	0	0	0	15	21	0	0	36	9	79	41	3	132	197
07:15 AM	0	20	9	5	34	0	0	0	5	5	11	36	0	7	54	17	67	28	1	113	206
07:30 AM	0	43	6	8	57	0	0	0	1	1	10	44	0	11	65	10	84	48	3	145	268
07:45 AM	0	47	11	6	64	0	0	0	4	4	8	38	0	18	64	17	92	41	3	153	285
Total	0	126	38	20	184	0	0	0	10	10	44	139	0	36	219	53	322	158	10	543	956
08:00 AM	0	63	14	4	81	0	0	0	5	5	9	38	0	5	52	13	78	47	3	141	279
08:15 AM	0	36	10	2	48	0	0	0	3	3	11	23	0	2	36	17	74	37	0	128	215
08:30 AM	0	48	8	2	58	0	0	0	2	2	18	28	0	8	54	12	80	34	1	127	241
08:45 AM	0	11	15	2	28	0	0	0	1	1	20	18	0	10	48	9	86	41	7	143	220
Total	0	158	47	10	215	0	0	0	11	11	58	107	0	25	190	51	318	159	11	539	955
Grand Total	0	284	85	30	399	0	0	0	21	21	102	246	0	61	409	104	640	317	21	1082	1911
Apprch %	0	71.2	21.3	7.5		0	0	0	100		24.9	60.1	0	14.9		9.6	59.1	29.3	1.9		
Total %	0	14.9	4.4	1.6	20.9	0	0	0	1.1	1.1	5.3	12.9	0	3.2	21.4	5.4	33.5	16.6	1.1	56.6	
Lights	0	279	80	0	359	0	0	0	0	0	97	236	0	0	333	99	605	304	0	1008	1700
% Lights	0	98.2	94.1	0	90	0	0	0	0	0	95.1	95.9	0	0	81.4	95.2	94.5	95.9	0	93.2	89
Other Vehicles	0	5	5	0	10	0	0	0	0	0	5	10	0	0	15	5	34	13	0	52	77
% Other Vehicles	0	1.8	5.9	0	2.5	0	0	0	0	0	4.9	4.1	0	0	3.7	4.8	5.3	4.1	0	4.8	4
Bicycles on Road	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1
% Bicycles on Road	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.2	0	0	0.1	0.1
Bicycles on Crosswalk	0	0	0	0	0	0	0	0	1	1	0	0	0	2	2	0	0	0	0	0	3
% Bicycles on Crosswalk	0	0	0	0	0	0	0	0	4.8	4.8	0	0	0	3.3	0.5	0	0	0	0	0	0.2
Pedestrians	0	0	0	30	30	0	0	0	20	20	0	0	0	59	59	0	0	0	21	21	130
% Pedestrians	0	0	0	100	7.5	0	0	0	95.2	95.2	0	0	0	96.7	14.4	0	0	0	100	1.9	6.8

File Name : Delaware_Ave_at_S_Chapel_St_AM_297613_03-08-2016

Site Code : Location 7 AM

Start Date : 3/8/2016

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File Name : Delaware_Ave_at_S_Chapel_St_AM_297613_03-08-2016

Site Code : Location 7 AM

Start Date : 3/8/2016

Page No : 3

	S Chapel St From North					Delaware Ave From East					S Chapel St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 07:30 AM																					
07:30 AM	0	43	6	8	57	0	0	0	1	1	10	44	0	11	65	10	84	48	3	145	268
07:45 AM	0	47	11	6	64	0	0	0	4	4	8	38	0	18	64	17	92	41	3	153	285
08:00 AM	0	63	14	4	81	0	0	0	5	5	9	38	0	5	52	13	78	47	3	141	279
08:15 AM	0	36	10	2	48	0	0	0	3	3	11	23	0	2	36	17	74	37	0	128	215
Total Volume	0	189	41	20	250	0	0	0	13	13	38	143	0	36	217	57	328	173	9	567	1047
% App. Total	0	75.6	16.4	8		0	0	0	100		17.5	65.9	0	16.6		10.1	57.8	30.5	1.6		
PHF	.000	.750	.732	.625	.772	.000	.000	.000	.650	.650	.864	.813	.000	.500	.835	.838	.891	.901	.750	.926	.918

File Name : Delaware_Ave_at_S_Chapel_St_PM_297614_03-08-2016

Site Code : Location 7 PM

Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

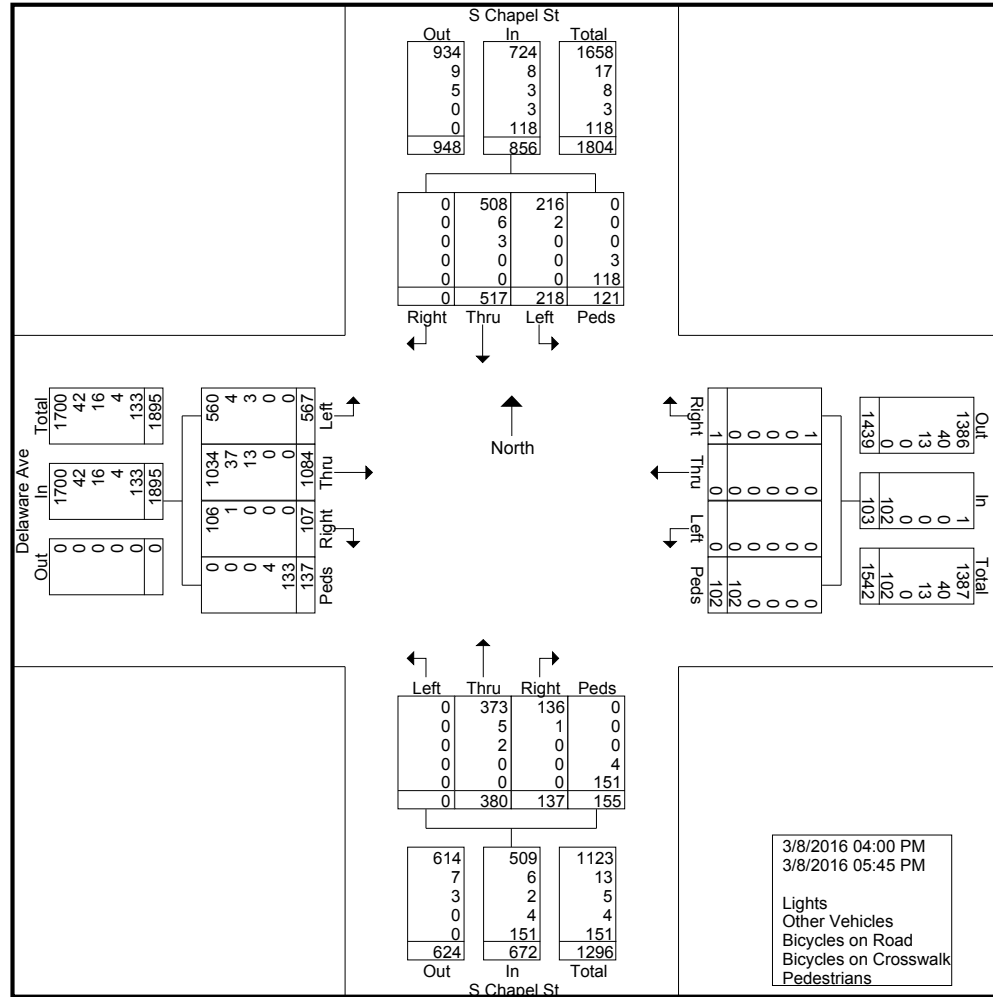
	S Chapel St From North					Delaware Ave From East					S Chapel St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
04:00 PM	0	52	32	19	103	1	0	0	12	13	12	52	0	12	76	10	112	80	18	220	412
04:15 PM	0	50	28	17	95	0	0	0	13	13	22	43	0	21	86	12	119	66	14	211	405
04:30 PM	0	69	29	9	107	0	0	0	15	15	17	67	0	20	104	8	120	74	7	209	435
04:45 PM	0	63	25	27	115	0	0	0	9	9	15	56	0	41	112	16	141	68	43	268	504
Total	0	234	114	72	420	1	0	0	49	50	66	218	0	94	378	46	492	288	82	908	1756
05:00 PM	0	81	31	17	129	0	0	0	19	19	13	47	0	12	72	17	175	72	15	279	499
05:15 PM	0	61	27	20	108	0	0	0	10	10	23	35	0	8	66	18	143	80	17	258	442
05:30 PM	0	77	24	7	108	0	0	0	9	9	16	53	0	14	83	10	139	62	12	223	423
05:45 PM	0	64	22	5	91	0	0	0	15	15	19	27	0	27	73	16	135	65	11	227	406
Total	0	283	104	49	436	0	0	0	53	53	71	162	0	61	294	61	592	279	55	987	1770
Grand Total	0	517	218	121	856	1	0	0	102	103	137	380	0	155	672	107	1084	567	137	1895	3526
Apprch %	0	60.4	25.5	14.1		1	0	0	99		20.4	56.5	0	23.1		5.6	57.2	29.9	7.2		
Total %	0	14.7	6.2	3.4	24.3	0	0	0	2.9	2.9	3.9	10.8	0	4.4	19.1	3	30.7	16.1	3.9	53.7	
Lights	0	508	216	0	724	1	0	0	0	1	136	373	0	0	509	106	1034	560	0	1700	2934
% Lights	0	98.3	99.1	0	84.6	100	0	0	0	1	99.3	98.2	0	0	75.7	99.1	95.4	98.8	0	89.7	83.2
Other Vehicles	0	6	2	0	8	0	0	0	0	0	1	5	0	0	6	1	37	4	0	42	56
% Other Vehicles	0	1.2	0.9	0	0.9	0	0	0	0	0	0.7	1.3	0	0	0.9	0.9	3.4	0.7	0	2.2	1.6
Bicycles on Road	0	3	0	0	3	0	0	0	0	0	0	2	0	0	2	0	13	3	0	16	21
% Bicycles on Road	0	0.6	0	0	0.4	0	0	0	0	0	0	0.5	0	0	0.3	0	1.2	0.5	0	0.8	0.6
Bicycles on Crosswalk	0	0	0	3	3	0	0	0	0	0	0	0	0	4	4	0	0	0	4	4	11
% Bicycles on Crosswalk	0	0	0	2.5	0.4	0	0	0	0	0	0	0	0	2.6	0.6	0	0	0	2.9	0.2	0.3
Pedestrians	0	0	0	118	118	0	0	0	102	102	0	0	0	151	151	0	0	0	133	133	504
% Pedestrians	0	0	0	97.5	13.8	0	0	0	100	99	0	0	0	97.4	22.5	0	0	0	97.1	7	14.3

File Name : Delaware_Ave_at_S_Chapel_St_PM_297614_03-08-2016

Site Code : Location 7 PM

Start Date : 3/8/2016

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File Name : Delaware_Ave_at_S_Chapel_St_PM_297614_03-08-2016

Site Code : Location 7 PM

Start Date : 3/8/2016

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	S Chapel St From North					Delaware Ave From East					S Chapel St From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 04:30 PM																					
04:30 PM	0	69	29	9	107	0	0	0	15	15	17	67	0	20	104	8	120	74	7	209	435
04:45 PM	0	63	25	27	115	0	0	0	9	9	15	56	0	41	112	16	141	68	43	268	504
05:00 PM	0	81	31	17	129	0	0	0	19	19	13	47	0	12	72	17	175	72	15	279	499
05:15 PM	0	61	27	20	108	0	0	0	10	10	23	35	0	8	66	18	143	80	17	258	442
Total Volume	0	274	112	73	459	0	0	0	53	53	68	205	0	81	354	59	579	294	82	1014	1880
% App. Total	0	59.7	24.4	15.9		0	0	0	100		19.2	57.9	0	22.9		5.8	57.1	29	8.1		
PHF	.000	.846	.903	.676	.890	.000	.000	.000	.697	.697	.739	.765	.000	.494	.790	.819	.827	.919	.477	.909	.933

File Name : Delaware_Ave_at_Pomeroy_Trail_AM_297615_03-08-2016

Site Code : Location * AM

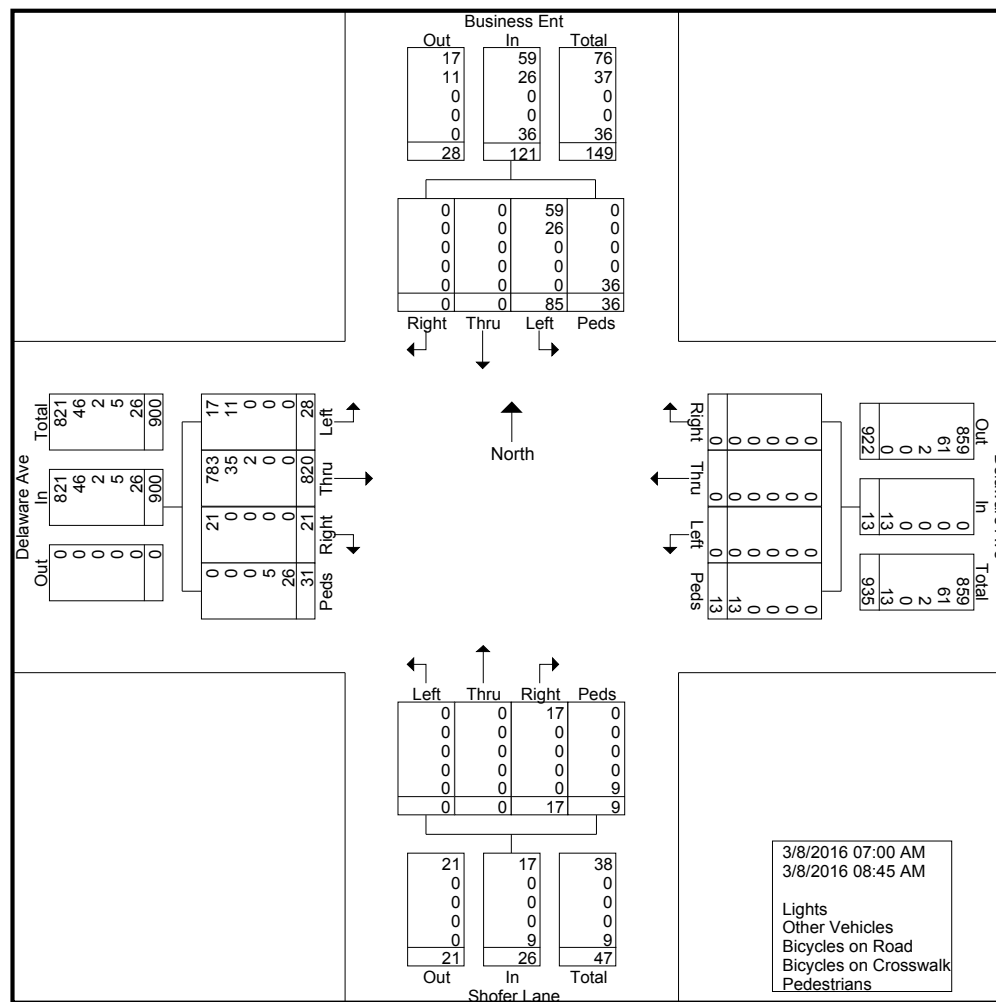
Start Date : 3/8/2016

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Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

	Business Ent From North					Delaware Ave From East					Shofer Lane From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
07:00 AM	0	0	11	12	23	0	0	0	0	0	3	0	0	0	3	0	116	4	3	123	149
07:15 AM	0	0	7	7	14	0	0	0	3	3	1	0	0	0	1	2	80	1	2	85	103
07:30 AM	0	0	11	3	14	0	0	0	2	2	4	0	0	1	5	2	106	3	9	120	141
07:45 AM	0	0	11	3	14	0	0	0	0	0	3	0	0	6	9	5	103	3	7	118	141
Total	0	0	40	25	65	0	0	0	5	5	11	0	0	7	18	9	405	11	21	446	534
08:00 AM	0	0	12	3	15	0	0	0	2	2	1	0	0	1	2	4	99	5	3	111	130
08:15 AM	0	0	9	1	10	0	0	0	0	0	2	0	0	0	2	1	95	4	2	102	114
08:30 AM	0	0	10	2	12	0	0	0	2	2	1	0	0	1	2	2	108	3	1	114	130
08:45 AM	0	0	14	5	19	0	0	0	4	4	2	0	0	0	2	5	113	5	4	127	152
Total	0	0	45	11	56	0	0	0	8	8	6	0	0	2	8	12	415	17	10	454	526
Grand Total	0	0	85	36	121	0	0	0	13	13	17	0	0	9	26	21	820	28	31	900	1060
Apprch %	0	0	70.2	29.8		0	0	0	100		65.4	0	0	34.6		2.3	91.1	3.1	3.4		
Total %	0	0	8	3.4	11.4	0	0	0	1.2	1.2	1.6	0	0	0.8	2.5	2	77.4	2.6	2.9	84.9	
Lights	0	0	59	0	59	0	0	0	0	0	17	0	0	0	17	21	783	17	0	821	897
% Lights	0	0	69.4	0	48.8	0	0	0	0	0	100	0	0	0	65.4	100	95.5	60.7	0	91.2	84.6
Other Vehicles	0	0	26	0	26	0	0	0	0	0	0	0	0	0	0	0	35	11	0	46	72
% Other Vehicles	0	0	30.6	0	21.5	0	0	0	0	0	0	0	0	0	0	0	4.3	39.3	0	5.1	6.8
Bicycles on Road	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	2	2
% Bicycles on Road	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.2	0	0	0.2	0.2
Bicycles on Crosswalk	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	5	5
% Bicycles on Crosswalk	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	16.1	0.6	0.5
Pedestrians	0	0	0	36	36	0	0	0	13	13	0	0	0	9	9	0	0	0	26	26	84
% Pedestrians	0	0	0	100	29.8	0	0	0	100	100	0	0	0	100	34.6	0	0	0	83.9	2.9	7.9

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File Name : Delaware_Ave_at_Pomeroy_Trail_AM_297615_03-08-2016

Site Code : Location * AM

Start Date : 3/8/2016

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	Business Ent From North					Delaware Ave From East					Shofer Lane From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 07:00 AM																					
07:00 AM	0	0	11	12	23	0	0	0	0	0	3	0	0	0	3	0	116	4	3	123	149
07:15 AM	0	0	7	7	14	0	0	0	3	3	1	0	0	0	1	2	80	1	2	85	103
07:30 AM	0	0	11	3	14	0	0	0	2	2	4	0	0	1	5	2	106	3	9	120	141
07:45 AM	0	0	11	3	14	0	0	0	0	0	3	0	0	6	9	5	103	3	7	118	141
Total Volume	0	0	40	25	65	0	0	0	5	5	11	0	0	7	18	9	405	11	21	446	534
% App. Total	0	0	61.5	38.5		0	0	0	100		61.1	0	0	38.9		2	90.8	2.5	4.7		
PHF	.000	.000	.909	.521	.707	.000	.000	.000	.417	.417	.688	.000	.000	.292	.500	.450	.873	.688	.583	.907	.896

File Name : Delaware_Ave_at_Pomeroy_Trail_PM_297616_03-08-2016

Site Code : Location 7 PM

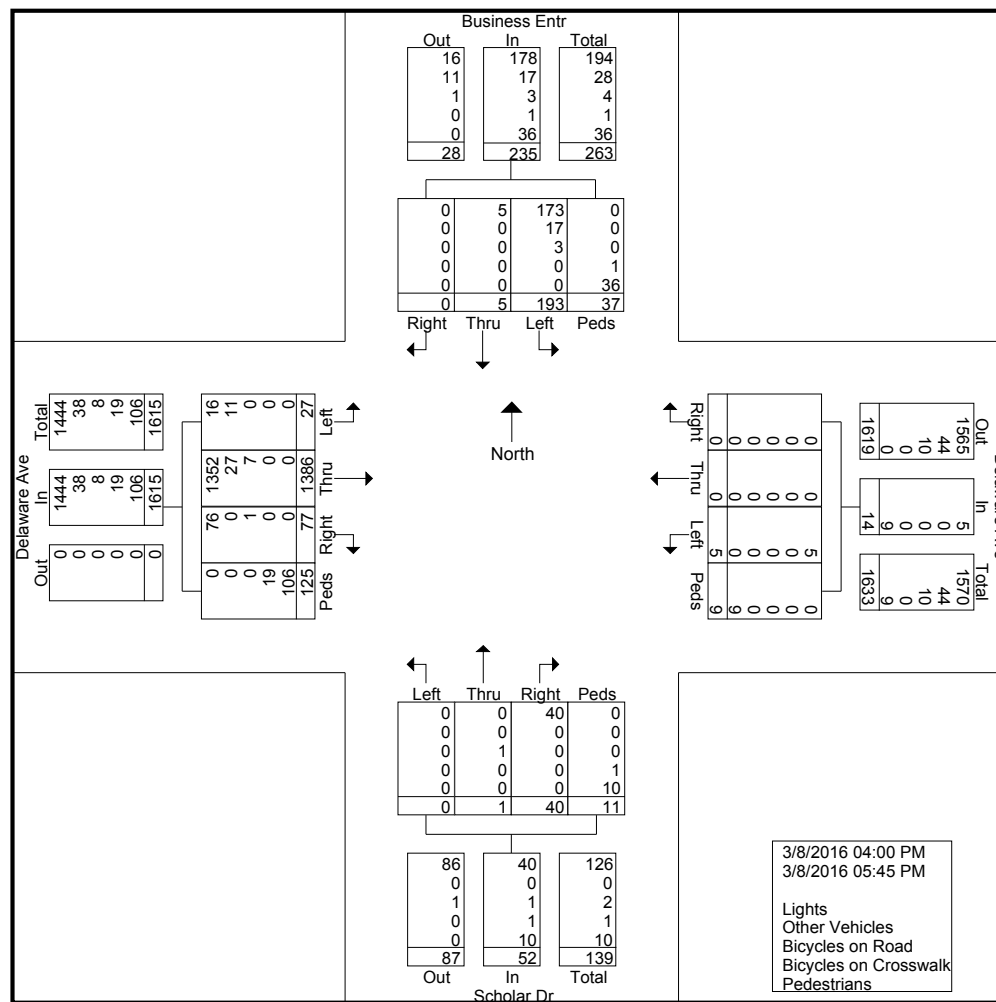
Start Date : 3/8/2016

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Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

	Business Entr From North					Delaware Ave From East					Scholar Dr From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
04:00 PM	0	0	16	6	22	0	0	0	1	1	1	0	0	2	3	4	152	3	14	173	199
04:15 PM	0	0	29	16	45	0	0	0	0	0	2	1	0	3	6	6	160	4	14	184	235
04:30 PM	0	0	22	3	25	0	0	0	1	1	5	0	0	4	9	8	158	2	11	179	214
04:45 PM	0	0	15	5	20	0	0	0	1	1	4	0	0	1	5	8	178	6	16	208	234
Total	0	0	82	30	112	0	0	0	3	3	12	1	0	10	23	26	648	15	55	744	882
05:00 PM	0	2	26	1	29	0	0	2	3	5	9	0	0	0	9	19	216	5	19	259	302
05:15 PM	0	1	27	2	30	0	0	1	0	1	5	0	0	0	5	9	174	1	26	210	246
05:30 PM	0	0	28	2	30	0	0	0	2	2	7	0	0	0	7	11	183	2	11	207	246
05:45 PM	0	2	30	2	34	0	0	2	1	3	7	0	0	1	8	12	165	4	14	195	240
Total	0	5	111	7	123	0	0	5	6	11	28	0	0	1	29	51	738	12	70	871	1034
Grand Total	0	5	193	37	235	0	0	5	9	14	40	1	0	11	52	77	1386	27	125	1615	1916
Apprch %	0	2.1	82.1	15.7		0	0	35.7	64.3		76.9	1.9	0	21.2		4.8	85.8	1.7	7.7		
Total %	0	0.3	10.1	1.9	12.3	0	0	0.3	0.5	0.7	2.1	0.1	0	0.6	2.7	4	72.3	1.4	6.5	84.3	
Lights	0	5	173	0	178	0	0	5	0	5	40	0	0	0	40	76	1352	16	0	1444	1667
% Lights	0	100	89.6	0	75.7	0	0	100	0	35.7	100	0	0	0	76.9	98.7	97.5	59.3	0	89.4	87
Other Vehicles	0	0	17	0	17	0	0	0	0	0	0	0	0	0	0	0	27	11	0	38	55
% Other Vehicles	0	0	8.8	0	7.2	0	0	0	0	0	0	0	0	0	0	0	1.9	40.7	0	2.4	2.9
Bicycles on Road	0	0	3	0	3	0	0	0	0	0	0	1	0	0	1	1	7	0	0	8	12
% Bicycles on Road	0	0	1.6	0	1.3	0	0	0	0	0	0	100	0	0	1.9	1.3	0.5	0	0	0.5	0.6
Bicycles on Crosswalk	0	0	0	1	1	0	0	0	0	0	0	0	0	1	1	0	0	0	19	19	21
% Bicycles on Crosswalk	0	0	0	2.7	0.4	0	0	0	0	0	0	0	0	9.1	1.9	0	0	0	15.2	1.2	1.1
Pedestrians	0	0	0	36	36	0	0	0	9	9	0	0	0	10	10	0	0	0	106	106	161
% Pedestrians	0	0	0	97.3	15.3	0	0	0	100	64.3	0	0	0	90.9	19.2	0	0	0	84.8	6.6	8.4

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File Name : Delaware_Ave_at_Pomeroy_Trail_PM_297616_03-08-2016

Site Code : Location 7 PM

Start Date : 3/8/2016

Page No : 3

	Business Entr From North					Delaware Ave From East					Scholar Dr From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 05:00 PM																					
05:00 PM	0	2	26	1	29	0	0	2	3	5	9	0	0	0	9	19	216	5	19	259	302
05:15 PM	0	1	27	2	30	0	0	1	0	1	5	0	0	0	5	9	174	1	26	210	246
05:30 PM	0	0	28	2	30	0	0	0	2	2	7	0	0	0	7	11	183	2	11	207	246
05:45 PM	0	2	30	2	34	0	0	2	1	3	7	0	0	1	8	12	165	4	14	195	240
Total Volume	0	5	111	7	123	0	0	5	6	11	28	0	0	1	29	51	738	12	70	871	1034
% App. Total	0	4.1	90.2	5.7		0	0	45.5	54.5		96.6	0	0	3.4		5.9	84.7	1.4	8		
PHF	.000	.625	.925	.875	.904	.000	.000	.625	.500	.550	.778	.000	.000	.250	.806	.671	.854	.600	.673	.841	.856

File Name : Delaware_Ave_at_Tyre_Ave_AM_297617_03-08-2016

Site Code : Location 9 AM

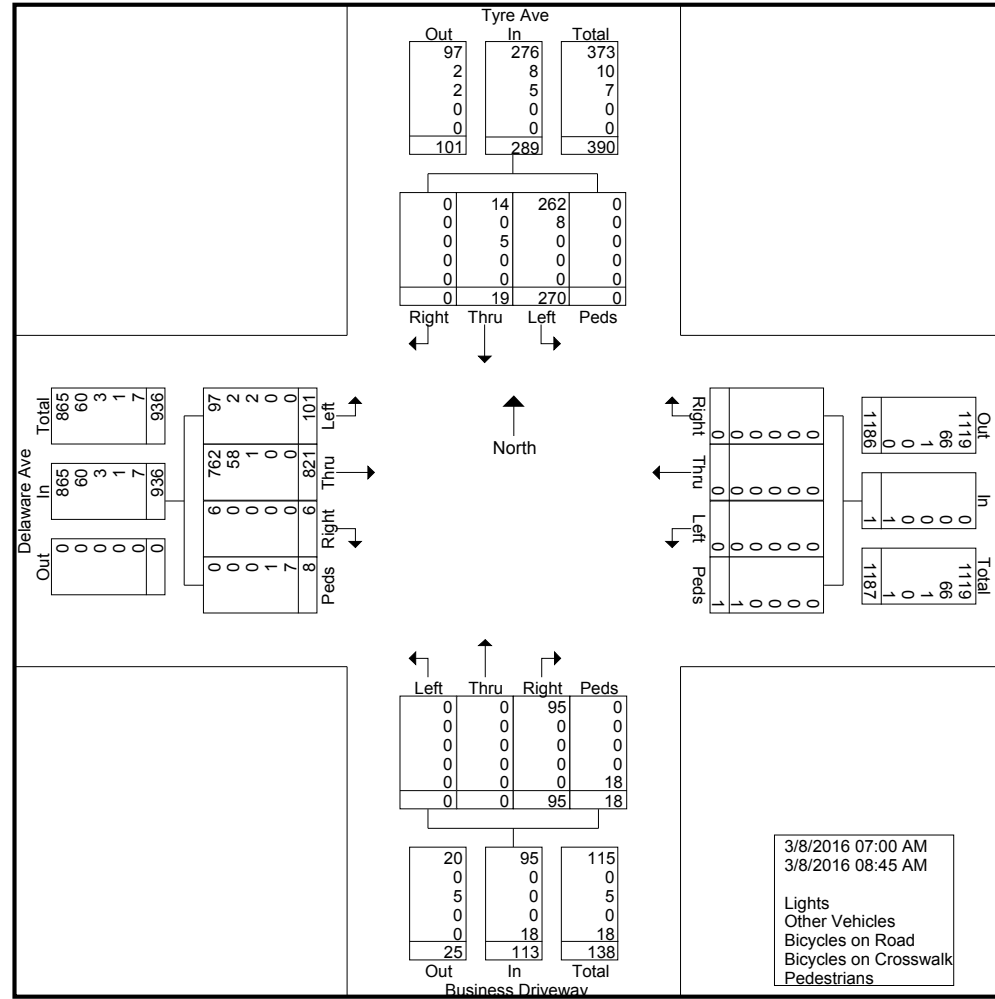
Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

	Tyre Ave From North					Delaware Ave From East					Business Driveway From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
07:00 AM	0	1	109	0	110	0	0	0	0	0	56	0	0	2	58	1	128	6	0	135	303
07:15 AM	0	4	44	0	48	0	0	0	0	0	18	0	0	0	18	0	85	8	2	95	161
07:30 AM	0	2	16	0	18	0	0	0	0	0	2	0	0	7	9	0	107	13	0	120	147
07:45 AM	0	1	23	0	24	0	0	0	0	0	3	0	0	4	7	0	98	16	2	116	147
Total	0	8	192	0	200	0	0	0	0	0	79	0	0	13	92	1	418	43	4	466	758
08:00 AM	0	2	11	0	13	0	0	0	0	0	4	0	0	1	5	1	96	15	0	112	130
08:15 AM	0	1	22	0	23	0	0	0	0	0	3	0	0	0	3	2	93	12	1	108	134
08:30 AM	0	3	17	0	20	0	0	0	1	1	8	0	0	2	10	1	108	11	3	123	154
08:45 AM	0	5	28	0	33	0	0	0	0	0	1	0	0	2	3	1	106	20	0	127	163
Total	0	11	78	0	89	0	0	0	1	1	16	0	0	5	21	5	403	58	4	470	581
Grand Total	0	19	270	0	289	0	0	0	1	1	95	0	0	18	113	6	821	101	8	936	1339
Apprch %	0	6.6	93.4	0		0	0	0	100		84.1	0	0	15.9		0.6	87.7	10.8	0.9		
Total %	0	1.4	20.2	0	21.6	0	0	0	0.1	0.1	7.1	0	0	1.3	8.4	0.4	61.3	7.5	0.6	69.9	
Lights	0	14	262	0	276	0	0	0	0	0	95	0	0	0	95	6	762	97	0	865	1236
% Lights	0	73.7	97	0	95.5	0	0	0	0	0	100	0	0	0	84.1	100	92.8	96	0	92.4	92.3
Other Vehicles	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	58	2	0	60	68
% Other Vehicles	0	0	3	0	2.8	0	0	0	0	0	0	0	0	0	0	0	7.1	2	0	6.4	5.1
Bicycles on Road	0	5	0	0	5	0	0	0	0	0	0	0	0	0	0	0	1	2	0	3	8
% Bicycles on Road	0	26.3	0	0	1.7	0	0	0	0	0	0	0	0	0	0	0	0.1	2	0	0.3	0.6
Bicycles on Crosswalk	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
% Bicycles on Crosswalk	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12.5	0.1	0.1
Pedestrians	0	0	0	0	0	0	0	0	1	1	0	0	0	18	18	0	0	0	7	7	26
% Pedestrians	0	0	0	0	0	0	0	0	100	100	0	0	0	100	15.9	0	0	0	87.5	0.7	1.9

Page No : 2



File Name : Delaware_Ave_at_Tyre_Ave_AM_297617_03-08-2016

Site Code : Location 9 AM

Start Date : 3/8/2016

Page No : 3

	Tyre Ave From North					Delaware Ave From East					Business Driveway From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 07:00 AM																					
07:00 AM	0	1	109	0	110	0	0	0	0	0	56	0	0	2	58	1	128	6	0	135	303
07:15 AM	0	4	44	0	48	0	0	0	0	0	18	0	0	0	18	0	85	8	2	95	161
07:30 AM	0	2	16	0	18	0	0	0	0	0	2	0	0	7	9	0	107	13	0	120	147
07:45 AM	0	1	23	0	24	0	0	0	0	0	3	0	0	4	7	0	98	16	2	116	147
Total Volume	0	8	192	0	200	0	0	0	0	0	79	0	0	13	92	1	418	43	4	466	758
% App. Total	0	4	96	0		0	0	0	0		85.9	0	0	14.1		0.2	89.7	9.2	0.9		
PHF	.000	.500	.440	.000	.455	.000	.000	.000	.000	.000	.353	.000	.000	.464	.397	.250	.816	.672	.500	.863	.625

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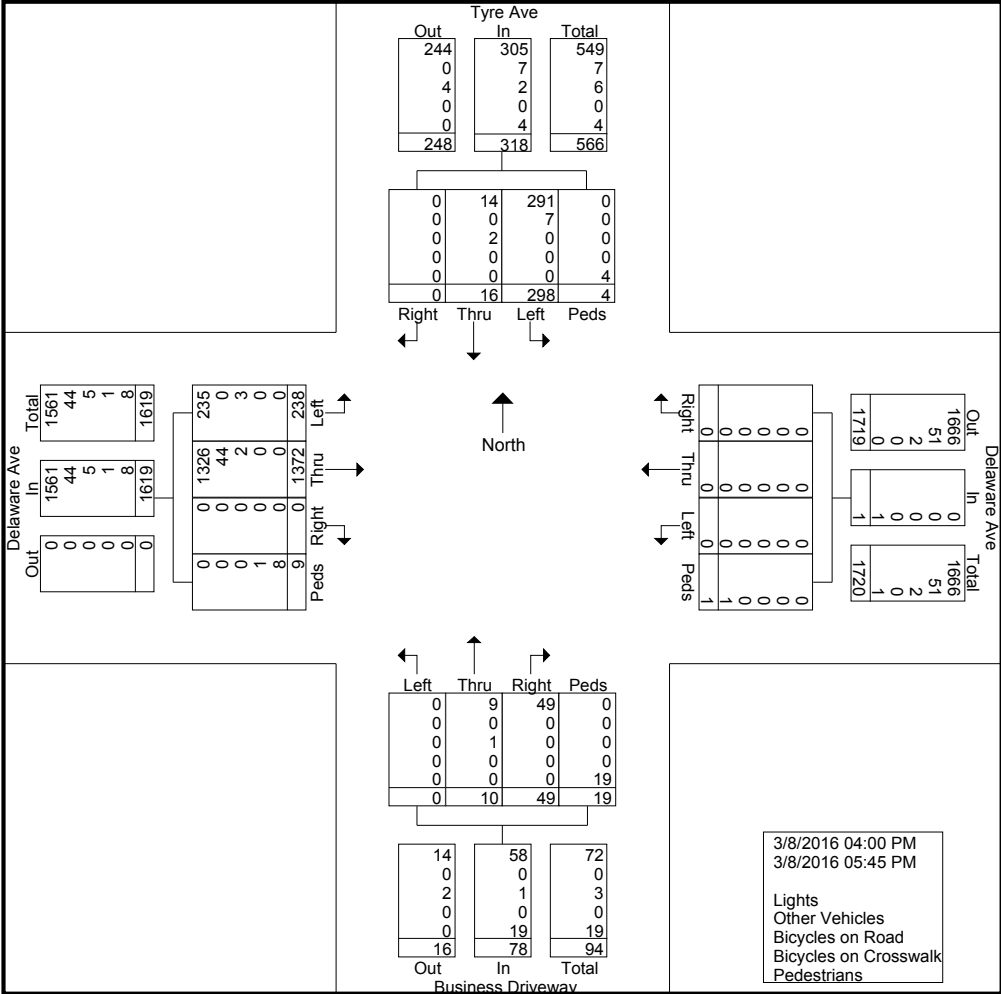
Site Code : Location 9 PM

Start Date : 3/8/2016

Page No : 1

Groups Printed- Lights - Other Vehicles - Bicycles on Road - Bicycles on Crosswalk - Pedestrians

	Tyre Ave From North					Delaware Ave From East					Business Driveway From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
04:00 PM	0	0	31	0	31	0	0	0	0	0	6	0	0	2	8	0	154	24	0	178	217
04:15 PM	0	3	40	0	43	0	0	0	0	0	4	0	0	6	10	0	142	41	0	183	236
04:30 PM	0	2	37	0	39	0	0	0	1	1	9	3	0	5	17	0	162	25	0	187	244
04:45 PM	0	3	49	1	53	0	0	0	0	0	9	2	0	1	12	0	166	22	1	189	254
Total	0	8	157	1	166	0	0	0	1	1	28	5	0	14	47	0	624	112	1	737	951
05:00 PM	0	4	48	0	52	0	0	0	0	0	6	1	0	1	8	0	225	35	2	262	322
05:15 PM	0	0	28	2	30	0	0	0	0	0	4	0	0	2	6	0	166	35	2	203	239
05:30 PM	0	0	36	1	37	0	0	0	0	0	6	1	0	1	8	0	188	31	2	221	266
05:45 PM	0	4	29	0	33	0	0	0	0	0	5	3	0	1	9	0	169	25	2	196	238
Total	0	8	141	3	152	0	0	0	0	0	21	5	0	5	31	0	748	126	8	882	1065
Grand Total	0	16	298	4	318	0	0	0	1	1	49	10	0	19	78	0	1372	238	9	1619	2016
Apprch %	0	5	93.7	1.3		0	0	0	100		62.8	12.8	0	24.4		0	84.7	14.7	0.6		
Total %	0	0.8	14.8	0.2	15.8	0	0	0	0	0	2.4	0.5	0	0.9	3.9	0	68.1	11.8	0.4	80.3	
Lights	0	14	291	0	305	0	0	0	0	0	49	9	0	0	58	0	1326	235	0	1561	1924
% Lights	0	87.5	97.7	0	95.9	0	0	0	0	0	100	90	0	0	74.4	0	96.6	98.7	0	96.4	95.4
Other Vehicles	0	0	7	0	7	0	0	0	0	0	0	0	0	0	0	0	44	0	0	44	51
% Other Vehicles	0	0	2.3	0	2.2	0	0	0	0	0	0	0	0	0	0	0	3.2	0	0	2.7	2.5
Bicycles on Road	0	2	0	0	2	0	0	0	0	0	0	1	0	0	1	0	2	3	0	5	8
% Bicycles on Road	0	12.5	0	0	0.6	0	0	0	0	0	0	10	0	0	1.3	0	0.1	1.3	0	0.3	0.4
Bicycles on Crosswalk	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
% Bicycles on Crosswalk	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11.1	0.1	0
Pedestrians	0	0	0	4	4	0	0	0	1	1	0	0	0	19	19	0	0	0	8	8	32
% Pedestrians	0	0	0	100	1.3	0	0	0	100	100	0	0	0	100	24.4	0	0	0	88.9	0.5	1.6



File Name : Delaware_Ave_at_Tyre_Ave_PM_297618_03-08-2016

Site Code : Location 9 PM

Start Date : 3/8/2016

Page No : 3

	Tyre Ave From North					Delaware Ave From East					Business Driveway From South					Delaware Ave From West					
Start Time	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Right	Thru	Left	Peds	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																					
Peak Hour for Entire Intersection Begins at 04:45 PM																					
04:45 PM	0	3	49	1	53	0	0	0	0	0	9	2	0	1	12	0	166	22	1	189	254
05:00 PM	0	4	48	0	52	0	0	0	0	0	6	1	0	1	8	0	225	35	2	262	322
05:15 PM	0	0	28	2	30	0	0	0	0	0	4	0	0	2	6	0	166	35	2	203	239
05:30 PM	0	0	36	1	37	0	0	0	0	0	6	1	0	1	8	0	188	31	2	221	266
Total Volume	0	7	161	4	172	0	0	0	0	0	25	4	0	5	34	0	745	123	7	875	1081
% App. Total	0	4.1	93.6	2.3		0	0	0	0		73.5	11.8	0	14.7		0	85.1	14.1	0.8		
PHF	.000	.438	.821	.500	.811	.000	.000	.000	.000	.000	.694	.500	.000	.625	.708	.000	.828	.879	.875	.835	.839

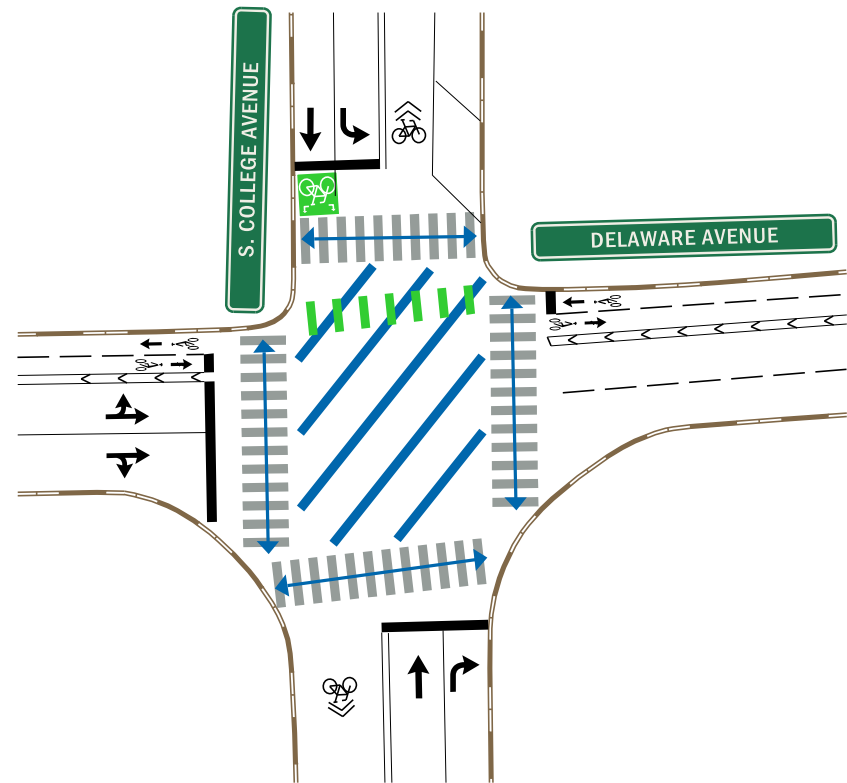


APPENDIX B

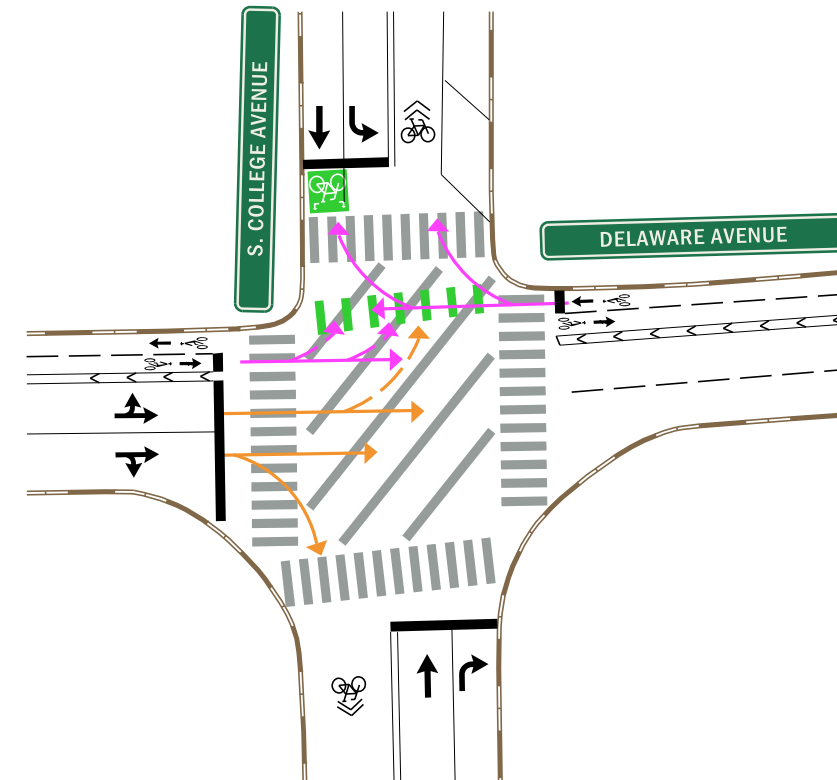
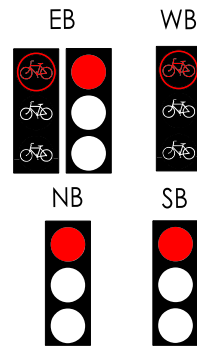
GRAPHICAL FIGURES DEPICTING THE OPERATION OF EACH ALTERNATIVE

DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

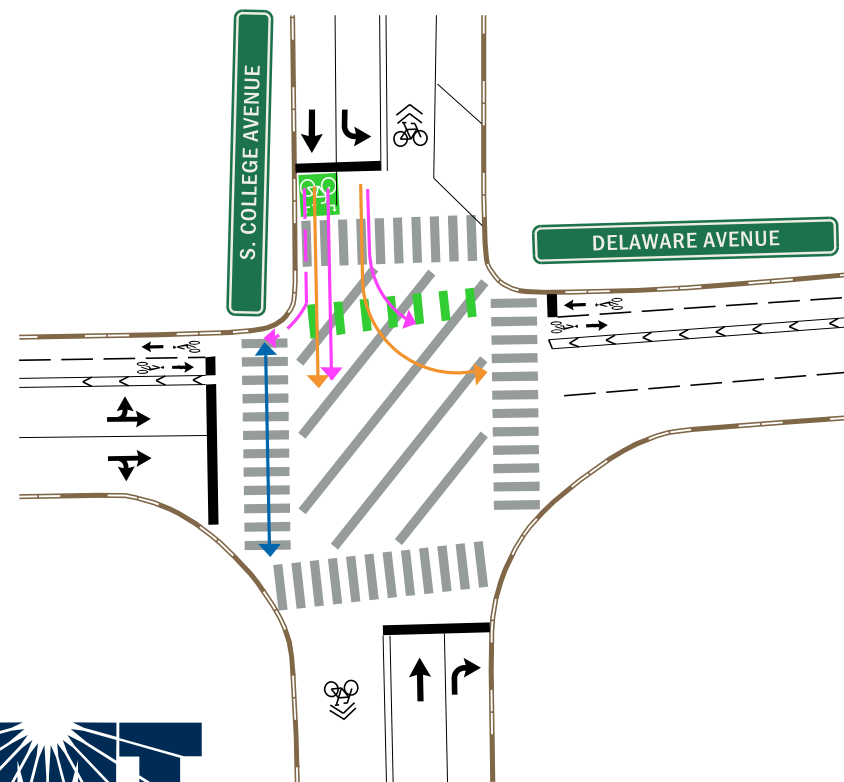
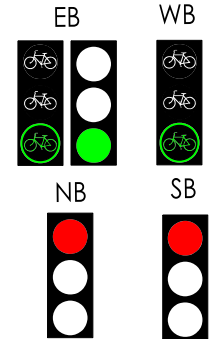
ALTERNATIVE 1- PERMISSIVE WITH TURN QUEUE BOX



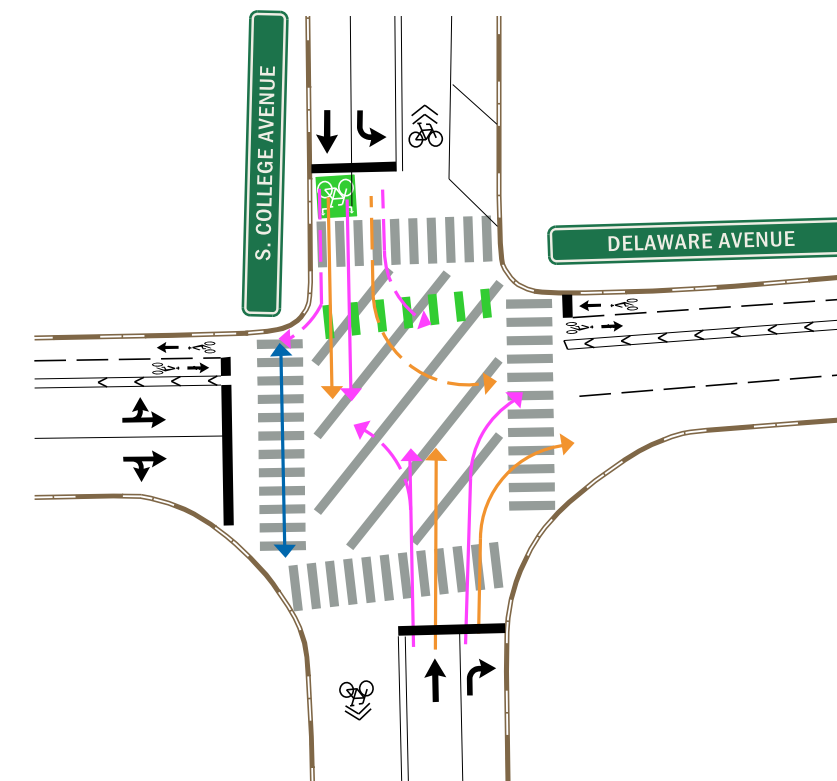
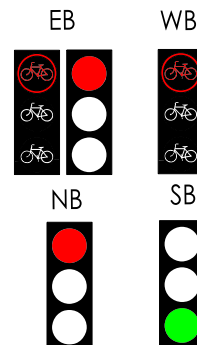
PHASE 1



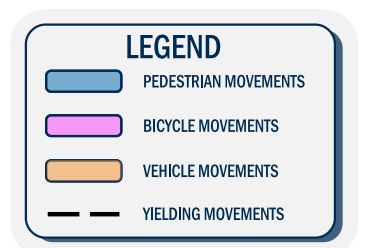
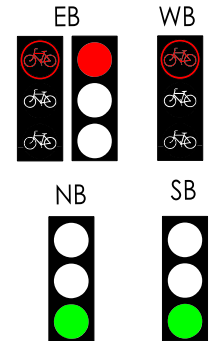
PHASE 2



PHASE 3

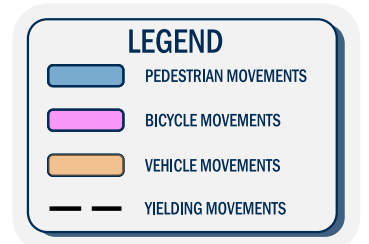
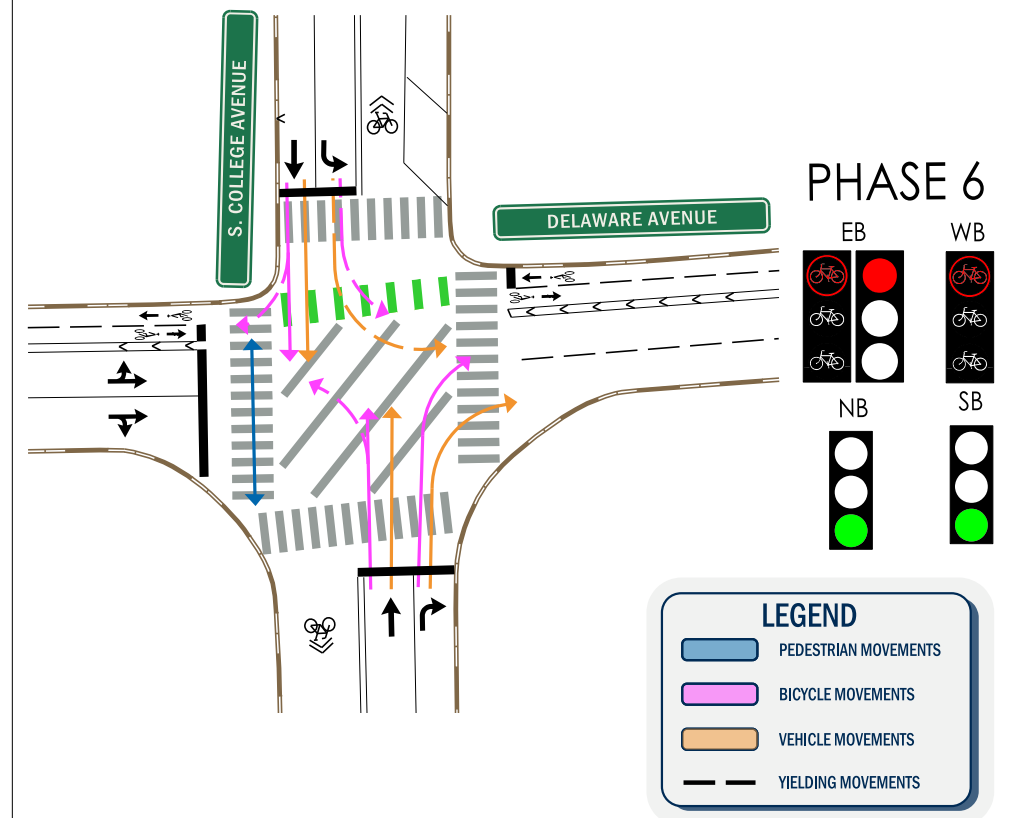
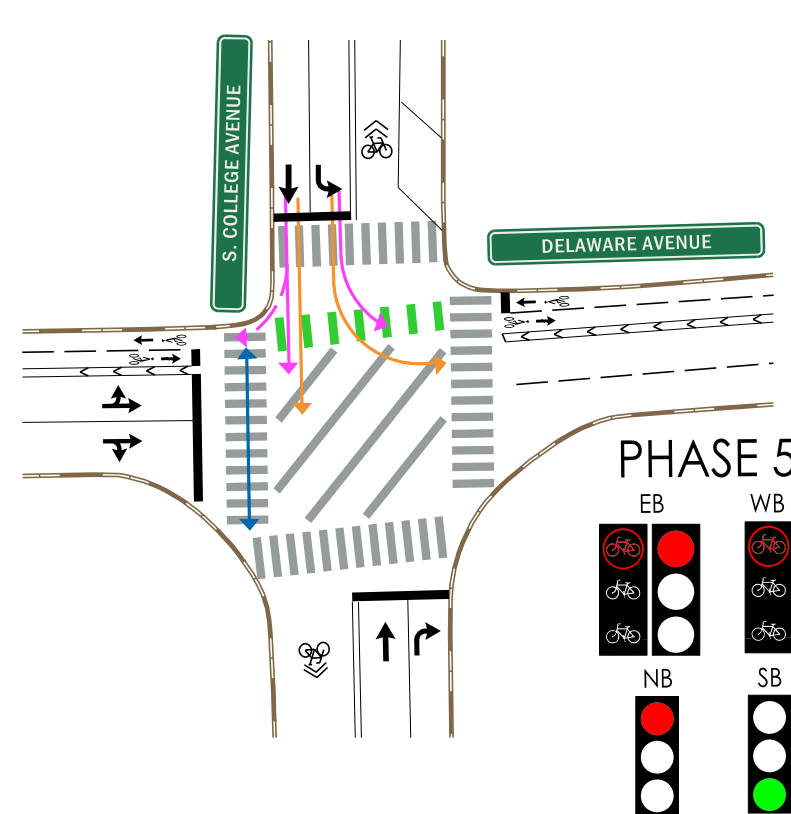
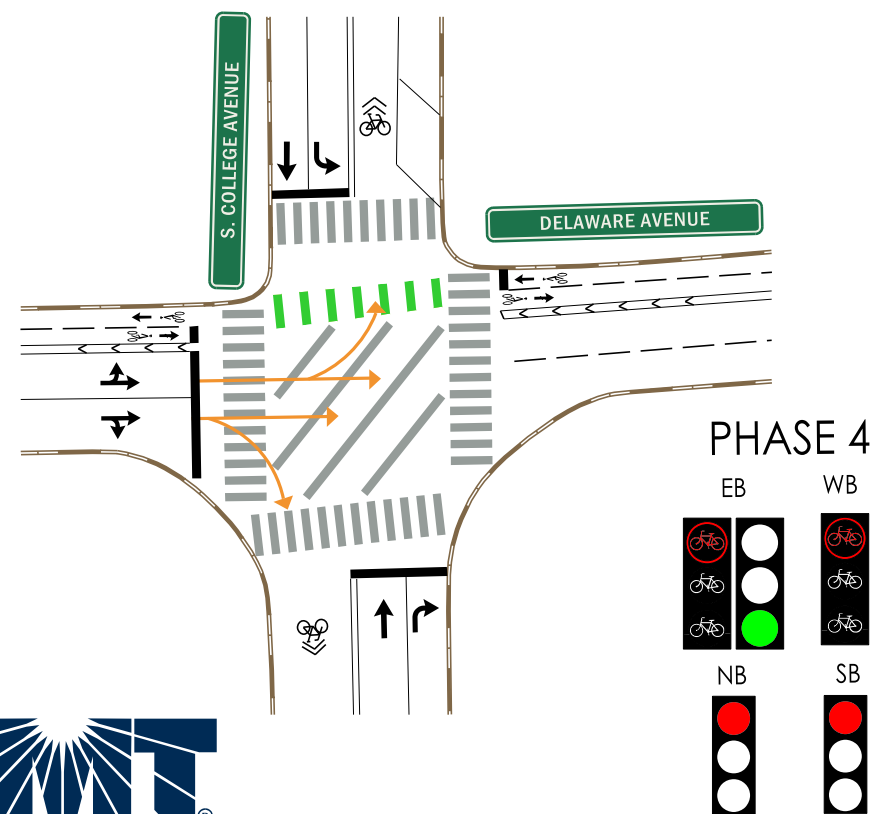
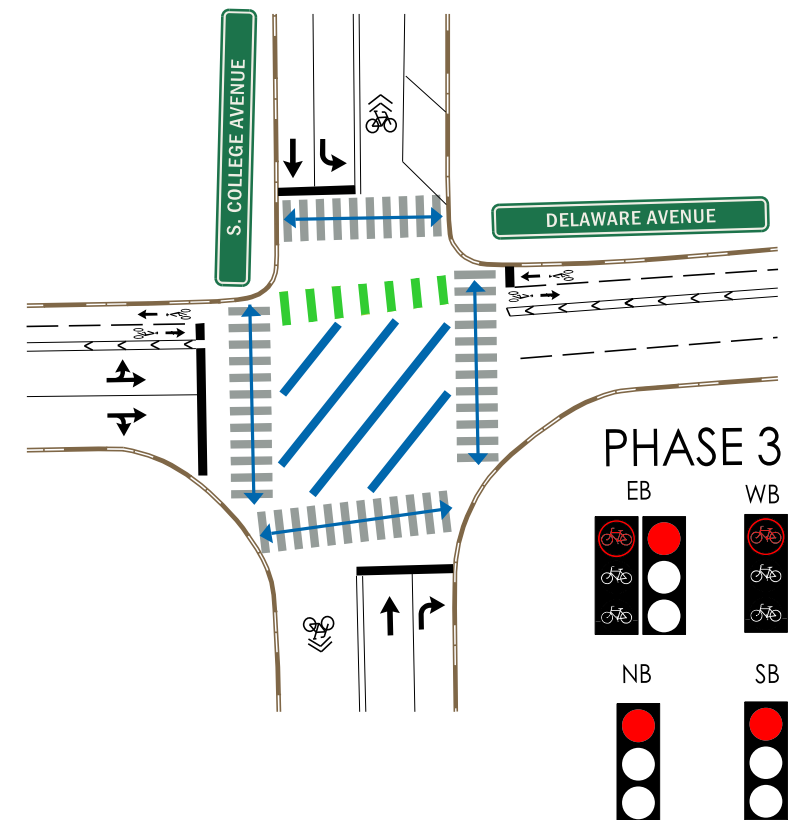
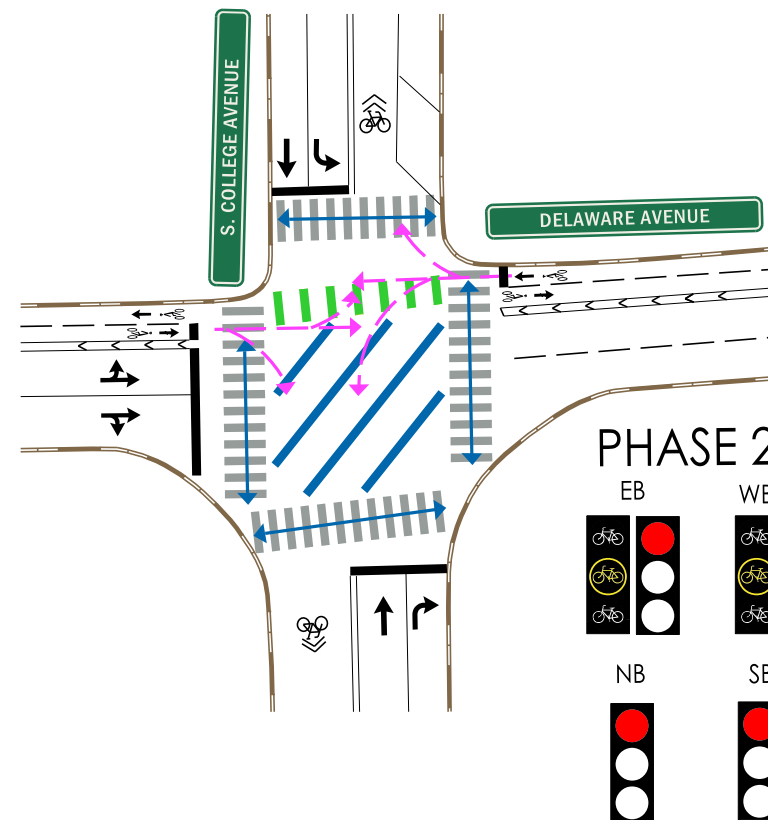
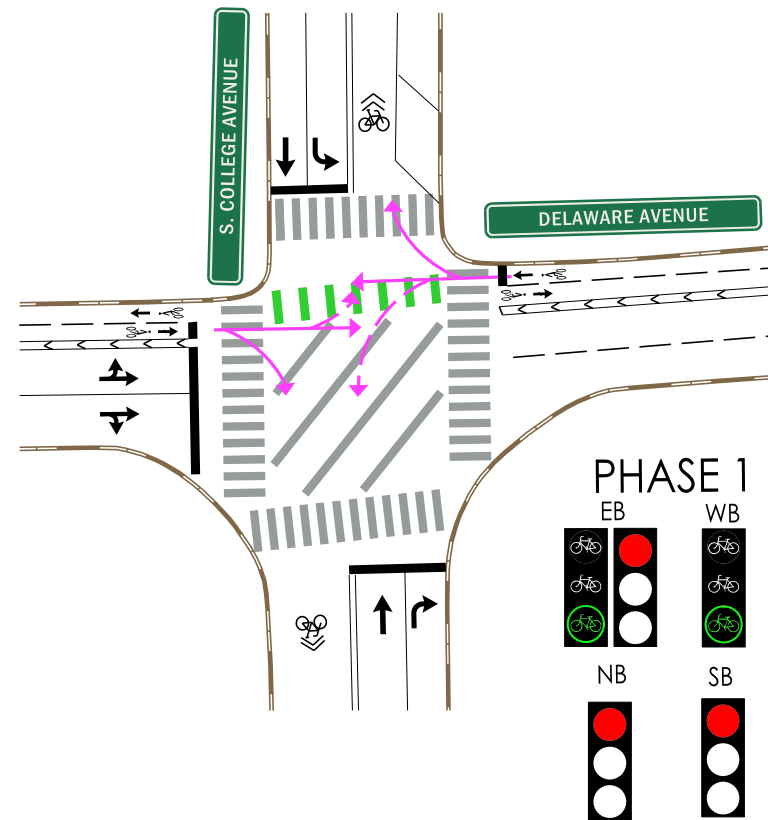


PHASE 4



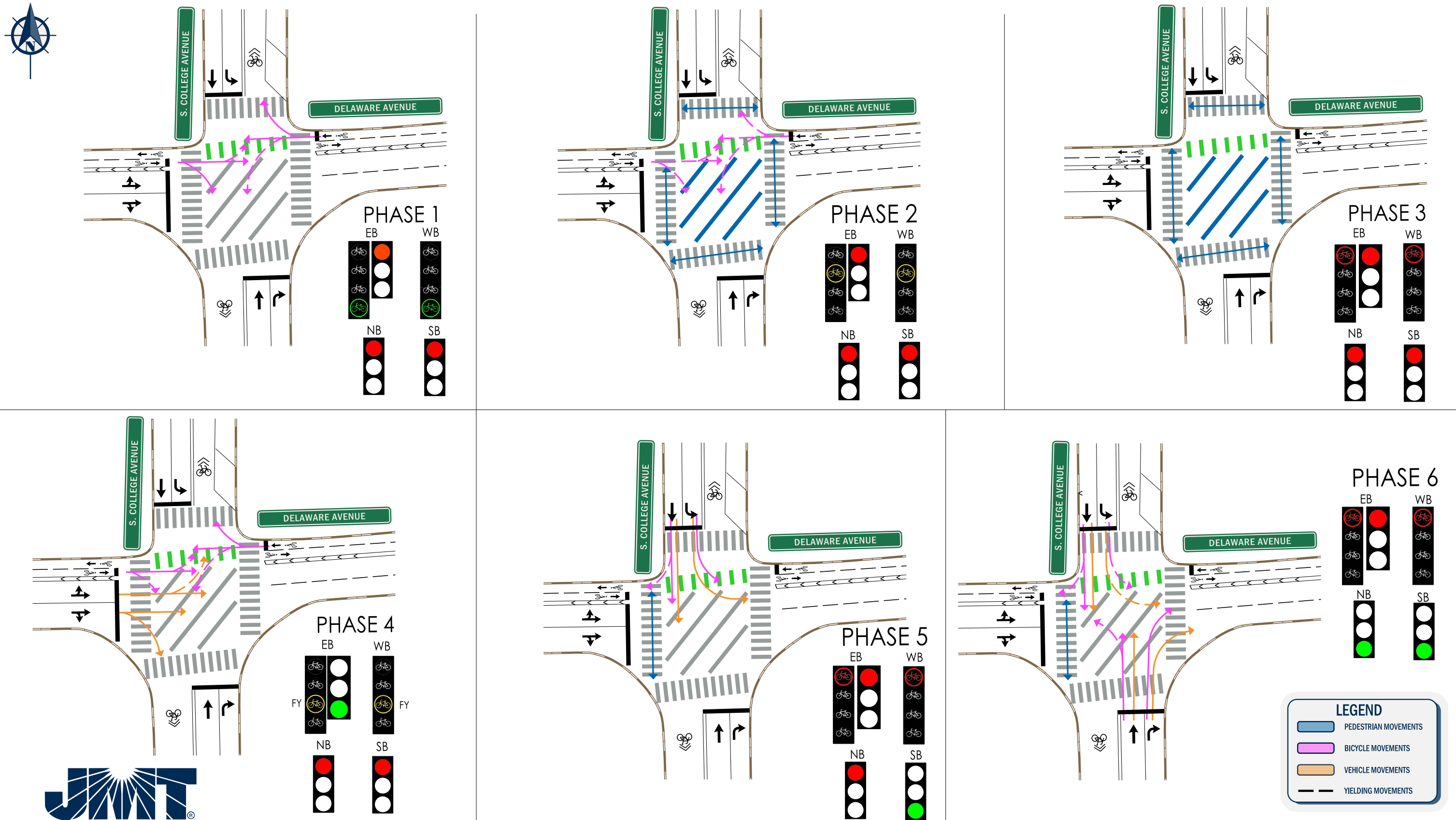
DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

ALTERNATIVE 2- PROTECTED ONLY



DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

ALTERNATIVE 3- PROTECTED AND PERMISSIVE



ALTERNATIVE 3- FEBRUARY 2018

FIGURE B-3

FEBRUARY 2018



APPENDIX C

VISSIM CALIBRATION TABLES

Existing PM Vissim Model Calibration Comparison

PM Existing Model Results

Travel Time Run Results

Start Location	End Location	Field Results (sec)	Vissim Results (sec)	Percent Change	Within 10%?
Orchard Rd	Tyre Ave	256	260	1.4%	yes

Volume Results

Intersection	Field Results (Veh)	Vissim Results (Veh)	Percent Change	Within 10%?
Orchard EBT	799	758	-5.1%	yes
S College EBT	615	600	-2.4%	yes
Academy St EBT	743	715	-3.8%	yes
Chapel St EBT	602	577	-4.2%	yes
Pomeroy EBT	736	721	-2.0%	yes
Tyre Ave EBT	753	728	-3.3%	yes
East of Tyre Ave	939	935	-0.4%	yes

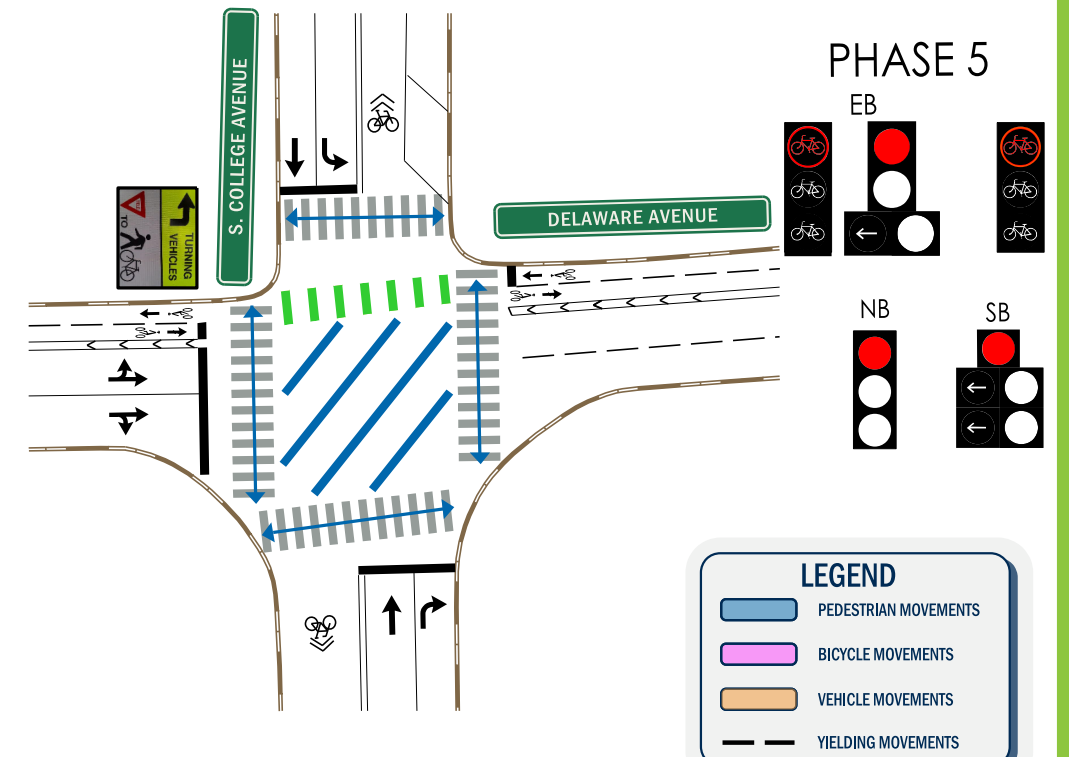
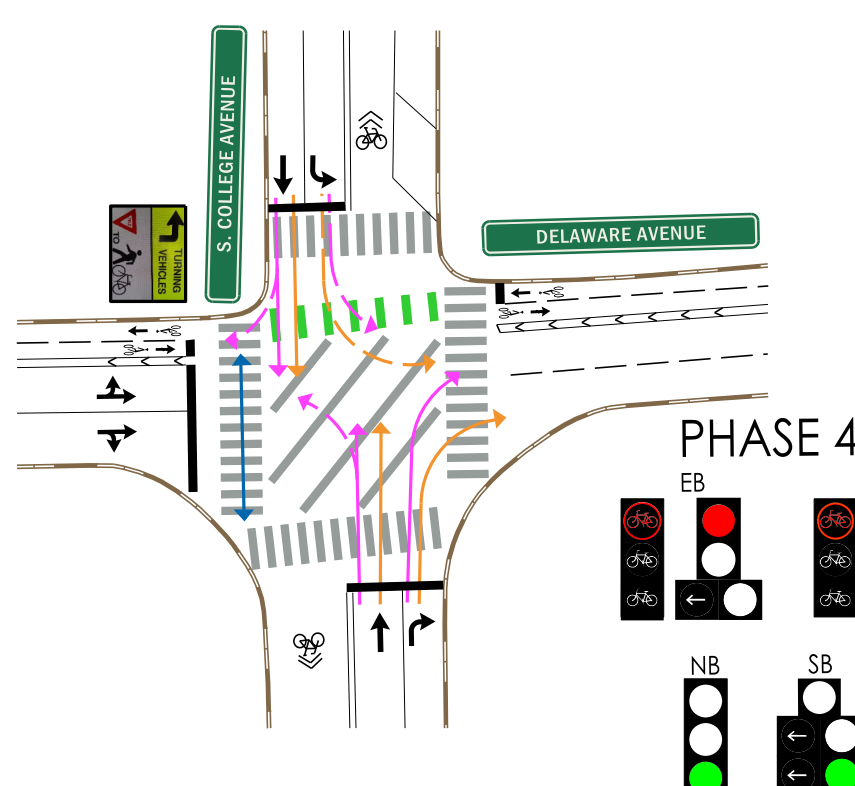
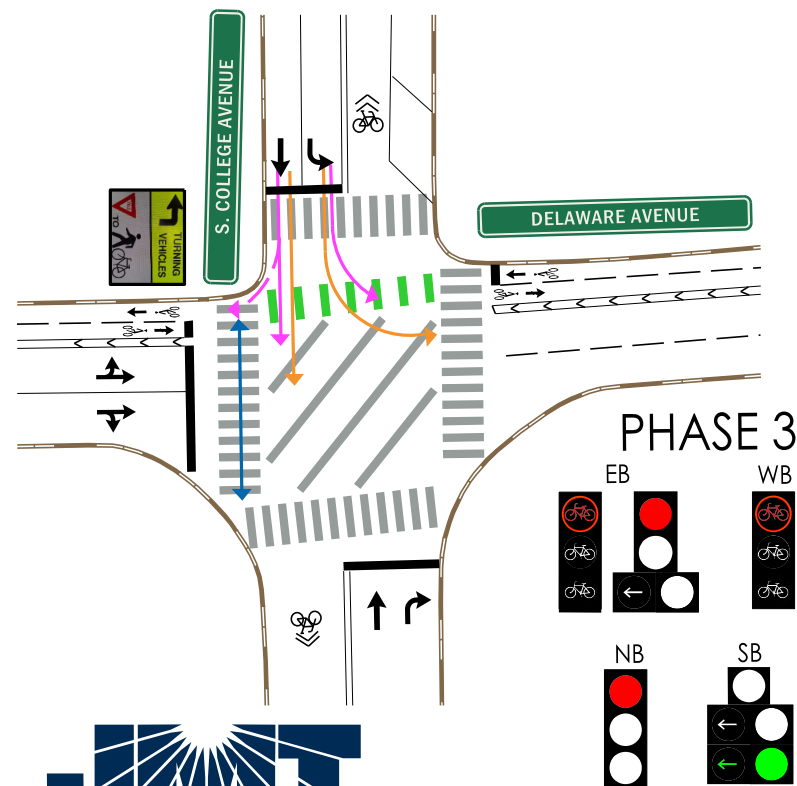
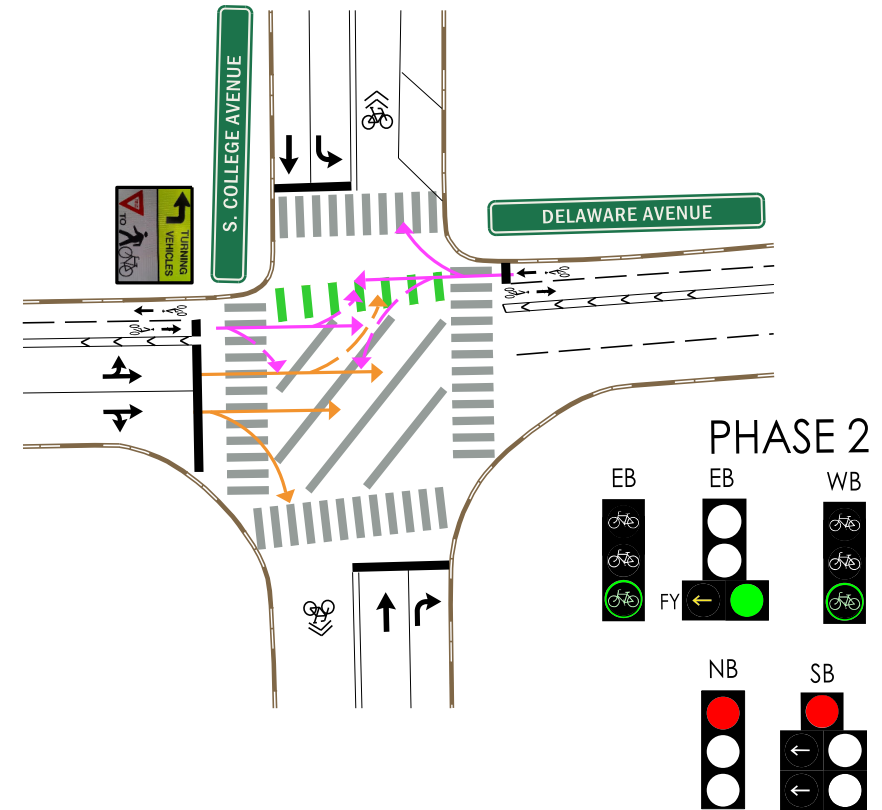
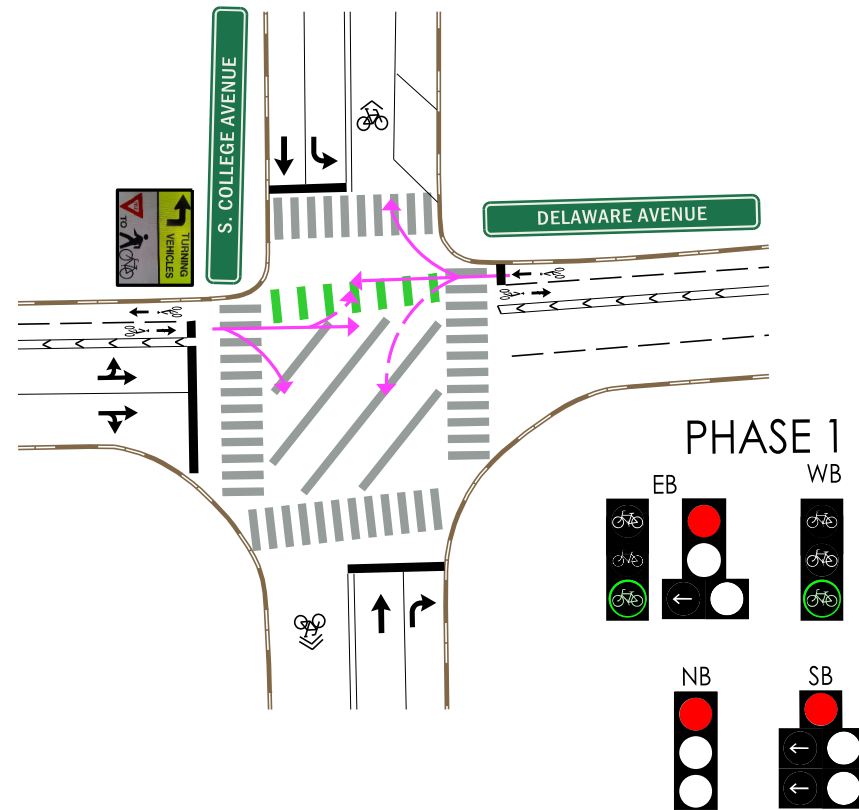


APPENDIX D

GRAPHICAL FIGURES DEPICTING OPERATION OPTIONS FOR ALTERNATIVE 3

DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

OPTION 1- PROTECTED AND PERMISSIVE



LEGEND

- PEDESTRIAN MOVEMENTS
- BIKEWAY MOVEMENTS
- VEHICLE MOVEMENTS
- YIELDING MOVEMENTS

FIGURE D-1

FEBRUARY 2018

DELAWARE AVENUE SEPARATED BIKEWAY CONCEPT

OPTION 2- PROTECTED AND PERMISSIVE

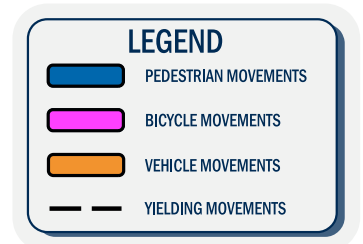
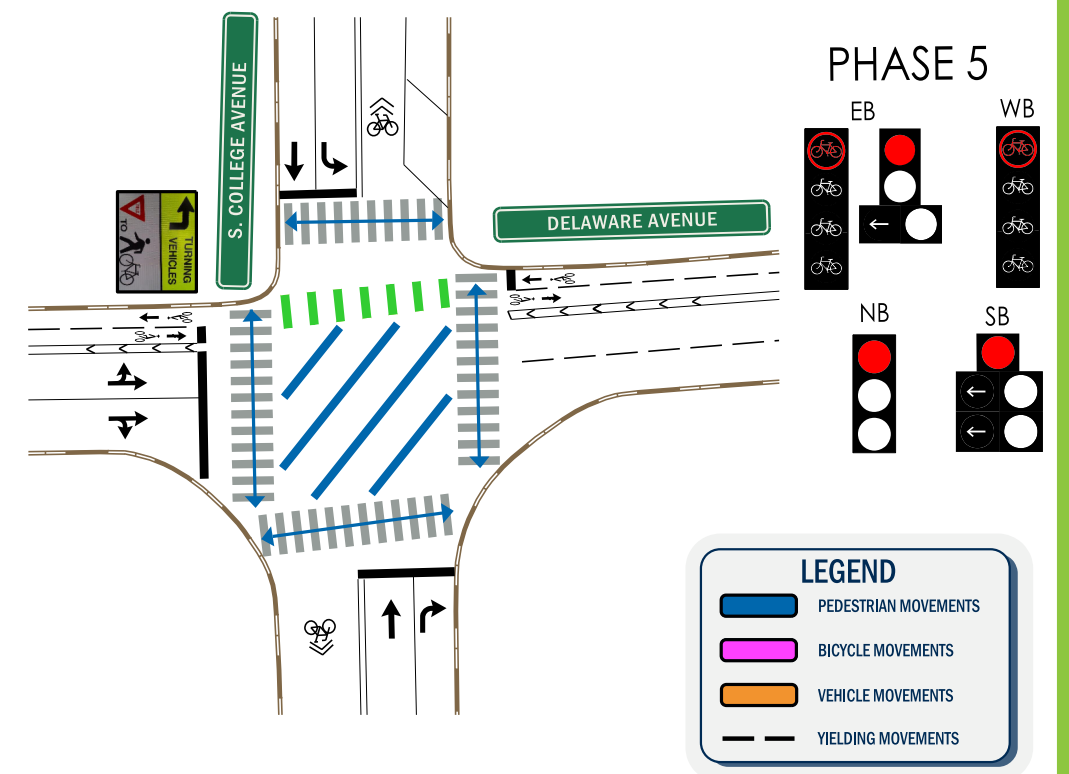
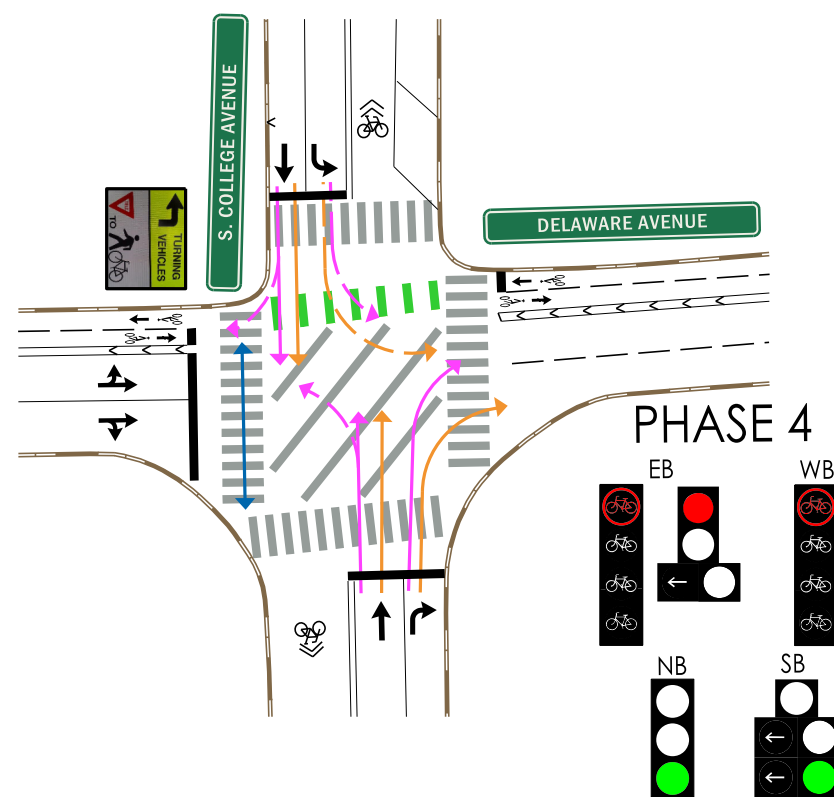
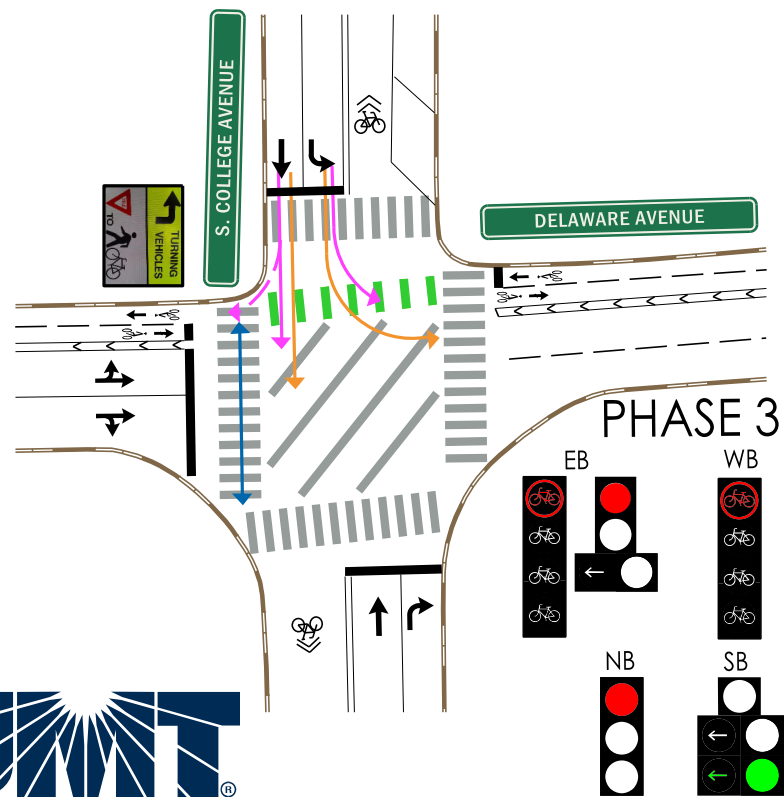
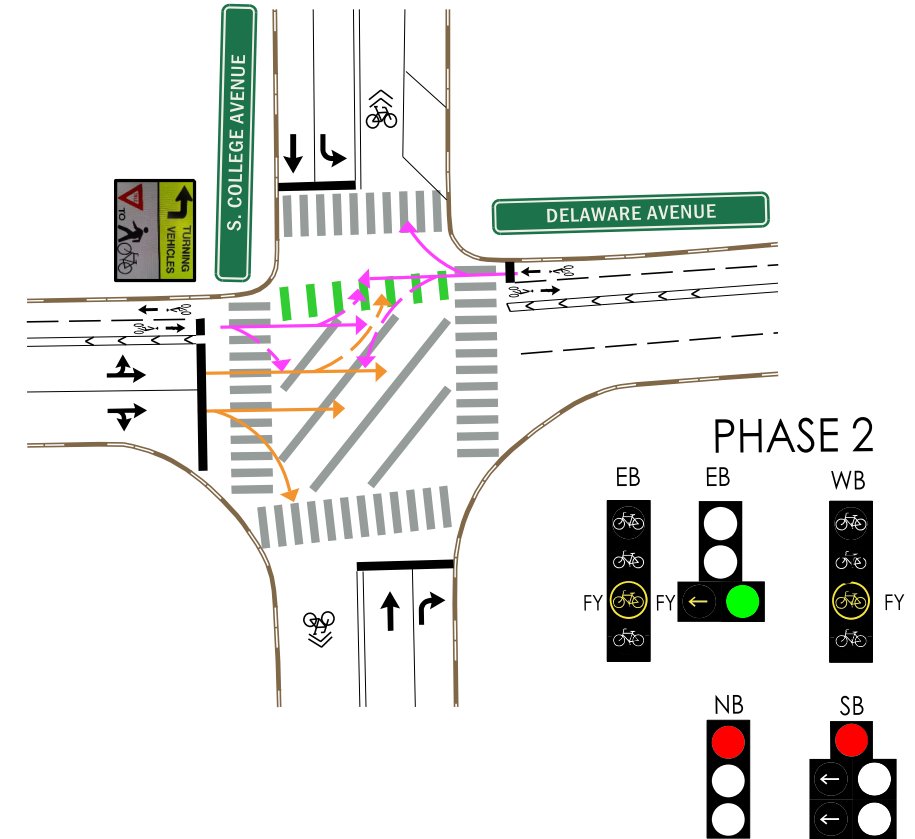
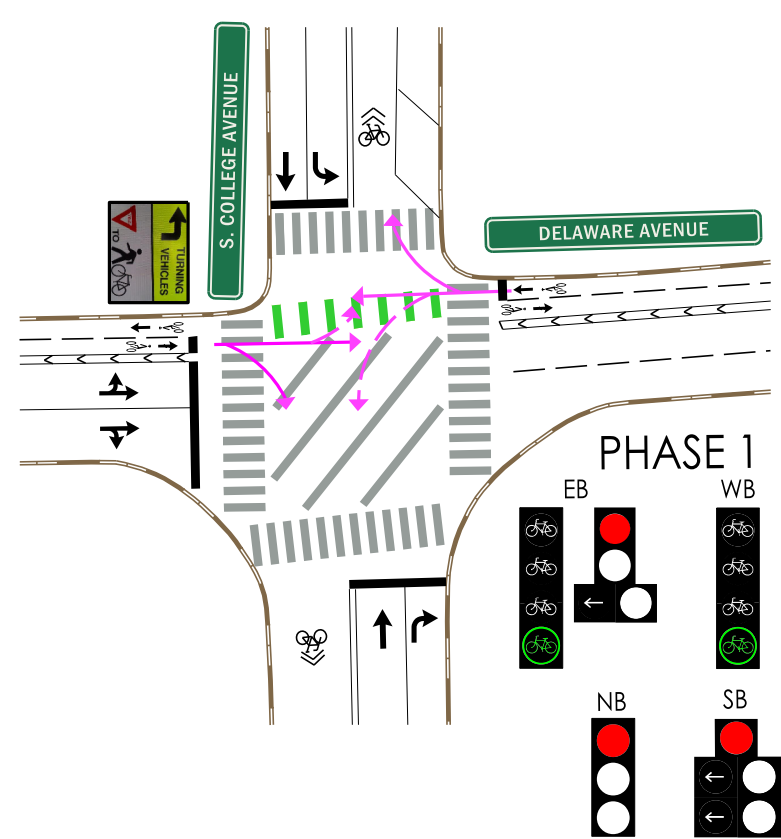


FIGURE D-2

FEBRUARY 2018



APPENDIX E

RESEARCH SUMMARY



May 2017

DELAWARE AVENUE SEPARATED BIKEWAY

Research Summary



PURPOSE

JMT conducted research on bicycle facilities and bicycle signals being used throughout the country to determine best practices for signal operation of two-way separated bike lanes on urban roadways. The purpose of this memorandum is to summarize what was learned during this research. The information found in this document is focused on common approaches, what has worked, and the challenges experienced by agencies.

CASE STUDIES EXAMINED

- Eugene, OR – City of Eugene
- Pittsburgh, PA – City of Pittsburgh
- Washington DC - DDOT
- Salt Lake, UT- Salt Lake County
- Madison, WI – City of Madison
- Denver, CO – City and County of Denver
- Minneapolis, MN – Hennepin County
- Columbus, OH – City of Columbus

EUGENE, OR

ALDER STREET

In 2011 the City of Eugene Oregon installed a two-way bike lane along Alder Street (from E Broadway to E 19th Street). The 12' wide two-way bike lane is separated by a 2' wide thermoplastic buffer containing white stripes and chevrons, with on-street parking between the bike lane and traffic lanes at limited locations.

INTERSECTION TREATMENTS

The bicycle facility includes nine intersections, three are signalized and six are stop-controlled. Each of the intersections have signage facing towards the intersecting side streets indicating that it is a one-way street with two-way bike traffic. Photo 1 shows an example of this signage.

The facility is not striped through any of the signalized intersections but it is striped through the stop controlled intersections using green spray-on thermoplastic and a dashed white bike lane line (see Photo 2).

Of the three signalized intersections, 18th Street is the only location of bi-directional bicycle signals (see Photo 3). These signals operate on a protected green bicycle phase. This allows bikes to control the intersection for any through or turning movements. The pedestrian signals are timed simultaneous with the traffic signals.



Photo 1: Alder Street and E 16th Avenue intersection Eugene, OR. (Source: Google)



Photo 2: Alder Street and E 17th Avenue intersection Eugene, OR. (Source: Google)



Photo 3: Alder Street and E 18th Avenue intersection Eugene, OR. (Source: Google)

The intersections at E 13th Avenue and E 11th Avenue provide a standard traffic signal for both contraflow and southbound cyclists on the two-way bike lane. Additionally, a green bike box is provided on Alder Street for southbound cyclists to position themselves in front of vehicles. This provides a better visual of people on bicycles as they travel through the intersection and turn onto intersecting streets (see Photo 4). Two-way two-stage turn queue boxes were not provided for northbound (contra-flow) cyclists to make left turns off the facility onto the intersecting streets, therefore cyclists use the pedestrian signals or turn during gaps in traffic.

TRANSITIONS

Alder Street transitions at E 19th Avenue from a one-way southbound street to a one-way northbound street for motor vehicles, while remaining a two-way facility for bicycles. “DO NOT ENTER EXCEPT BICYCLES” signs are placed at the transition (see Photo 5).

The opposite end of the facility transitions across Broadway onto a wide sidewalk that runs adjacent to Broadway. The bicycle crossing on Broadway is located adjacent to a pedestrian crosswalk with a bicycle signal and pedestrian signal located on the far end of the intersection (see Photo 6). The bicycle and pedestrian signals are phased together regardless of whether a pedestrian activates the signal using the pushbutton or if the inductive bicycle loop detectors trigger the signal actuation. Bicyclists transitioning from the sidewalk on Broadway to the two-way bike lane must press a pushbutton located on the bicycle signal pole to activate the traffic signal to cross Broadway.



Photo 4: Alder Street and E 13th Avenue intersection. (Source: Google)

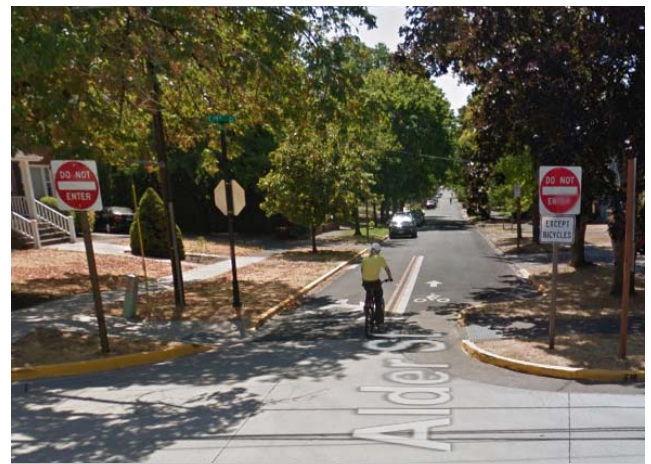


Photo 5: Alder Street and E 19th Avenue transition. (Source: Google)



Photo 6: Alder Street and Broadway transition. (Source: Google)

COST

This project was funded as part of a larger \$2.3 million street rehabilitation project for Alder and 13th Avenue. The two-way bike lanes were primarily paid for by an Oregon Department of Transportation (ODOT) grant for \$706,998, the City of Eugene provided \$78,554 in local match from the City's System Development Charge funds. These are often referred to as development impact fees in other states. These are one-time charges collected from developers for financing new infrastructure construction and operations.

LESSONS LEARNED

The City of Eugene has seen little compliance with the bicycle signals located along the facility. The inductive loop detectors require a cyclist to position themselves directly on the loop detector symbol painted on the roadway to trigger a green light for bikes, there are few cyclists that do so. Many cyclists seem to lack the understanding of how the intersection operates and will instead use the pedestrian signals to cross at intersections. The City has looked at ways to address the lack of compliance and increase bicycle users understanding of how the facility operates through PSA videos, partnering with the University to provide information to students, and even "handlebar messaging", which includes tying incentive coupons paired with educational messaging to bicycle handlebars.

The current bicycle signals are located at the same height as vehicle signals. This places the bicycle signals above the typical visual range of a person on a bicycle. The City is now investigating opportunities to increase the visibility of the bicycle signals by angling the signals downward towards bicycle traffic, or installing new bicycle signals that are mounted lower than the existing signals. Mounting the

bicycle signals at no less than 8 Ft. above the ground with brighter yellow paint will hopefully heighten visibility and awareness of the bicycle signals and lead to better compliance from cyclists. Duplicate near sided bicycle signals are also being considered to increase visibility of bicycle signals at intersections.

Blue light indicators and yellow countdown bars, such as the signal indication devices being used in Canada, are also being considered by the City of Eugene. The purpose of such devices being that cyclist will receive an indication that they have been detected by the signal and will be more likely to wait for the signal.

The intersection at 18th Avenue, in particular, is in need of a signal reconfiguration. The issue being that when initially installed the signal timing did not fully account for late arriving cyclists. The advanced loop detectors were used so that the late arriving cyclists would be accounted for during the signal phase but not allow enough time for the cyclist to travel from the advanced detector through the intersection. To accommodate the late arriving cyclists a slower speed should be accounted for in the signal re-timing.

Limited post project evaluation was required on this project. The evaluation that was done found that initially crashes along Alder Street increased although bicycle counts remained relatively unchanged. Over time, however, the crashes did decrease. This trend was attributed to the fact that this was the first facility of this type in Eugene. Drivers and cyclists alike initially lacked familiarity with the facility, which could explain the initial spike in crashes. Over time all users became more accustomed to the facility and began to navigate it more safely. To date, there have been no fatal or serious injury bicycle crashes reported along the facility.



This facility received negative feedback from the public initially, but once construction was completed the public perception regarding the facility changed. Alder Street bike-way has become a model facility to the citizens in Eugene. The City regularly receives requests from the public to replicate the facility throughout the streets of Eugene.

To date, there have not been any maintenance activities performed on the facility. The thermoplastic striping and green spray on thermoplastic has held up well for the first 5 years, but will require maintenance in the next few years. The city is considering installing flexible delineators to better delineate the bike lanes, but since cars have generally respected the facility, delineators may not be used in order to avoid higher maintenance costs.

Since the facility is located adjacent to a college campus, there is an increased volume of skateboarders and long boarders that use the two-way bike lanes. Currently, this is against city code which categorizes skateboarders and long boarders as pedestrians, thus restricting them from using the facility. This code is not enforced by the City, and the City is working to change the code to give people on skateboards and longboards bicycle status allowing them to legally use the two-way bike lane.

PITTSBURGH, PA

PENN AVENUE



Photo 7: Cyclist crossing signalized intersection along Penn Avenue Cycle Track

A two-way separated bike lane was installed along Penn Avenue (from Stanwix Street to 16th Street). Penn Avenue is a city street that was once a two-way multilane street with no dedicated bicycle facility. The street has since been converted to a one-way street with on-street parking along the right side of the road, varying one-way southbound vehicle lanes throughout the corridor (1-2 lanes), and a two-way separated bike lane.

The 12' wide and 0.5 mile long protected two-way bike lane installed along this corridor is separated from vehicular traffic by a 2 foot buffer containing two striped white lines and flexible delineators.

This section of Penn Avenue was redesigned with the intention to be more attractive and

conducive to bicycling. Multiple decorative bike racks have been provided throughout the corridor, as well as a bike repair station and bike share rental stations.



Photo 8: Green pavement pattern used at minor intersection along Penn Avenue.

Green colored pavement (as seen in Photo 8) was used along the facility at conflict points such as; commercial driveways, parking garage entrances, minor intersections, and transition areas.

INTERSECTION TREATMENTS

The two-way separated bike lane crosses several intersections along Penn Avenue, seven of which are signalized intersections. The facility is striped through both signalized and non-signalized intersections using a yellow dashed centerline, white double chevrons indicating bike lane direction, and dashed white lines outlining the width of the facility (See Photo 7).

When Penn Avenue was converted from a vehicular two-way street to a vehicular one-

way southbound street, the traffic signals were left in operation. The northbound vehicle traffic signals now serves as the signals for contraflow bicycle traffic. These signals are timed simultaneously with the one-way southbound vehicular signals. “BIKES USE VEHICLE SIGNAL” signing was installed at the intersections along Penn Avenue to encourage people on bikes to use the repurposed vehicle signals.



Photo 9: “TURNING VEHICLES YIELD TO BIKES” intersection signage.

There is also “TURNING VEHICLES YIELD TO BIKES” signage installed at the intersections instructing vehicles to yield the right-of-way to bicycles (See Photo 9). These signs are installed along the left side of the roadway at the far side of the intersections approximately 7’ off the ground. This puts the signage at the bicycle/pedestrian height where motorists are likely to be looking when turning off Penn Avenue. The signage is not installed consecutively at every intersection. Size of the signs also varies throughout the corridor.

Bicycle turning movements from Penn Avenue onto side streets were not accounted for in the design of this facility. A few of the intersections have “BIKES USE PED SIGNAL” signage, located below the pedestrian signal facing both directions of bicycle traffic on Penn Avenue. The purpose of this signage is to instruct cyclists turning across Penn Avenue to dismount their bikes and cross at the crosswalks using the pedestrian signals.

Directing turning cyclists to use the pedestrian signal and crosswalks becomes even more problematic as there is no waiting area for bikes at the intersections thus bicyclists remain in the cycle track blocking all other through bicycle movements. As a result many cyclists make turns by either merging into the vehicle lanes before the intersection to turn along with vehicles or by turning across Penn Avenue during gaps in traffic.

TRANSITIONS

Penn Avenue transitions at 16th Street for northbound cyclists to a “DO NOT ENTER” one-way street. There is “BIKE LANE ENDS” signage installed prior to the intersection along with bicycle route wayfinding indicating to turn right to the Smallman Street Bicycle Route. The transition area at 16th Street for southbound cyclists contains sharrows approaching the intersection and a short center green bike lane just before the intersection (See Photo 10). The bike lane is placed in the middle of the roadway to better position bikes to transition into the left side two-way separated bike facility on Penn Avenue.

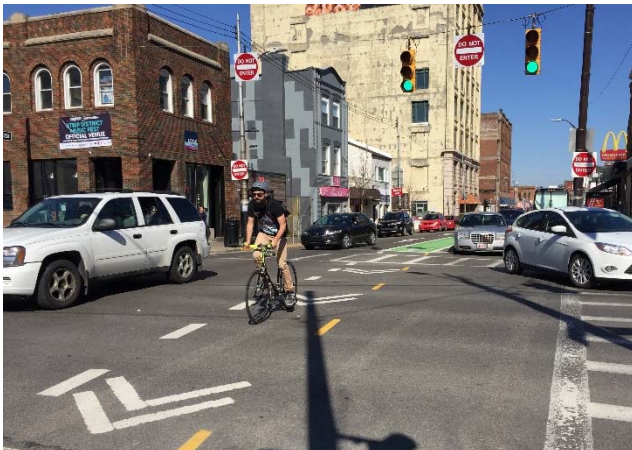


Photo 10: Southbound transition onto two-way separated facility at 16th Street.

The transition at Stanwix Street for southbound cyclists is designed to direct bicycle traffic to exit the two-way separated facility and merge into vehicular traffic on Penn Avenue. There are sharrows on the far right lane indicating that cyclists should position themselves in the far right lane for crossing Stanwix Street where Penn Avenue becomes a 3 lane one-way road. There is not a transition for northbound cyclists at Stanwix Street as there are no dedicated bicycle facilities on Stanwix Street in which cyclists would be transitioning onto the two-way separated bike lane.



Photo 11: Southbound transition into shared lanes at Stanwix Street intersection.

LESSONS LEARNED

The simultaneously timed signals used for the separated bike lane movements and one-way vehicular movements along Penn Avenue facilitates through moving bicycle traffic but does not provide protection for turning cyclists. Likewise, having cyclists use the pedestrian signals to cross Penn Avenue is not ideal in functionality and does not yield high compliance. Providing two-way two-stage turn queue boxes at the signalized intersections to assist in turning movements for cyclists could be beneficial to this facility and address the turning movements of cyclists. Additionally, an advanced green signal phase for bicyclists could assist in turning movements by holding all conflicting vehicle turning movements. Installation of some form of advanced bicycle detection could also address the late arriving cyclists by holding the light long enough for them to maneuver through the intersection.

The striped buffer and flexible delineators used for this facility discourage vehicles from encroaching into the facility but does not completely prevent them from entering. The areas in which the flexible delineators are not present (commercial driveways, parking garage entrances, minor intersections, and transition areas) present an opportunity for vehicles to enter the facility. In Photo 12 a moving van is parked in the two-way separated bike lane completely blocking the bicycle facility, forcing people on bicycles into motor vehicle traffic.

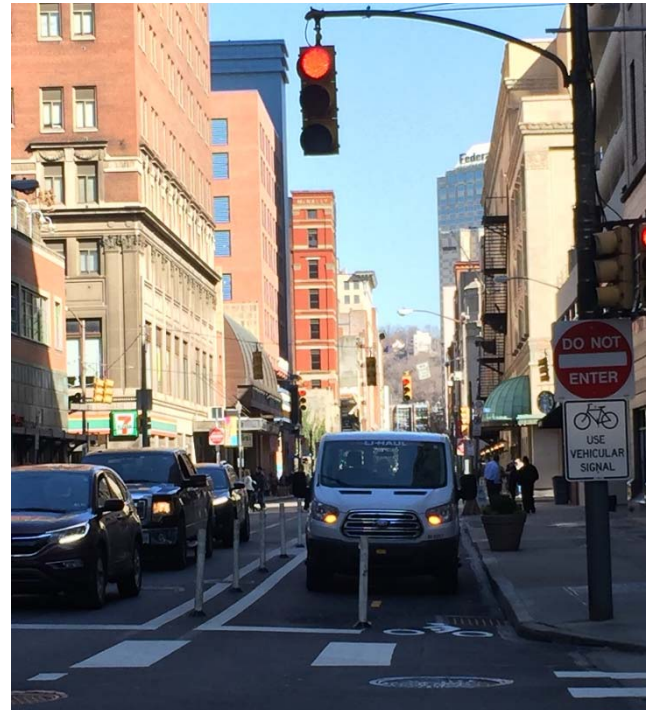


Photo 12: Moving van parked in two-way separated bike lane blocking bicycle traffic.

WASHINGTON, DC

FIRST STREET NE



Photo 13: Intersection of First St. NE and L Street
(Source: Google)

First Street NE in Washington, DC is a two-way street containing an 8' wide two-way separated bike lane (installed from M Street to G Street). The facility spans a 0.5 mile section and intersects two signalized intersections. This facility is buffered from traffic by a 2' wide beveled concrete curb from M Street to midblock of L and K Street where it transitions to painted white lines with no vertical buffer. At K street the buffer switches to a single striped white line with flexible delineators until the terminus point at G Street. The entire facility (except at intersections) has been painted solid green with a yellow dashed center line and bike symbols indicating direction of bicycle traffic.

INTERSECTION TREATMENTS

The facility is striped through the intersections using white double chevrons indicating bike

lane direction and dashed white lines outlining the width of the facility

The signalized intersections along the facility do not use any bicycle signals or two-stage turn queue boxes to direct bicycle turning movements. Instead "TURNING BIKES USE CROSSWALKS" signage is provided to instruct cyclists to dismount their bicycle and cross at the crosswalks using the pedestrian signals. Bicyclists traveling in either of the through directions follow the traffic signals with the vehicular traffic. The District has observed conflicts with this signal set up. There are future plans to evaluate the intersections along this facility and potentially ban right turns for vehicles to decrease the possibility of the right hook.

OTHER BICYCLE SIGNAL LOCATIONS

DDOT has installed bicycle signals at a variety of other intersections and bicycle facilities throughout the district. The following lists a few of the intersections that have had bicycle signals recently installed:

- 15th Street Northwest and Pennsylvania Avenue – Uses a protected bicycle phase simultaneous with pedestrian phase.
- First Street Northeast by Union Station – First Street was converted from two-way to one-way street so the bicycle signals are repurposed traffic signals that were left in place. The signal uses a protected bicycle phase to allow bicycles control of the intersection while vehicles are held on red.
- New Hampshire Ave NW, 16th Street, and U Street NW – one-way bike lanes have bike boxes and dedicated bicycle signals for cyclists to navigate intersections. The intersection is timed to give a 12 second

protected bicycle phase for bicycles to control the intersection while all vehicles are held on a red signal. "DON'T BLOCK THE BOX" signage is installed on the traffic mast arms indicating vehicles are not to block the bike boxes.

LESSONS LEARNED

DDOT performed a before and after study in 2012 to evaluate the bicycle facilities and facility features within the district. They found that using signs to direct bicyclists to use pedestrian signals is not an effective intersection solution. Using bicycle signals allows for more flexibility in signal timing for vehicles and bicycles which can lead to less delay. On Pennsylvania Avenue, for example, the same signal indications control both through vehicle and through bicyclist movements. As a result, the through vehicles receive a red indication during protected left-turns to prevent conflicts between left-turning vehicles and through-cyclists (even when there are no conflicting movements). Bicycle signals would allow through vehicles to advance through the intersection with left-turning vehicles while through bicycle movements remained stopped. The use of bicycle signals would allow for more efficient signal operations and reduce delay for vehicles. It is also preferential to include cyclist's progression speed as a factor in signal timing.

DC has used green colored pavement for various purposes but has decided that it should only be used for conflict areas, to more effectively caution both cyclists and motorists. This also reduces the cost and maintenance requirements of facilities. Likewise, the use of flexible delineators has been found a nuisance in terms of maintenance because it is difficult to proactively replace damaged delineators. The replacement process has become a complaint driven process in DC and is not effective.

When determining facility treatments it is essential to provide consistent traffic controls along a particular facility. In addition, instituting an education and enforcement campaign to encourage compliance with traffic signals is vital to the successful utilization of bicycle infrastructure.

SALT LAKE COUNTY, UT

3900 SOUTH

In 2016, Salt Lake County installed 8 bicycle signals along a one-way bike lane on county road 3900 South. The bicycle signals were retrofitted onto existing signal poles by running conduit from the traffic signals along the pole to the bicycle signals, which are vertically mounted on the side of the traffic signal pole.

The signals use actuated radar technology to detect cyclists as they approach the intersections and activate the bicycle signal. The first radar eye detects cyclists 100 feet before the intersection and activates the detection cycle. The technology uses zone detection with five separate radar detection eyes to track cyclists as they approach the intersection.

The Wavetronix radar technology used for each of the signals additionally counts bicycles, notes their speed, direction of travel, and turning movements. The drawback of the radar technology being used is that the signal compliance cannot be assessed through this technology.

LESSONS LEARNED

When setting up the radar detection eyes, the County found that putting the detection at a side angle of the intersection was problematic when vehicles were present. The radar would detect the vehicle and would not be able to detect the bicycle, thus not activating the bicycle signal. They found that a straight on alignment eliminated this issue, 10 degree angles being the threshold. They also have found that the technology will only count one

bicycle if two bicycles are riding close together. The radar technology has the advantage of easy manipulation of zones and configuration, unlike inductive loop technology.

Radar technology was used in Salt Lake County instead of video technology, because at the time the equipment was purchased, the video technology was not able to detect bicycles in snowstorms or fog. Since then, the detection capabilities of the video detection technology have been increased with a higher resolution lens. The County plans to experiment with the video technology to assess compliance in addition to counts. Additionally, there are plans to monitor the bicycle counts and compliance at the signals to evaluate if blue light indicators could be justified for installation.

MADISON, WI

UNIVERSITY AVENUE



Photo 14: Intersection of North Mills Street and University Avenue in Madison, WI. (Source: Google)

University Avenue stretches through the University of Wisconsin-Madison campus as a one-way roadway for vehicles and two-way roadway for bicycles. It was designed (originally in 1984) to allow three lanes of westbound vehicular movement as well as a (right side) dedicated westbound bicycle lane and outside transit lane. The roadway also contains a separated eastbound bicycle lane located on the left side of the roadway to allow contraflow bicycle movement.

The contraflow bike lane is buffered by a raised curb and signed as a bike lane. “BIKE WRONG WAY” signs are placed at every intersection to discourage westbound bicyclists from traveling in the wrong direction in the eastbound separated bike lane. The contraflow bike lane is striped through intersections using a dashed yellow line outlining the edge of the bike lane adjacent to the vehicle lanes.

INTERSECTION TREATMENTS

The intersections were designed and signalized similar to a typical two-way street. The signals used for the contraflow bicycle traffic are the same design that is used for the motor vehicle traffic signals (see Photo 14). The contraflow bicycle lane signal is timed simultaneous with the through movement of the westbound vehicular and bicycle movements. The exception being a few locations that have heavy vehicle turning movements onto the side streets. These intersections give a longer turning vehicle movement phase while the contraflow bike lane movements are restricted using a red signal.

The locations where side streets approach University Avenue are all signed with one-way street signs and “NO RIGHT TURNS EXCEPT BIKES” signage (see Photo 15).



Photo 15: “Intersection of North Park Street and University Avenue. (Source: Google)



If a person on a bicycle wishes to turn off University Avenue onto one of the side streets they may do so by turning just as any other vehicle and yield the right-of-way to pedestrians and the through moving vehicles. Or they can use the pedestrian signals and cross using the crosswalks.

LESSONS LEARNED

This facility was designed before the use of bicycle signals was established in the US. Regardless, the City does not plan to switch these signals into the MUTCD experimental bicycle signals. This is due to the current FHWA requirement that bicycle signals cannot run concurrent with any vehicle turning movements. This would significantly increase the signal phasing required at each intersection, which could lead to traffic congestion. Currently, signage at the intersections indicate that turning vehicles are to yield the right of way to through moving bicycles, for both eastbound and westbound bicycles. This however does not eliminate the possibility of a vehicle turning into a bicycle.

DENVER, CO

LAWRENCE STREET AND 15TH STREET



Photo 16: Lawrence and 15th Street intersection
Denver, CO. (Source: Google)

15th Street in Denver, Colorado is a multi-lane one-way street with a one-way protected bike lane along the left-side of the street. The intersection at Lawrence Street is comprised of three northbound vehicle lanes and a one-way northbound cycle track buffered by a single yellow line and flexible delineators.

Green colored pavement (as seen in Photo 16) was used at commercial driveways and intersections along the facility.

INTERSECTION TRANSITION TREATMENTS

There are two bicycle signals provided at the intersection with Lawrence Street to transition cyclists continuing along 15th Street northbound (See Photo 17). The signals use an 8" lens with the bicycle symbol.

The left signal, mounted horizontally on the signal mast arm, is for cyclists crossing Lawrence and turning at the following intersection (Larimer Street) where the bike lane ends. The bicycle signal, mounted vertically on the signal pole, located on the right side of the street is to encourage cyclists traveling further than Larimer Street to merge across traffic to the right most travel lane. This merge is important as 15th Street becomes a triple Left-Turn-Only further down the corridor. Which would force cyclists into making a left turn and prevent them from continuing along the right side of 15th Street.

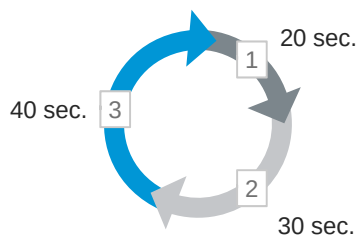


Photo 17: Lawrence and 15th Street intersection bicycle signals. (Source: Google)

“BIKE SIGNAL” signage and directional arrows are used at the intersection to distinguish the signals and bicycle movements associated with each signal.

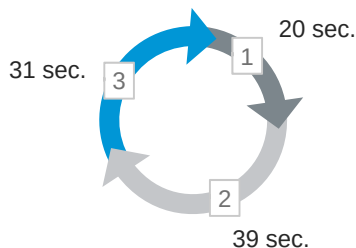
The timed signal phasing for the Lawrence Street intersection is (these timings do not include the yellow or red phases or any pedestrian phases):

AM Signal Timing



- 1- 20 seconds for both bike movements down 15th St (all vehicle movements are held)
- 2- 30 seconds for vehicle/bike movement down 15th St
- 3- 40 seconds for vehicle/bike movements down Lawrence St

PM Signal Timing



- 1- 20 seconds for both bike movements down 15th St (all vehicle movements are held)
- 2- 39 seconds for vehicle/bike movement down 15th St
- 3- 31 seconds for vehicle/bike movements down Lawrence St

A two-stage turn queue box is marked in front of the crosswalk on Lawrence Street for cyclists turning right into the bike lane along Lawrence (see Photo 18).



Photo 18: Two-stage turn queue box located at Lawrence and 15th Street intersection Denver, CO.
(Source: Google)

LESSONS LEARNED

The City of Denver has installed multiple bike signals throughout the City. The preferential design is to use an 8" lens with a bicycle signal face as well as mounting the signal at cyclist's eye level. The horizontal alignment is used if the signal has to be mounted at the same level as the traffic signals. This helps to distinguish that the signal is for bicycle use only.

The intersection of 15th Street and Lawrence Street is a problematic intersection for the City of Denver. The two-way two-stage turn queue box is not often utilized by cyclists despite public outreach and education that has been performed. Most cyclists will cut across traffic or use the pedestrian phasing to cross 15th Street. Additional permanent signage instructing right-turning cyclists to use the two-stage turn queue boxes could yield a higher compliance. Furthermore, the city is considering ways to simplify the transition at Lawrence Street or relocate the bike lane to the right side of the street to eliminate the need for the existing complex bicycle transition.

HENNEPIN COUNTY, MN

WASHINGTON AVENUE SOUTH (COUNTY ROAD 152)

Hennepin County, in coordination with the City of Minneapolis, is reconstructing Washington Avenue South (County Road 152) between Hennepin Avenue and Fifth Avenue South. Construction began in August 2016. With completed project construction anticipated for 2017.

Once construction is completed, Washington Avenue will be a two-way street with two eastbound and two westbound motor vehicle lanes as well as turning lanes at intersections. There will be two one-way separated bicycle lanes installed on either side of the roadway. The bike lane on the northern side of the roadway will be for westbound bicycle traffic and the lane installed on the southern side of Washington Avenue will be for eastbound bicycle traffic. Each of the separated bike lanes will be installed at the curb/sidewalk elevation. The bicycle lanes will drop down to the roadway elevation at the intersections. The bicycle signals will be activated using actuated signal detection technology. The separated bike lanes are buffered from the sidewalks using urban street-scaping and street furnishings.

INTERSECTION TREATMENTS

The intersections will each be striped with green dashed thermoplastic milled into the pavement indicating the location of the bike lanes through the intersections.

The county worked with FHWA to experiment using a four section bicycle signal head mounted above the separated bike lanes. This four section bicycle signal head will include a red bicycle symbol, yellow bicycle symbol, flashing yellow bicycle symbol, and green bicycle symbol indications with yellow backplates. The use of a flashing yellow bicycle signal phase is not listed as an allowable bicycle indication operation of FHWA Interim Approval. Likewise, the use of a yellow signal head back plate is not listed as an allowable device in the MUTCD. Consequently, Hennepin County requested approval of a FHWA Request to Experiment with a four-section bicycle signal head with flashing yellow bicycle operation and a yellow back plate

This proposed signal configuration will give people on bicycles a protected crossing phase entering the intersection before interacting with motor vehicle traffic during a permissive signal phase. It is proposed to use a four second protected green bicycle phase before the signal switches to a yellow bicycle signal for three seconds. The signal would then turn to a flashing yellow bicycle signal simultaneous with a green motor vehicle signal. The purpose of this shared signal phase would be to allow bicycles and motor vehicles to move simultaneously through the intersection and indicate that bicycles are not operating in a protected phase. The signal will then convert to a solid yellow signal followed by a solid red signal for both bicycles and motor vehicles traveling along Washington Avenue.

This signal will only include the bicycle phasing when bicycles are detected using the inductive loop detection technology at the

intersections. The county would prefer to use video technology but since the City will be operating the signals they requested that this project follow the City standards, which require inductive pavement loops.

As part of FHWA's acceptance of this signal configuration they have requested the County to install a variable messaging sign that will activate a NO RIGHT TURN message for motor vehicles while bicycles are operating on the protected green bicycle signal phase. FHWA would also like this sign to show a TURNING VEHICLES YIELD TO BIKES message when the protected bicycle phase is not activated. However, the County and City are trying to limit the amount of signing installed at each of the intersections and are working with FHWA to resolve the intersection signing. In an effort to reduce the signage, the county opted out of using the traditional black and white "BIKE SIGNAL" sign. Instead they are proposing to use a yellow back plate on the bicycle signal head with "BIKE SIGNAL" in black letters printed on the back plate above the signal head.

The initial intersection design included two-stage turn queue boxes for bicycle traffic to turn across Washington Avenue. FHWA would only approve the use of the two-stage turn queue boxes if additional signage would be installed to compliment them. In the effort to reduce signage at the intersections, the County decided to remove the two-stage turn queue boxes from the design. The City also follows a standard rule that two-stage turn queue boxes will only be used on facilities that are directing turning bicycles onto a receiving roadway that contains some type of bicycle facility. For these reasons the two-stage turn

queue boxes were not included in the design. This means that bicycles must use the crosswalks or cut through traffic to cross Washington Avenue. The intersections are planned to have extended gutter pans to provide a visual cue to people in motor vehicles to avoid the corners of the street. This is intended as a space for people on bicycles to wait for an opportunity to cross Washington Avenue.

COLUMBUS, OH

SUMMIT STREET

The City of Columbus constructed a two-way protected bike lane facility along Summit Street between 11th Avenue and East Hudson Street. This project is still under construction but is operating for public use. Summit Street is a one-way multi-lane roadway. The separated bicycle facility is located along the left side of roadway. Green colored pavement was used only for two-stage turn queue boxes. The buffer type and width varies throughout the corridor using a mixture of on-street parking, transit bus islands, thermoplastic striping, plastic bollards, and raised plastic lane separators.

INTERSECTION TREATMENTS

The two-way separated bike lane crosses several intersections along Summit Street. The facility is striped through both signalized and non-signalized intersections using a yellow dashed centerline, and dashed white lines outlining the width of the facility.

There are protected bicycle signal heads used at the signalized intersections along the facility. The bicycle signals were mounted on the signal poles approximately 7 feet off the ground. Supplemental “BICYCLE SIGNAL” signs have been placed directly below the signals to indicate their purpose. Both near-sided and far sided bicycle signals heads were used throughout the corridor (See Image 19). The bicycle signals currently operate under timed, protected bicycle only phasing. The City has not yet determined the precise timings that will be used but expect there to be at least two or three different bicycle signal timings. These

various signal timings will be programmed into the signals to account for the varied traffic flows that occur throughout a daily 24-hour period.

Two-stage turn queue boxes were also provided at the signalized intersections. Their placement varied depending on whether on-street parking was present. In locations with on-street parking, the two-stage turn queue boxes were placed between the separated bike lanes and the standard motor vehicle lanes. In locations that did not have on-street parking the two-stage turn queue boxes were located between the crosswalk and separated bike lanes.

The intersections along Summit St. are signed as “ONE WAY EXCEPT BIKE LANE” throughout the corridor to increase driver expectancy of contraflow bicycle traffic.



Photo 19: Near Sided and Far Sided Bicycle Signal Heads

TRANSITIONS

The southern transition of the two-way protected bike lane is located at 11th Street. The facility transitions from a two-way facility to a one-way facility as it continues south along Summit Street. As bicycles approach the facility from the west on 11th Street, a green bicycle lane appears approximately 40 feet before the stop bar between the standard vehicle lanes. The bike lane is located to the left of the westbound vehicle lane so that people on bikes may enter the two-way facility on Summit Street without conflicting with westbound and right turning motor vehicles.

LESSONS LEARNED

This project is still under the contractors control so the City of Columbus has not been able to gather significant findings as to its performance. From the initial feedback that has been received from the public, the facility is operating efficiently and is a comfortable bicycle route. The only negative criticism has been due to concern over sight distance at the areas with on-street parking.

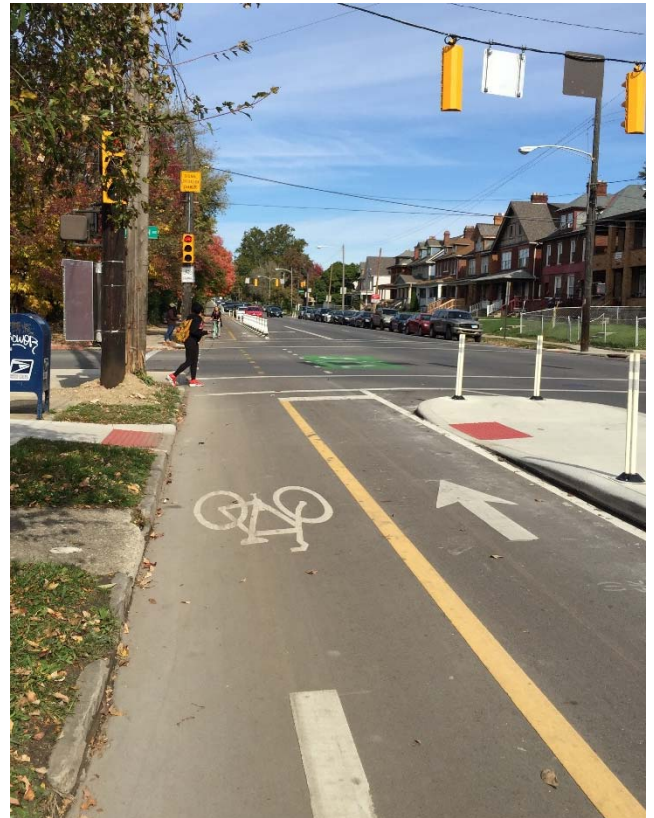


Photo 20: Summit Street Two-Way Protected Bicycle Lanes

CONCLUSIONS

INTERSECTION SIGNALIZATION

Two-way separated bicycle facilities are being installed throughout the United States by State, City, County, and local governments. Because these types of facilities are relatively new in the United States, approaches to intersection design, signalization, pavement markings and signage varies between agencies. Many facilities have been installed along City owned roadways and the MUTCD experimental approval process is not always formally documented. This leads to variation in design of the facilities but also limited post-implementation evaluation data.

This research effort revealed some common approaches when converting a two-way street to a one-way street with a two-way bike lane, as in the facilities in Pittsburgh and Madison. Likewise, the Washington, DC example retained a two-way street with the addition of a two-way separated bike lane. In all three of these examples the signal faces are left in place to function for both the vehicular and bicycle traffic as well as the contraflow bicycle traffic. This signal configuration uses a timed signal for both directions of bicycle traffic simultaneous with vehicular traffic. In terms of intersection efficiency for motor vehicles this allows for a decreased amount of delay but does not provide adequate protection for bicyclists. This shared signal does not offer protection to cyclists from turning vehicles or establish a mechanism for bicyclists to turn onto side streets.

Some of the case studies included signage at the intersections to direct bicyclists to use the pedestrian signals. This method has revealed low compliance with such a movement and the agency officials that were interviewed do not view this treatment as ideal.

Installing actuated bicycle signalization with protected bicycle signals and two-stage turn queue boxes can help increase the protection of bicycles at intersections and improve the intersection functionality. If timed appropriately, this can result in little impact on intersection delay. This is typically the preferable intersection configuration for two-way separated bike lanes but because of constraints such as budget, right-of-way, roadway capacity, bicycle delay, and traffic volumes it is not always feasible to install.

When determining bicycle signal detection and automation needs, it is key to understand the impact the bicycle signal could have on traffic queuing and the volume of bicycles at the intersection. It is not ideal to have a protected bicycle green phase during every signal cycle if bicycle traffic at the intersection is infrequent. Using automated signal detection technology can remove the protected bicycle phase from the intersection signal operations when bikes are not present. Likewise, the bicycle signal will only be activated when a bicycle is present at the intersection. When using actuation, it is helpful to install advanced signal detection to trigger the signal phasing before a bicycle reaches the intersection but also to account for the late arriving cyclist. Signal indication technology that communicates to cyclists that their presence has been detected by the signal is also helpful.

This typically leads to a higher signal compliance from people riding bicycles and more predictable bicycle-vehicle interactions.

Actuated bicycle signal technology could be any of the following; push buttons, inductive loops, radar, and video technology. Push buttons are often not preferable as cyclists have to dismount their bicycles to reach the signal or a foot rest needs to be installed so that a person can remain balanced on their bike when activating the signal. Inductive loops do not allow the same flexibility as radar and video technology as they are permanently installed in the pavement. Additionally, inductive loops narrow the detection area for bicycles creating actuation issues, as observed in the example in Eugene, OR. An advantage of both radar and video technology is that bike counts and turning movements can be automatically collected through the equipment. The main advantage video detection has over radar is the ability to assess signal compliance. This can be done through either a manual visual review of footage or by using a combination of video and radar technology. However, in the case of the Delaware Avenue project, it would be up to DelDOT regarding the detection used for bikes.

When pairing video and radar detection for bicyclists at an intersection, the video detection is typically focused at the stop bar location. The radar is used to detect a cyclist as they approach the intersection as well as movements made after the stop bar through the intersection. The pairing of technology offers the ability to program signal compliance assessment based on bicycle location relative to signal timing.

INTERSECTION DESIGN TREATMENTS

The intersections of each of the case studies varied in regards to pavement markings and signage. Many of the facilities examined provided some type of pavement delineation through intersections, typically reserving green colored pavement for conflict areas such as; stop-controlled intersections, transit stops, commercial driveways, and parking garage approaches.

The signage provided at intersections typically conveyed a message to motorists to be aware of two-way bicycle traffic on a one-way street, although sign design and messaging varied. The presence and design of signage at each of the facilities varied not only between case studies but also within some of the projects themselves. This type of variation in messaging can lead to confusion for all users and result in issues with compliance and safety. Therefore, it is vital to maintain identical intersection and signage throughout a facility.

The project being installed in Hennepin County Minnesota is proposing the use of a bicycle signal that offers a protected bicycle phase along with a permissive bicycle and vehicle phase. Once installed, this will be the first of its kind in the United States, but could offer an ideal intersection design that could be replicated on similar facilities throughout the country.

FACILITY WIDTH

Retrofitting a roadway to include a two-way separated bike lane can become a balancing act in regards to roadway space allocation. The desired minimum width of the two-way facility is 12 feet. However, narrower facilities have been installed effectively for short distances, such as the 0.5 mile long and 8' wide two-way facility on First Street NE in Washington, DC. Likewise, when using a striped buffer with flexible delineators, it is not recommended to use a width less than 3 feet. However, Washington, DC, Pittsburgh, PA, and Eugene, OR have used narrower widths successfully. This is not ideal but given the roadway constraints it was the option that worked best when balancing all of the needs of the project.